PW
System Requirement Specification

Business Partner Recognition & Verification Platform (NFT / Blockchain)

Version: 1.0

Vietnam, 2026

INTRODUCTION

Purpose

This System Requirement Specification (SRS) is the common reference for the PW platform. It turns the business processes, calculation logic (Class-based pricing, loyalty discount, reward points, ranking) and control rules (KYC, one-time transfer, soulbound Awards, Admin-only revocation) defined in the BRD and BPMN into detailed, testable use cases. It gives the Development and QA teams an exact basis to build and test the system, gives the business mentor and Product Owner a single place to agree on the solution, and serves as the baseline for acceptance and change management throughout the project.

Project scope

PW is an international web platform that uses NFT / blockchain to recognize and authenticate business-partner relationships. PW acts as the issuing authority: Companies and Individuals can purchase a Title (Philanthropist, Investor, Bridge) for themselves or as a gift, which is issued as a Distinction NFT, or be granted an Award by PW, which is issued as a soulbound Award NFT. Every recognition is publicly listed in the Public Directory and can be verified on-chain by anyone.

- Estimation for the project is about 2 months (8 weeks), including: 01 PM, 01 BA, 01 UI/UX, 02 APP, 02 BE, 01 Tester.
- Elicitation & Design: 4–5 weeks (business analysis and BA deliverables, UI/UX design).
- Development: 3 weeks for development and bug fixing (proposed split).
- Test: 1 week (proposed).
- Total: 8 weeks.

In-Scope Item	Out-of-Scope Item
Item 1: Landing Page (Waitlist) and onboarding: Register Account, Submit KYC, Admin KYC review, Classify Business (Class 0–7).	Item 1: Deploying source code and operating blockchain nodes (Alchemy is used).
Item 2: Account & wallet: Log in, custodial wallet created on demand via Circle, View Wallet, View My NFTs, Link / Unlink non-custodial wallet.	Item 2: End-user training.
Item 3: Title purchase: Browse Titles, View Pricing by Class, Purchase for myself, Gift to another Member, Mint Distinction NFT.	Item 3: Building custom wallet / node / KYC infrastructure (Circle and Alchemy are used).
Item 4: Transfer: one-time transfer of a	Item 4: Accept / decline step for gifted or

In-Scope Item	Out-of-Scope Item
Distinction NFT (to own non-custodial wallet or to another party), custody tracking, periodic reconciliation.	granted NFTs (NFTs are minted directly and active).
Item 5: Points & pricing: price by Class, 20% Year-2 discount, reward points 10/10/10, point balance, use points.	Item 5: Manual "Create wallet" action for users; manual NFT minting screen.
Item 6: Public Directory, automatic Ranking and on-chain verification (Valid / Revoked / Invalid).	Item 6: Peer-to-peer mutual verification between companies (PW is always the issuer).
Item 7: Admin Portal: Review KYC, Grant Award (soulbound Award NFT), Revoke NFT, Adjust Points, Audit Logs.	Item 7: Trading / resale marketplace for NFTs.
Item 8: AI Assistant: Q&A, NFT lookup, platform guidance, Title recommendation.	

Conventions (Acronyms)

Acronym	Reference
PW	The PW platform (issuing authority of recognition NFTs)
NFT	Non-Fungible Token
KYC	Know Your Customer – identity verification
Title	Purchasable honor: Philanthropist, Investor, Bridge
Distinction NFT	NFT issued for a purchased Title; transferable once
Award NFT	NFT issued for an Award granted by PW; soulbound (non-transferable)
Member	Registered Company or Individual user
Class	Company tier 0–7 by headcount; determines the Title price
CHF	Swiss franc; 1 CHF = 1 Point (rounded up)
Custodial wallet	Wallet managed by PW via Circle on the Member's behalf
Non-custodial wallet	Wallet whose keys the Member controls

Acronym	Reference
UC / BR / NFR	Use Case / Business Rule / Non-functional Requirement
FR	Functional requirement in the BRD (FR-001 – FR-035)
BPMN D1–D6	PW TO-BE process diagrams
TBD	To be decided

Text in **[[red]]** marks an open item that needs a decision from the business mentor / Product Owner (see OPEN ISSUES). Text in **{{gold}}** marks an assumption made by the BA to keep a requirement testable; it must be confirmed.

References

Title	Reference	Description
FDD	PW Function Decomposition (see Overall structure)	Function decomposition of the project.
BPMN	[PW] BPMN TO-BE D1–D6	TO-BE business processes (source of the flows in this SRS).
BRD	[PW] Business Requirements Document	FR-001 – FR-035, NFR-001 – NFR-010.
Project brief	[MindX K18] ITBA Career Kickstart – PW business case	Original business problem and rules.
User story	PW_US_AC	36 user stories with acceptance criteria.
UI design	WF_PW_WIP (Figma export)	Screens referenced in the screen descriptions.

OVERALL DESCRIPTION

Background and origin

Today, recognition between business partners relies on word of mouth, paper certificates, letters or press articles. These proofs are not stored durably, cannot be verified publicly by a third party

and are easy to copy, edit or forge. This lowers trust in international business relationships and reduces the honor value of genuine partners.
PW digitalizes this process end-to-end: after registration and KYC, a business or individual can buy a Title or be granted an Award, and PW issues the recognition as an immutable NFT on the blockchain (via Circle custodial wallets and Alchemy). All recognitions are listed in a Public Directory, ranked automatically and verifiable on-chain as Valid, Revoked or Invalid. Reward points and a Year-2 loyalty discount encourage long-term relationships with PW.

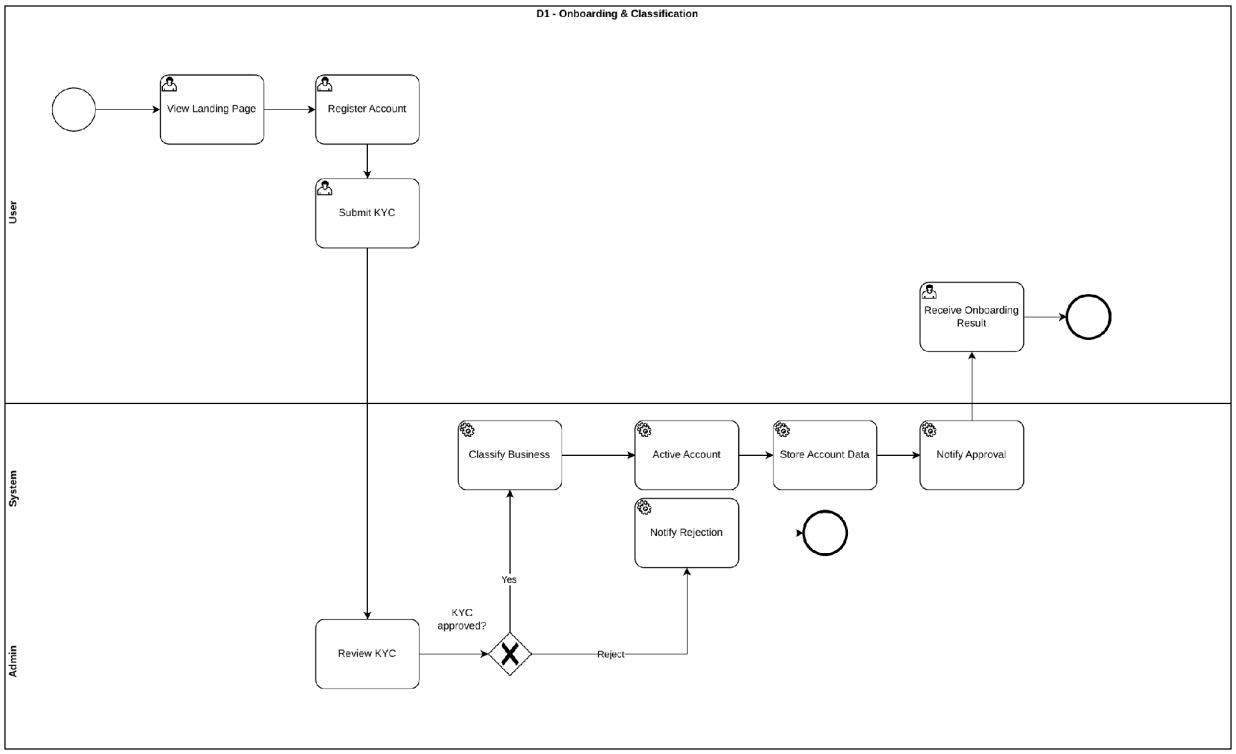
Overall structure

Function decomposition of PW (groups and use cases specified in this document):


PW Platform
1. Onboarding & Classification
UC01 Join Waitlist
UC02 Register Account
UC03 Log In
UC04 Submit KYC
UC05 Review KYC (Approve / Reject)
UC06 Classify Business into Class 0-7
2. Account & Wallet
UC07 Create Custodial Wallet on Demand
UC08 View Wallet
UC09 View My NFTs
UC10 Link Non-custodial Wallet
UC11 Unlink Wallet
3. Title Purchase & NFT Issuance
UC12 View Pricing by Class
UC13 Browse Titles
UC14 Purchase Title for Myself
UC15 Gift Title to Another Member
UC16 Calculate Price and Apply Discount
UC17 Mint NFT
UC18 Distribute Reward Points
4. NFT Transfer & Custody
UC19 Transfer NFT to Own Non-custodial Wallet (Self-custody)
UC20 Transfer Distinction NFT to Another Party
UC21 Track NFT Custody Status
UC22 Reconcile Linked Wallets On-chain
5. Points
UC23 View Point Balance and History
UC24 Use Points
6. Public Directory & Verification
UC25 Browse Public Directory
UC26 View Profile and Recognitions
UC27 Verify NFT On-chain
UC28 Calculate Ranking
7. Administration
UC29 Grant Award
UC30 Revoke NFT
UC31 Manually Adjust Points
UC32 View Audit Logs
8. AI Assistant
UC33 Ask AI Assistant

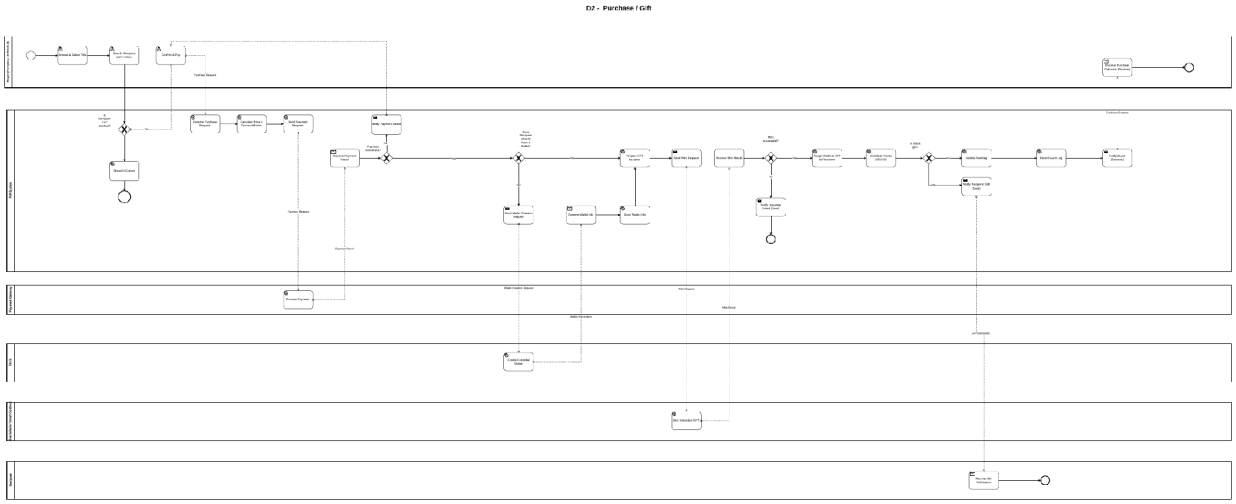
D1 – Onboarding & Classification

Register → Submit KYC → Admin review → (approved) Classify Business, activate account, notify / (rejected) notify rejection.



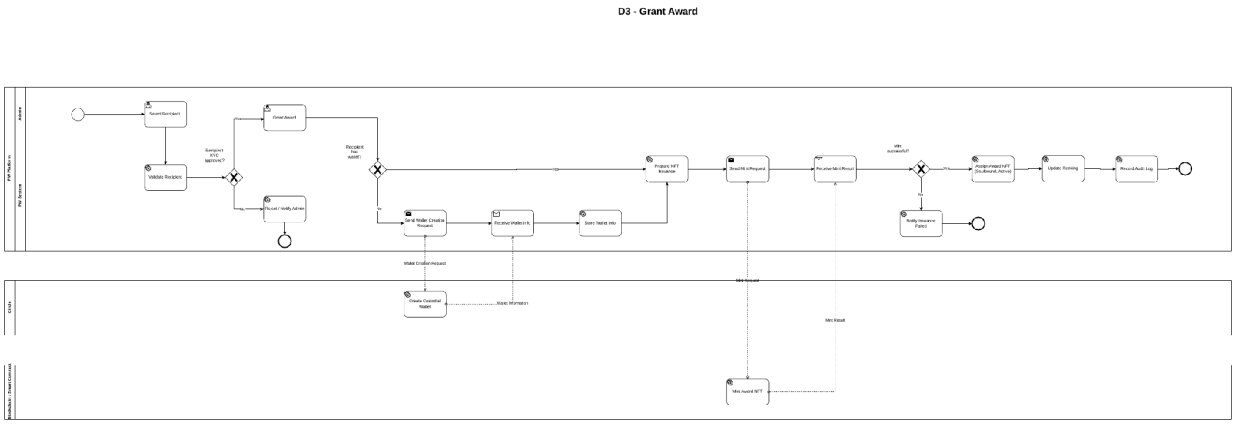
D2 – Purchase / Gift

Select Title → specify recipient (self / other) → recipient KYC check → Confirm & Pay → payment → wallet check / creation (Circle) → mint Distinction NFT → assign to recipient → points 10/10/10 → notify recipient (gift) → Ranking → Audit Log → notify buyer.



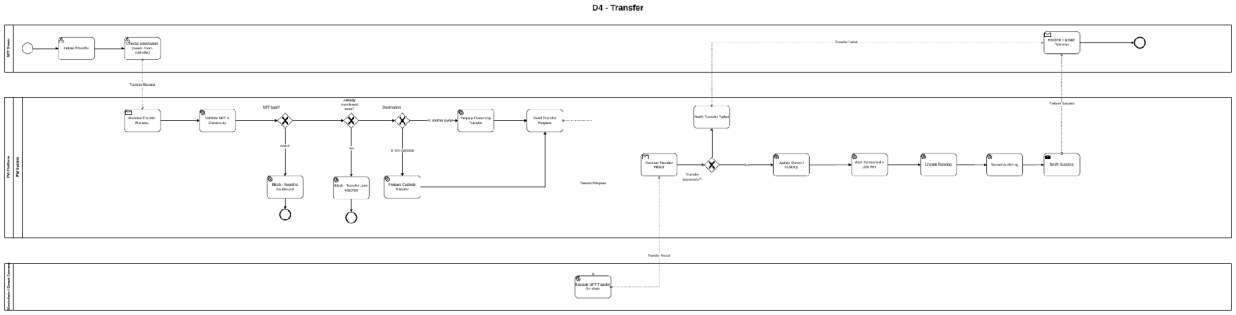
D3 – Grant Award

Admin selects recipient → KYC check → wallet check / creation → mint soulbound Award NFT → assign → Ranking → Audit Log.



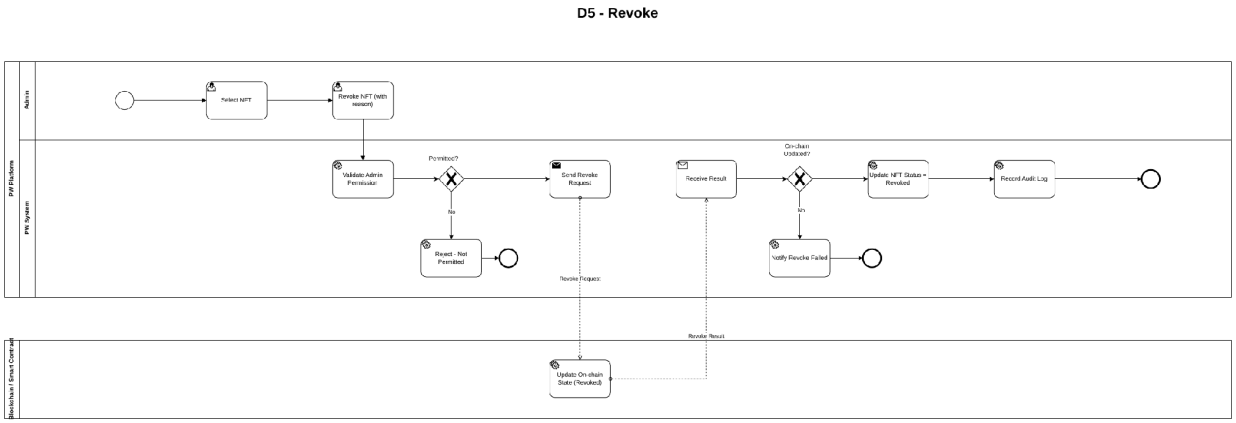
D4 – Transfer

Owner initiates transfer → NFT type check (Award blocked) → already transferred once? → destination (another owner / own non-custodial) → on-chain transfer → mark consumed + lock → Ranking → Audit Log.



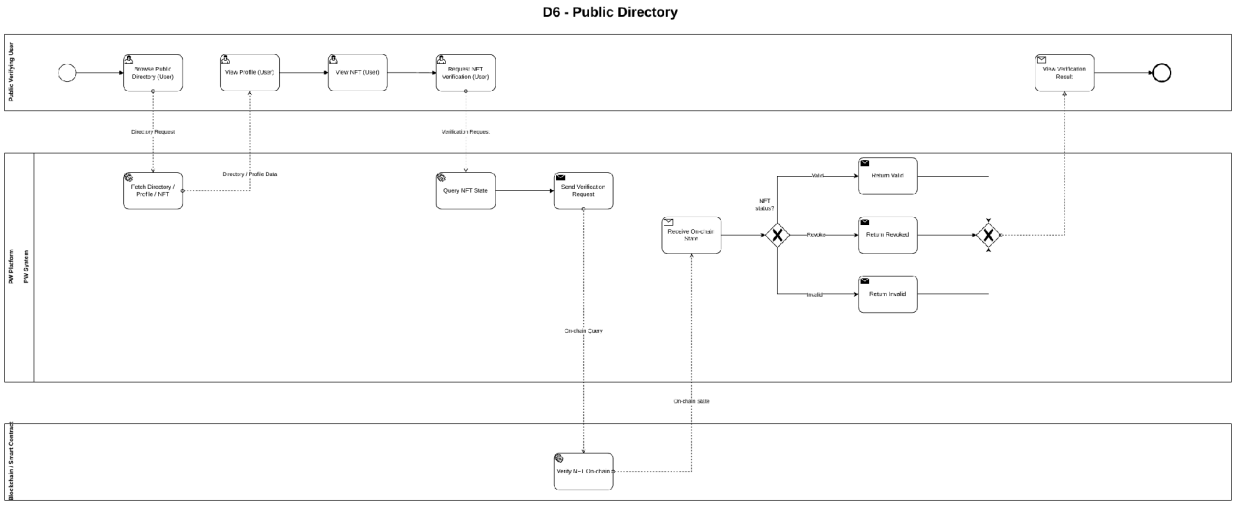
D5 – Revoke

Admin selects NFT with reason → permission check → on-chain revoke → status Revoked → Audit Log.



D6 – Public Directory & Verification

Browse Directory → view profile / NFT → request verification → on-chain query → Valid / Revoked / Invalid.



Users and permissions

Users

Role code	Role name	Description
R01	Visitor / Public Verifying User	Anyone without login: joins the waitlist, views pricing, browses the Directory, verifies NFTs, asks general AI questions.
R02	Company	Registered business Member: KYC, Class 0–7, buys/gifts Titles, holds and transfers NFTs, earns/uses points.

Role code	Role name	Description
R03	Individual / Public Figure	Registered personal Member with the same capabilities as a Company except headcount-based Class.
R04	PW Admin	PW operator: reviews KYC, corrects Class, grants Awards, revokes NFTs, adjusts points, views Audit Logs.
R05	PW System	Automated capabilities: classification, wallet creation, pricing, minting, points, ranking, custody tracking, reconciliation.

External systems (not user roles): Circle (custodial wallets), Blockchain / Smart Contract via Alchemy, Payment Gateway, AI Service, Email service.

Permission matrix

Use Case	R01	R02	R03	R04	R05	Note
1/ Onboarding & Classification						
UC01 Join Waitlist	✅	❌	❌	❌	❌	Anyone, no login
UC02 Register Account	✅	❌	❌	❌	❌	Visitor becomes R02/R03
UC03 Log In	❌	✅	✅	✅	❌	
UC04 Submit KYC	❌	✅*	✅*	❌	❌	Own account only
UC05 Review KYC (Approve / Reject)	❌	❌	❌	✅	❌	
UC06 Classify Business into Class 0–7	❌	❌	❌	✅*	✅*	System auto-assigns; Admin corrects during KYC review
2/ Account & Wallet						
UC07 Create Custodial Wallet on Demand	❌	❌	❌	❌	✅*	Automatic on first issuance
UC08 View Wallet	❌	✅*	✅*	❌	❌	Own wallet only
UC09 View My NFTs	❌	✅*	✅*	❌	❌	Own NFTs only
UC10 Link Non-custodial Wallet	❌	✅*	✅*	❌	❌	KYC-approved; max 1

Use Case	R01	R02	R03	R04	R05	Note
						linked wallet
UC11 Unlink Wallet	❌	✅*	✅*	❌	❌	Own linked wallet
3/ Title Purchase & NFT Issuance						
UC12 View Pricing by Class	✅	✅	✅	✅	❌	
UC13 Browse Titles	❌	✅*	✅*	❌	❌	Purchase actions only when KYC Approved
UC14 Purchase Title for Myself	❌	✅*	✅*	❌	❌	KYC Approved
UC15 Gift Title to Another Member	❌	✅*	✅*	❌	❌	Buyer and recipient KYC-verified
UC16 Calculate Price and Apply Discount	❌	❌	❌	❌	✅*	Automatic
UC17 Mint NFT	❌	❌	❌	❌	✅*	Triggered only by payment or grant
UC18 Distribute Reward Points	❌	❌	❌	❌	✅*	Paid transactions only
4/ NFT Transfer & Custody						
UC19 Transfer NFT to Own Non-custodial Wallet (Self-custody)	❌	✅*	✅*	❌	❌	Own Active Distinction, transfer unused
UC20 Transfer Distinction NFT to Another Party	❌	✅*	✅*	❌	❌	Own Active Distinction, transfer unused
UC21 Track NFT Custody Status	❌	❌	❌	❌	✅*	Automatic
UC22 Reconcile Linked Wallets On-chain	❌	❌	❌	❌	✅*	Scheduler
5/ Points						
UC23 View Point Balance and History	❌	✅*	✅*	❌	❌	Own points only
UC24 Use Points	❌	✅*	✅*	❌	❌	Within balance; mechanism TBD
6/ Public Directory & Verification						

Use Case	R01	R02	R03	R04	R05	Note
UC25 Browse Public Directory	✅	✅	✅	✅	❌	Public data only
UC26 View Profile and Recognitions	✅	✅	✅	✅	❌	Public data only
UC27 Verify NFT On-chain	✅	✅	✅	✅	❌	
UC28 Calculate Ranking	❌	❌	❌	❌	✅*	Automatic
7/ Administration						
UC29 Grant Award	❌	❌	❌	✅	❌	
UC30 Revoke NFT	❌	❌	❌	✅	❌	
UC31 Manually Adjust Points	❌	❌	❌	✅	❌	
UC32 View Audit Logs	❌	❌	❌	✅	❌	Read-only
8/ AI Assistant						
UC33 Ask AI Assistant	✅*	✅	✅	❌	❌	Visitors: general questions only

Legend:

- ✅: Full permission
- ✅* : Permission with limits (see Note)
- ❌ : No permission

Assumptions and dependencies

Assumptions

The following are assumed to be ready or true for the use cases to work:

- **Authentication and authorization:** every user performing a use case other than the public ones (UC01, UC12, UC25–UC27, UC33) is authenticated and authorized according to the permission matrix (role-based access control).
- **Identity data:** Members complete KYC before buying, being gifted, being granted or owning an NFT; KYC data is kept up to date.
- **Recipient model:** a gift recipient is a KYC-verified PW Member selected by the buyer, and a custodial wallet is created for the recipient if missing (BRD To-Be and BPMN D2). **User**

stories US-TN03 / US-AW01 still describe entering a wallet address and must be aligned.

- **Wallet signing:** Members who want self-custody have a non-custodial wallet able to sign messages.
- **Configuration:** the Class table, the Title price table and the Year-2 threshold are configured before go-live (**values TBD**).

Dependencies

External services the platform depends on; an outage affects the listed use cases:

- **Circle** – custodial wallet creation (UC07, UC14, UC15, UC29).
- **Blockchain / Smart Contract via Alchemy** – mint, transfer, revoke, verification and reconciliation (UC17, UC19–UC22, UC27, UC30).
- **Payment Gateway** – Title payment (UC14, UC15).
- **AI Service** – AI Assistant (UC33).
- **Email service** – notifications (UC01, UC05, UC15, UC29, UC30).

USE CASE ANALYSIS AND MAIN BUSINESS FLOWS

The table below lists all use cases and their traceability to the BRD functional requirements and the BPMN diagrams.

UC ID	Use case	Primary actor	BRD FR	BPMN
1/ Onboarding & Classification				
UC01	Join Waitlist	Visitor (R01)	FR-001	D1 (entry)
UC02	Register Account	Visitor (R01) becoming a Company (R02) or Individual (R03)	FR-002, FR-005	D1 – Register Account
UC03	Log In	Company (R02), Individual (R03), PW Admin (R04)	FR-006	—
UC04	Submit KYC	Company (R02), Individual (R03)	FR-003	D1 – Submit KYC
UC05	Review KYC (Approve / Reject)	PW Admin (R04)	FR-004, FR-005	D1 – Review KYC / KYC approved?
UC06	Classify Business into Class 0–7	PW System (R05)	FR-005	D1 – Classify Business /

UC ID	Use case	Primary actor	BRD FR	BPMN
				Active Account
2/ Account & Wallet				
UC07	Create Custodial Wallet on Demand	PW System (R05)	FR-007	D2, D3 – Does recipient already have a wallet? / Circle
UC08	View Wallet	Company (R02), Individual (R03)	FR-008	—
UC09	View My NFTs	Company (R02), Individual (R03)	FR-009	—
UC10	Link Non-custodial Wallet	Company (R02), Individual (R03)	FR-031	D4 (supporting)
UC11	Unlink Wallet	Company (R02), Individual (R03)	FR-032	—
3/ Title Purchase & NFT Issuance				
UC12	View Pricing by Class	Visitor (R01), Company (R02), Individual (R03)	FR-015	—
UC13	Browse Titles	Company (R02), Individual (R03)	FR-011	D2 – Browse & Select Title
UC14	Purchase Title for Myself	Buyer – Company (R02) / Individual (R03)	FR-012, FR-013, FR-016, FR-017, FR-018	D2 – Purchase (self)
UC15	Gift Title to Another Member	Buyer – Company (R02) / Individual (R03)	FR-012, FR-013, FR-018	D2 – Purchase (gift) / Notify Recipient
UC16	Calculate Price and Apply Discount	PW System (R05)	FR-016, FR-017	D2 – Calculate Price + Discount/Points
UC17	Mint NFT	PW System (R05)	FR-013	D2, D3 – Prepare NFT Issuance / Mint
UC18	Distribute Reward Points	PW System (R05)	FR-018	D2 – Distribute Points

UC ID	Use case	Primary actor	BRD FR	BPMN
				10/10/10
4/ NFT Transfer & Custody				
UC19	Transfer NFT to Own Non-custodial Wallet (Self-custody)	NFT owner – Company (R02) / Individual (R03)	FR-010, FR-014, FR-031, FR-034	D4 – Transfer (to non-custodial)
UC20	Transfer Distinction NFT to Another Party	NFT owner – Company (R02) / Individual (R03)	FR-033, FR-014, FR-034	D4 – Transfer (to another owner)
UC21	Track NFT Custody Status	PW System (R05)	FR-034	D4 – Receive Transfer Result / Update Owner
UC22	Reconcile Linked Wallets On-chain	PW System (R05, scheduler)	FR-035	—
5/ Points				
UC23	View Point Balance and History	Company (R02), Individual (R03)	FR-019	—
UC24	Use Points	Company (R02), Individual (R03)	FR-020	D2 – Discount/Points
6/ Public Directory & Verification				
UC25	Browse Public Directory	Public Verifying User / Visitor (R01) (also Members and Admin)	FR-021	D6 – Browse Public Directory
UC26	View Profile and Recognitions	Public Verifying User / Visitor (R01)	FR-022, FR-023	D6 – View Profile / View NFT
UC27	Verify NFT On-chain	Public Verifying User / Visitor (R01)	FR-024	D6 – Request NFT Verification / NFT status?
UC28	Calculate Ranking	PW System (R05)	FR-025	D2, D3, D4 – Update Ranking
7/ Administration				
UC29	Grant Award	PW Admin (R04)	FR-026	D3 – Grant

UC ID	Use case	Primary actor	BRD FR	BPMN
				Award
UC30	Revoke NFT	PW Admin (R04)	FR-027	D5 – Revoke
UC31	Manually Adjust Points	PW Admin (R04)	FR-028	—
UC32	View Audit Logs	PW Admin (R04)	FR-029	D2–D5 – Record Audit Log
8/ AI Assistant				
UC33	Ask AI Assistant	Visitor (R01), Company (R02), Individual (R03)	FR-030	—

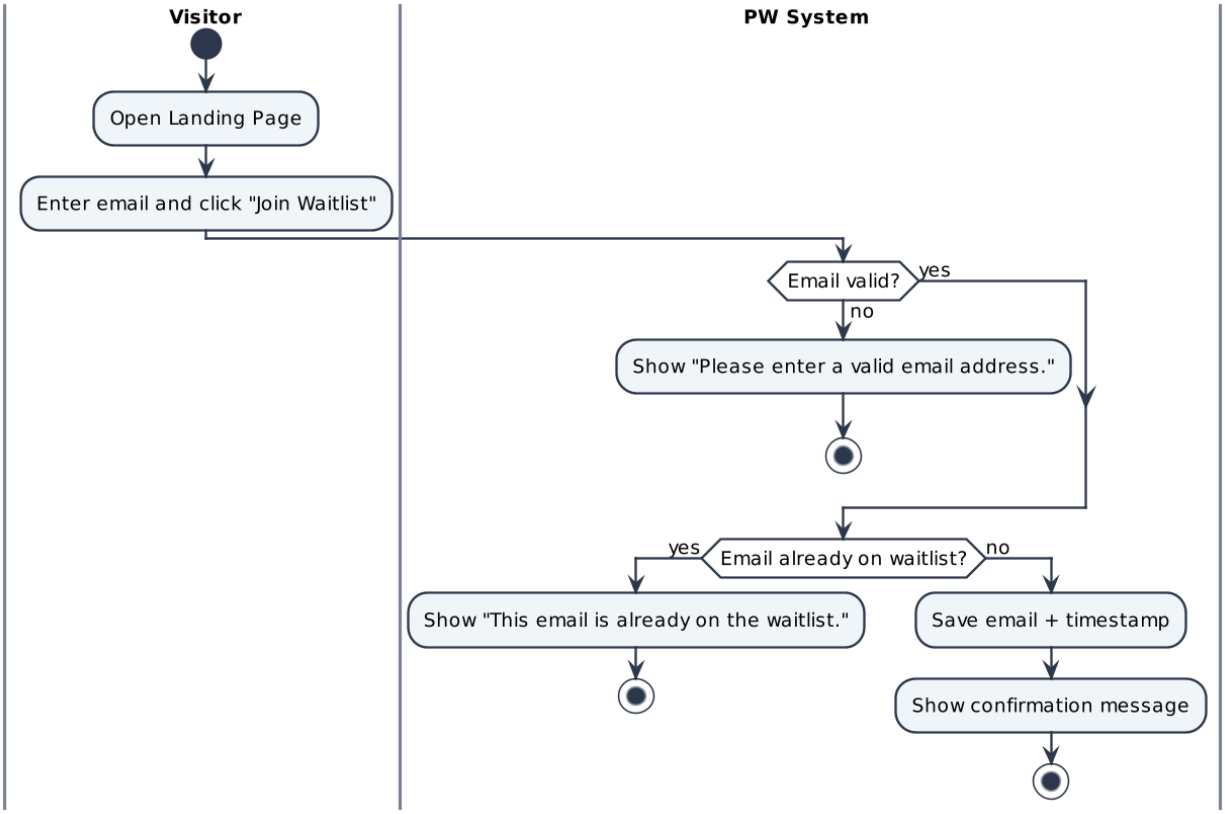
UC01: Join Waitlist

Use Case Description

UC ID	UC01
UC Name	Join Waitlist
Objective	Allow an anonymous visitor to leave an email address on the Landing Page so that PW can notify them when registration opens.
Actor	Visitor (R01); PW System (R05)
Trigger	The Visitor clicks "Join Waitlist" on the Landing Page.
Pre-condition	- The Landing Page is publicly accessible; no login is required.
Post-condition	**Success:** • The email is stored once in the waitlist with the submission timestamp. • The confirmation message is displayed. **Failure:** • No waitlist record is created; an inline error message is displayed.
Basic flow	1. Visitor opens the Landing Page. 2. System displays the waitlist section with the Email field and the "Join Waitlist" button. 3. Visitor enters an email address and clicks "Join Waitlist". 4. System validates that the email is not empty and has a valid format.

	5. System checks that the email does not already exist in the waitlist. 6. System saves the email with the submission timestamp. 7. System displays "You're on the list. We'll email you when PW opens."
Alternative flow	**A1 – Email already on the waitlist** 1. At step 5 the email already exists. 2. System does not create a duplicate record. 3. System displays "This email is already on the waitlist." 4. End of use case.
Exception flow	**E1 – Empty or invalid email** 1. At step 4 validation fails. 2. System displays "Please enter a valid email address." under the field. 3. The email is not saved; Visitor may correct it and resume at step 3. **E2 – System error** 1. Saving fails. 2. System displays a generic error "Something went wrong. Please try again." and no record is created.
Business rule	• BR01 – An email address can exist in the waitlist only once (comparison is case-insensitive – assumption). • BR02 – Joining the waitlist does not create a PW account.
Non-functional requirement	• NFR01 – The confirmation or error message is displayed within 2 seconds after clicking "Join Waitlist" (proposed).
Acceptance criteria	**AC01 – Join with a valid email** Given the visitor is on the Landing Page, when a valid email is submitted, then the email is saved and "You're on the list. We'll email you when PW opens." is displayed. **AC02 – Duplicate email** Given the email already exists in the waitlist, when it is submitted again, then no duplicate is created and "This email is already on the waitlist." is displayed. **AC03 – Invalid email** Given the field is empty or the format is wrong, when the visitor submits, then "Please enter a valid email address." is displayed and nothing is saved.
Traceability	BRD: FR-001 · BPMN: D1 (entry)

Activity Diagram



Screen Description

Screen 1: Landing Page – Waitlist section

[Screenshot: Figma frame "PW — Landing Page (Optimized Flow) / (Error) / (Confirmed)"]

#	Name	Control type	Data type	Required	Description
1	Email address	Textbox	String	Yes	Placeholder "Enter your email". Validated on submit: not empty, valid email format (max 254 characters – assumption).
2	Join Waitlist	Button	N/A		Submits the email. Disabled while the request is being processed.
3	Success message	Label	String		"You're on the list. We'll email you when PW opens." Shown after a successful save (Confirmed state).

#	Name	Control type	Data type	Required	Description
4	Error message	Label (inline)	String		"Please enter a valid email address." or "This email is already on the waitlist." (Error state).

UC02: Register Account

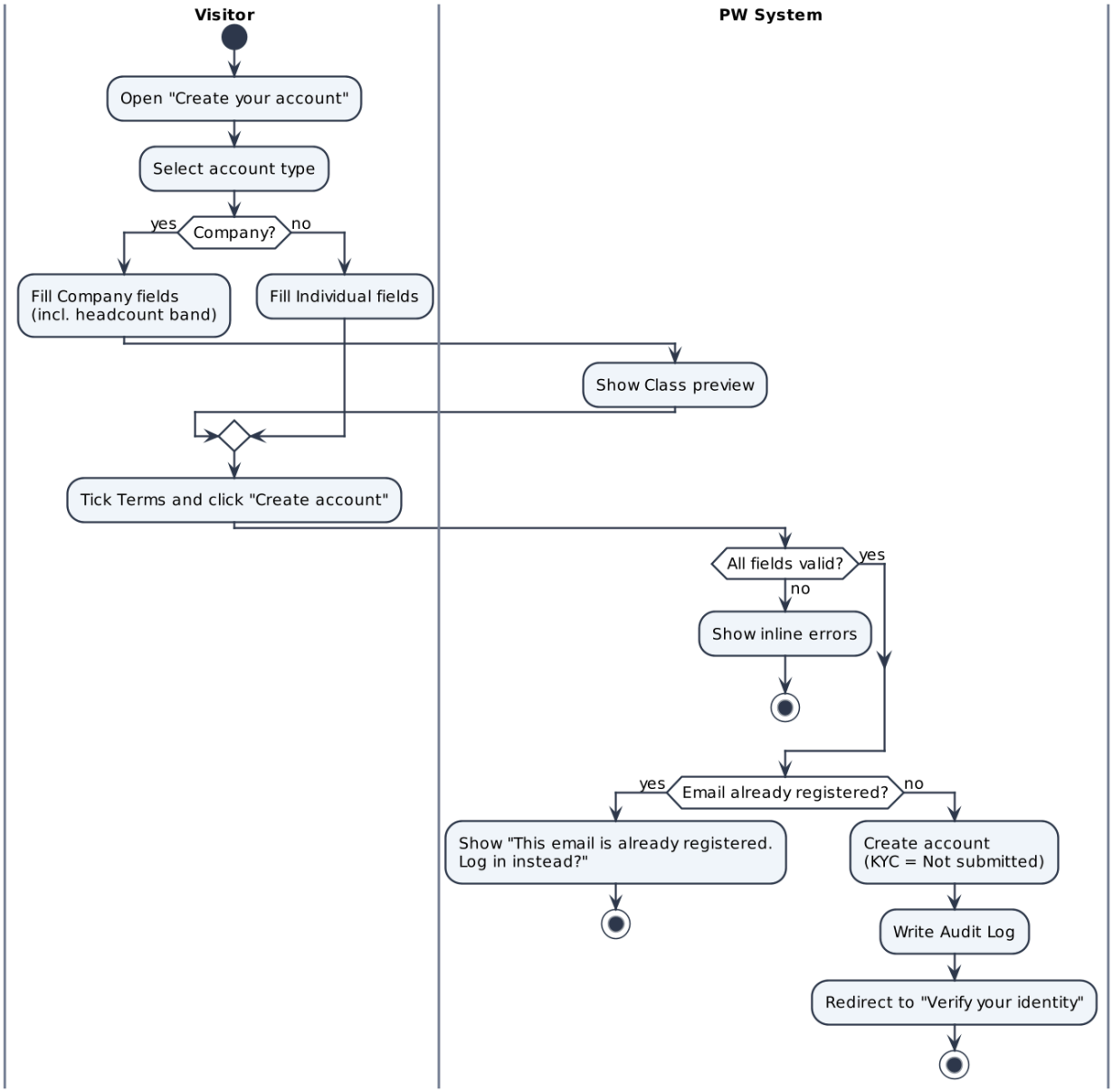
Use Case Description

UC ID	UC02
UC Name	Register Account
Objective	Allow a new user to create a PW account as a Company or an Individual (Public Figure). A Company declares its headcount band, which gives a provisional Class. After registration the user continues to KYC (UC04).
Actor	Visitor (R01) becoming a Company (R02) or Individual (R03); PW System (R05)
Trigger	The Visitor opens "Create your account" and clicks "Create account".
Pre-condition	- The Visitor is not logged in. - The email is not yet registered in PW.
Post-condition	**Success:** • An account is created with type = Company or Individual and KYC status = "Not submitted". • Company: the headcount band and provisional Class are stored. • The user is signed in and redirected to "Verify your identity" (UC04). **Failure:** • No account is created; errors are shown on the form.
Basic flow	1. Visitor opens "Create your account". 2. System displays the account type selector (Company selected by default (assumption)) and the Company form. 3. Visitor selects the account type. 4. Visitor fills in the fields of the selected type (see Screen 1). 5. Company only: when a headcount band is selected, System shows the live preview "Your class: Class X (based on headcount)" (UC06). 6. Visitor ticks "I agree to the Terms of Service and Privacy Policy". 7. Visitor clicks "Create account". 8. System validates all fields and checks that the email is not already

	registered. 9. System creates the account with KYC status = "Not submitted" (Company: stores headcount band and provisional Class). 10. System writes an Audit Log entry (account created) and redirects the user to "Verify your identity".
Alternative flow	**A1 – Switch account type** 1. At step 3/4 the Visitor switches between Company and Individual. 2. System shows only the fields of the selected type; values of hidden fields are not submitted. 3. Resume at step 4. **A2 – Individual account** 1. At step 3 Visitor selects Individual. 2. System shows Full name, Email, Country, Password, Confirm password; no headcount and no Class. 3. Resume at step 6; at step 9 the account is created with type = Individual and no Class.
Exception flow	**E1 – Email already registered** 1. At step 8 the email exists. 2. System displays "This email is already registered. Log in instead?" with a link to Log in. 3. No account is created. **E2 – Missing or invalid fields** 1. At step 8 a required field is empty, the email format is invalid, the password is shorter than 8 characters or has no number/symbol, or Confirm password does not match. 2. System shows an inline error under each invalid field; no account is created. **E3 – Terms not accepted** 1. The Terms checkbox is not ticked. 2. "Create account" stays disabled (or System shows "Please accept the Terms of Service and Privacy Policy.").
Business rule	• BR01 – One email address = one PW account. • BR02 – Password: at least 8 characters including at least one number/symbol (exact complexity rule to be confirmed); Confirm password must match. • BR03 – Headcount is chosen from a fixed list of bands; each band maps to exactly one Class 0–7. The Class is provisional until KYC is approved. • BR04 – An Individual / Public Figure is a full user (one entity with two views: account user and public Directory profile); Individuals have no headcount and no Class. • BR05 – Registration does not create a wallet (wallet is created on demand – UC07).
Non-functional	• NFR01 – The Class preview updates within 1 second after a band is

requirement	selected (proposed). • NFR02 – Passwords are stored only as a salted hash (NFR-003).
Acceptance criteria	**AC01 – Register Company** Given type = Company and all fields are valid, when the user clicks "Create account", then the account is created with KYC = "Not submitted", headcount and provisional Class saved, and the user is redirected to KYC. **AC02 – Register Individual** Given type = Individual and all fields are valid, when the user clicks "Create account", then the account is created with type = Individual, no Class, and the user is redirected to KYC. **AC03 – Duplicate email** Given the email already exists, when the user clicks "Create account", then "This email is already registered. Log in instead?" is displayed and no account is created. **AC04 – Invalid fields** Given any field breaks a rule, when the user clicks "Create account", then inline errors are shown and no account is created.
Traceability	BRD: FR-002, FR-005 · BPMN: D1 – Register Account

Activity Diagram



Screen Description

Screen 1: Create your account – Company

[Screenshot: Figma frame "PW — Create Account (Company)"]

#	Name	Control type	Data type	Required	Description
I. Account type					
1	Account type	Segmented	Enum	Yes	Company | Individual. Switching

#	Name	Control type	Data type	Required	Description
		control			shows only the fields of the selected type.
II. Company information					
1	Registered legal entity name	Textbox	String	Yes	Legal name of the company.
2	Authorized representative full name	Textbox	String	Yes	Person acting on behalf of the company.
3	Work email	Textbox	String	Yes	Valid email format; must be unique in PW (BR01). Used as login ID.
4	Organization headcount	Dropdown	Enum	Yes	Fixed headcount bands; each band maps to one Class 0–7 (**band ranges TBD**).
5	Class preview	Label	String		"Your class: Class X (based on headcount)" – read-only, updated on band change. Displayed as "Company Class".
6	Country of incorporation	Dropdown	String	Yes	Country list.
III. Security & consent					
1	Password	Textbox (masked)	String	Yes	Min. 8 characters incl. at least one number/symbol (BR02). Show/hide toggle.
2	Confirm password	Textbox (masked)	String	Yes	Must match Password.
3	I agree to the Terms of Service and Privacy Policy	Checkbox	Boolean	Yes	"Create account" is disabled until ticked.
4	Create account	Button	N/A		Validates and creates the account (steps 8–10).
5	Log in	Link	N/A		Navigates to the Log in screen (UC03).

Screen 2: Create your account – Individual

[Screenshot: Figma frame "PW — Create Account (Individual)"]

#	Name	Control type	Data type	Required	Description
1	Account type	Segmented control	Enum	Yes	Individual selected.
2	Full name	Textbox	String	Yes	Legal full name.
3	Email	Textbox	String	Yes	Valid format; unique (BR01).
4	Country	Dropdown	String	Yes	Country of residence.
5	Password / Confirm password	Textbox (masked)	String	Yes	Same rules as the Company form.
6	Terms checkbox	Checkbox	Boolean	Yes	Same rule as the Company form.
7	Create account	Button	N/A		Creates the Individual account (no headcount, no Class).

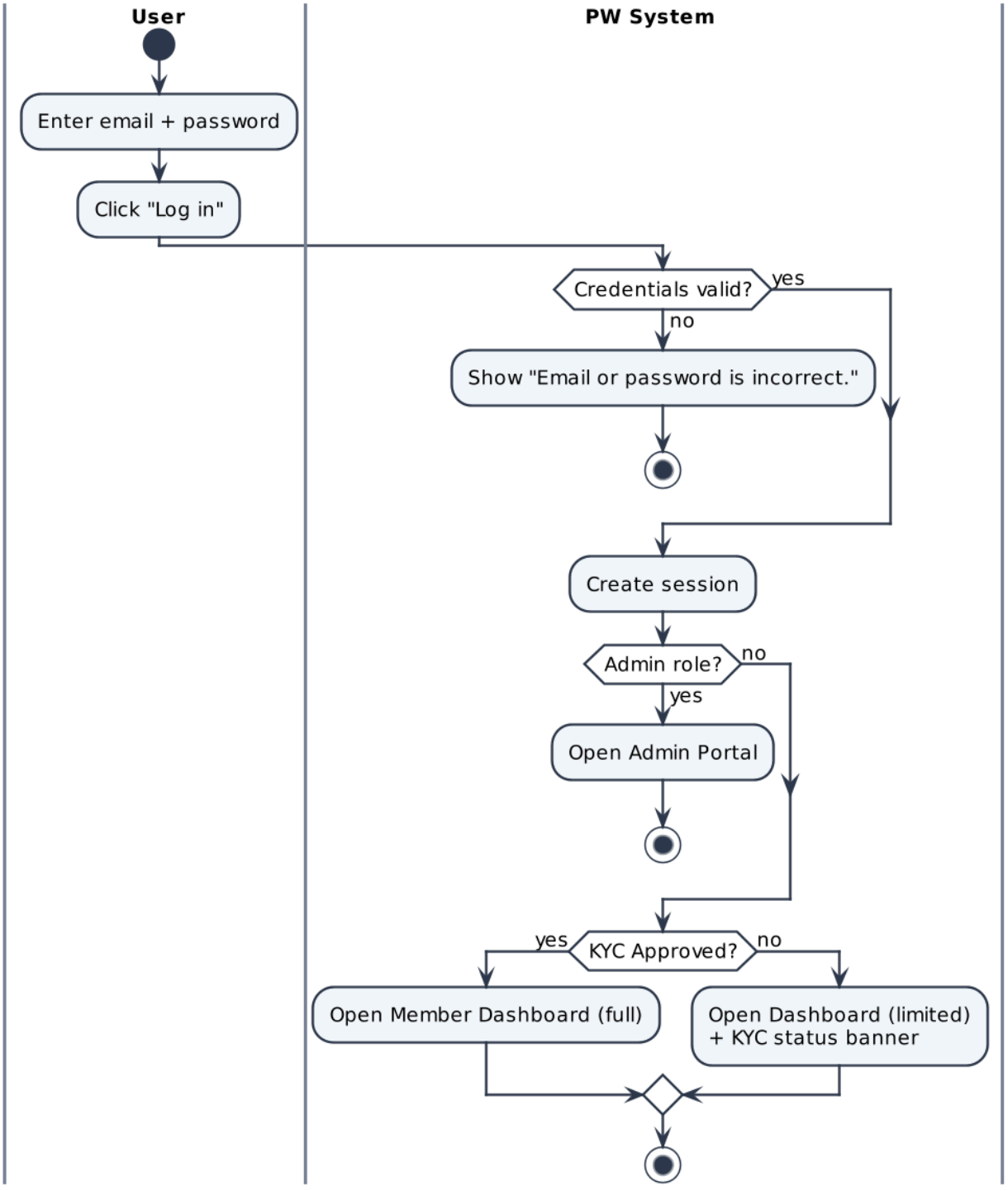
UC03: Log In

Use Case Description

UC ID	UC03
UC Name	Log In
Objective	Authenticate a registered Member or Admin with email and password and open the appropriate portal.
Actor	Company (R02), Individual (R03), PW Admin (R04); PW System (R05)
Trigger	The user clicks "Log in".
Pre-condition	- The user has a PW account.
Post-condition	**Success:** • An authenticated session is created. • Member → Member Dashboard (full mode if KYC Approved, otherwise limited mode). Admin → Admin Portal. **Failure:** • No session is created; an error message is displayed.

Basic flow	1. User opens the Log in screen. 2. User enters email and password and clicks "Log in". 3. System verifies the credentials. 4. System creates an authenticated session and identifies the user's role. 5. System opens the Member Dashboard (KYC status = "Approved").
Alternative flow	**A1 – KYC not yet approved** 1. At step 5 the KYC status is "Not submitted", "Pending review" or "Rejected". 2. System opens the Dashboard in limited mode with a banner showing the current KYC status and the next step (e.g. "Complete your identity verification"). 3. Purchase / Gift Title is disabled. **A2 – Admin user** 1. At step 4 the user holds the Admin role. 2. System opens the Admin Portal.
Exception flow	**E1 – Wrong credentials** 1. At step 3 verification fails. 2. System displays "Email or password is incorrect." and does not reveal which field is wrong. 3. User may retry from step 2.
Business rule	• BR01 – Error messages must not disclose whether the email exists. • BR02 – The Admin Portal is accessible only to users with the Admin role; a Member session can never open Admin screens.
Non-functional requirement	• NFR01 – Strong authentication and secure session handling (NFR-003). • NFR02 – Log in response within 2 seconds under normal load (proposed).
Acceptance criteria	**AC01 – Successful login** Given a Member with KYC Approved, when correct credentials are submitted, then the Member Dashboard opens. **AC02 – Wrong credentials** Given the user is on Log in, when a wrong email or password is submitted, then "Email or password is incorrect." is displayed. **AC03 – KYC not approved** Given KYC status ≠ Approved, when the user logs in, then the Dashboard opens in limited mode with a KYC banner.
Traceability	BRD: FR-006 · BPMN: —

Activity Diagram



Screen Description

Screen 1: Log in

[Screenshot: Figma frame "PW — Log In / Log In (Error)"]

#	Name	Control type	Data type	Required	Description
1	Email	Textbox	String	Yes	Registered email.
2	Password	Textbox (masked)	String	Yes	Show/hide toggle.
3	Log in	Button	N/A		Authenticates the user (steps 3–5).
4	Error message	Label	String		"Email or password is incorrect."
5	Create account	Link	N/A		Navigates to UC02.

UC04: Submit KYC

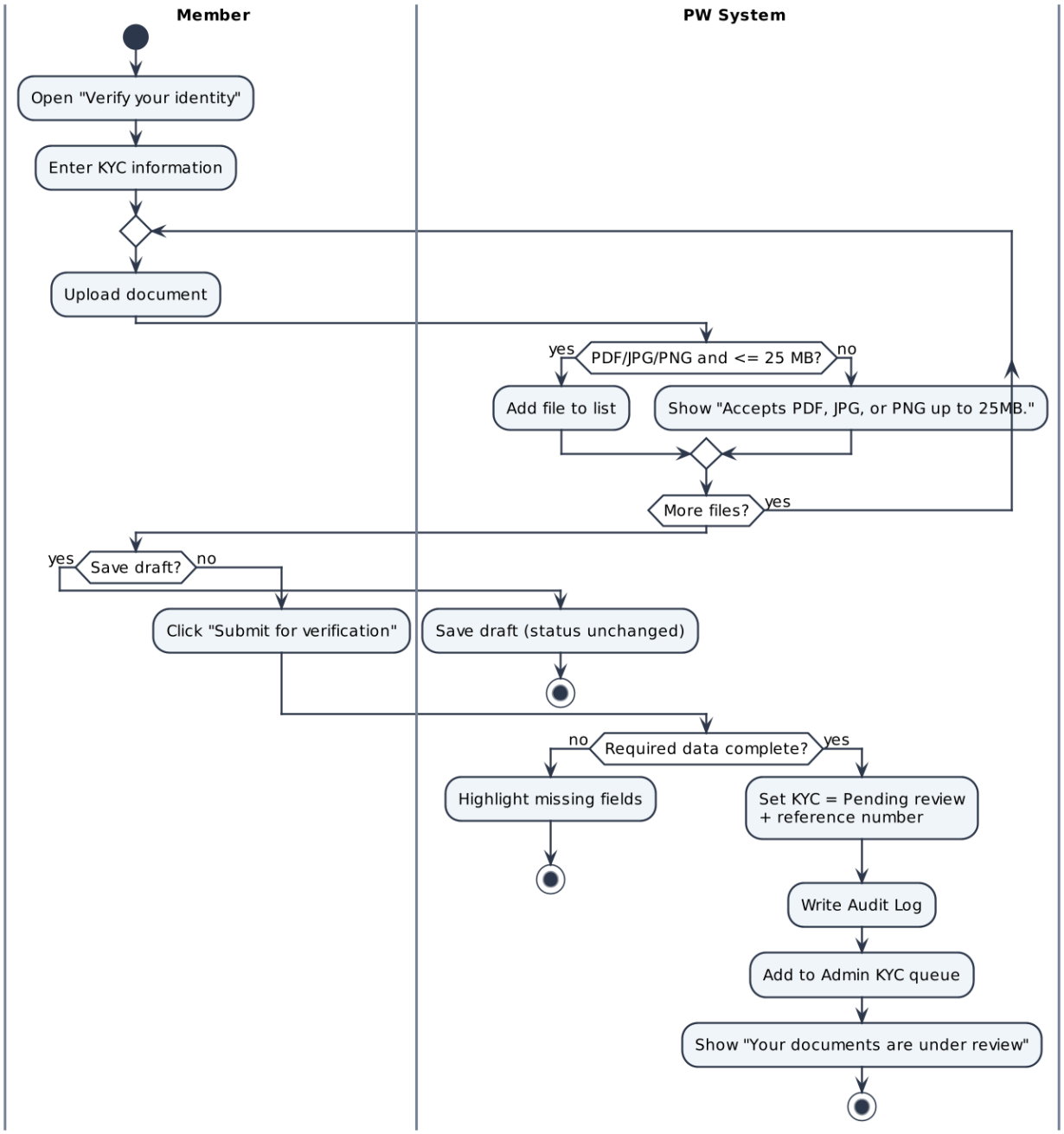
Use Case Description

UC ID	UC04
UC Name	Submit KYC
Objective	Allow a Member to submit identity information and supporting documents so that PW can verify the entity before it buys, receives or owns an NFT.
Actor	Company (R02), Individual (R03); PW System (R05)
Trigger	The Member clicks "Submit for verification" (or "Re-submit documents") on "Verify your identity".
Pre-condition	- The Member is logged in. - KYC status = "Not submitted" (first submission) or "Rejected" (re-submission).
Post-condition	**Success:** • KYC status = "Pending review"; a unique KYC reference number is generated. • The request appears in the Admin KYC queue (UC05). • An Audit Log entry is written. **Draft:**

	• Data is saved; KYC status remains "Not submitted". **Failure:** • KYC is not submitted; errors are shown.
Basic flow	1. Member opens "Verify your identity". 2. System displays the KYC form (pre-filled with the account name). 3. Member enters Legal entity or personal name, Verification document type and Document identification / Registry no. 4. Member uploads at least one document (PDF, JPG or PNG, max 25 MB each). 5. System validates each file on upload and lists accepted files. 6. Member clicks "Submit for verification". 7. System validates the required fields and that at least one document is attached. 8. System sets KYC status = "Pending review", generates the KYC reference number and writes an Audit Log entry. 9. System displays "Your documents are under review" with the reference number and the estimated review time (1–2 business days). 10. System adds the request to the Admin KYC queue.
Alternative flow	**A1 – Remove an uploaded file** 1. Before step 6 Member clicks the delete icon on a file. 2. System removes the file from the submission. **A2 – Save draft and exit** 1. Before step 6 Member clicks "Save Draft & Exit". 2. System saves the draft; KYC status stays "Not submitted". 3. When Member returns, the form is restored from the draft. **A3 – Re-submit after rejection** 1. KYC status = "Rejected". 2. At step 2 System also displays the rejection reason and the resubmission requirements. 3. Member updates information/documents and clicks "Re-submit documents"; continue at step 7.
Exception flow	**E1 – Missing required information** 1. At step 7 a required field is empty or no document is uploaded. 2. System highlights the missing field; KYC is not submitted. **E2 – Unsupported or oversized file** 1. At step 5 the file is not PDF/JPG/PNG or is larger than 25 MB. 2. System rejects the file with "Accepts PDF, JPG, or PNG up to 25MB." and does not add it. **E3 – Upload/network error** 1. Upload fails. 2. System marks the file as failed and allows retry.
Business rule	• BR01 – KYC status lifecycle: Not submitted → Pending review → Approved | Rejected; Rejected → Pending review on re-submission.

	• BR02 – Accepted files: PDF, JPG, PNG; max 25 MB per file; at least one document. • BR03 – While KYC ≠ Approved, Purchase / Gift Title is disabled and the Member cannot be selected as a gift or Award recipient. • BR04 – **Final list of KYC fields and accepted document types to be confirmed with Compliance.**
Non-functional requirement	• NFR01 – KYC documents are stored encrypted and accessible only to the owner and Admins (NFR-008). • NFR02 – File upload progress is shown; a 25 MB file uploads within 30 seconds on a 10 Mbps connection (proposed).
Acceptance criteria	**AC01 – Submit successfully** Given KYC = Not submitted and all required data + ≥1 valid file, when "Submit for verification" is clicked, then KYC = Pending review, a reference number and "Your documents are under review" (1–2 business days) are shown, and the request is in the Admin queue. **AC02 – Invalid file** Given a non-PDF/JPG/PNG or >25 MB file, when it is uploaded, then "Accepts PDF, JPG, or PNG up to 25MB." is shown and the file is not added. **AC03 – Save draft** Given some data is entered, when "Save Draft & Exit" is clicked, then the draft is saved and KYC stays Not submitted. **AC04 – Re-submit** Given KYC = Rejected, when updated documents are re-submitted, then the rejection reason was shown and KYC = Pending review.
Traceability	BRD: FR-003 · BPMN: D1 – Submit KYC

Activity Diagram



Screen Description

Screen 1: Verify your identity (KYC)

[Screenshot: Figma frame "PW — KYC Verification Hub / (In Review) / (Action Required)"]

#	Name	Control type	Data type	Required	Description
I. Identity information					
1	Legal entity or personal name	Textbox	String	Yes	Pre-filled from the account; editable.
2	Verification document type	Dropdown	Enum	Yes	e.g. Business registration certificate, Passport, National ID (**list TBD with Compliance**).
3	Document identification / Registry no.	Textbox	String	Yes	Number printed on the document.
II. Documents					
1	Upload area	File upload (drag & drop)	File	Yes	PDF, JPG, PNG; max 25 MB per file; ≥1 file. Helper text "Accepts PDF, JPG, or PNG up to 25MB."
2	Uploaded file list	List	File		File name, size, status; delete icon per file (A1).
III. Actions & status					
1	Save Draft & Exit	Button	N/A		Saves the draft (A2).
2	Submit for verification / Re-submit documents	Button	N/A		Submits the KYC (steps 6–10).
3	Rejection reason	Alert	String		Shown only when KYC = Rejected (A3).
4	Under-review panel	Panel	String		"Your documents are under review", KYC reference number, estimated review time 1–2 business days.

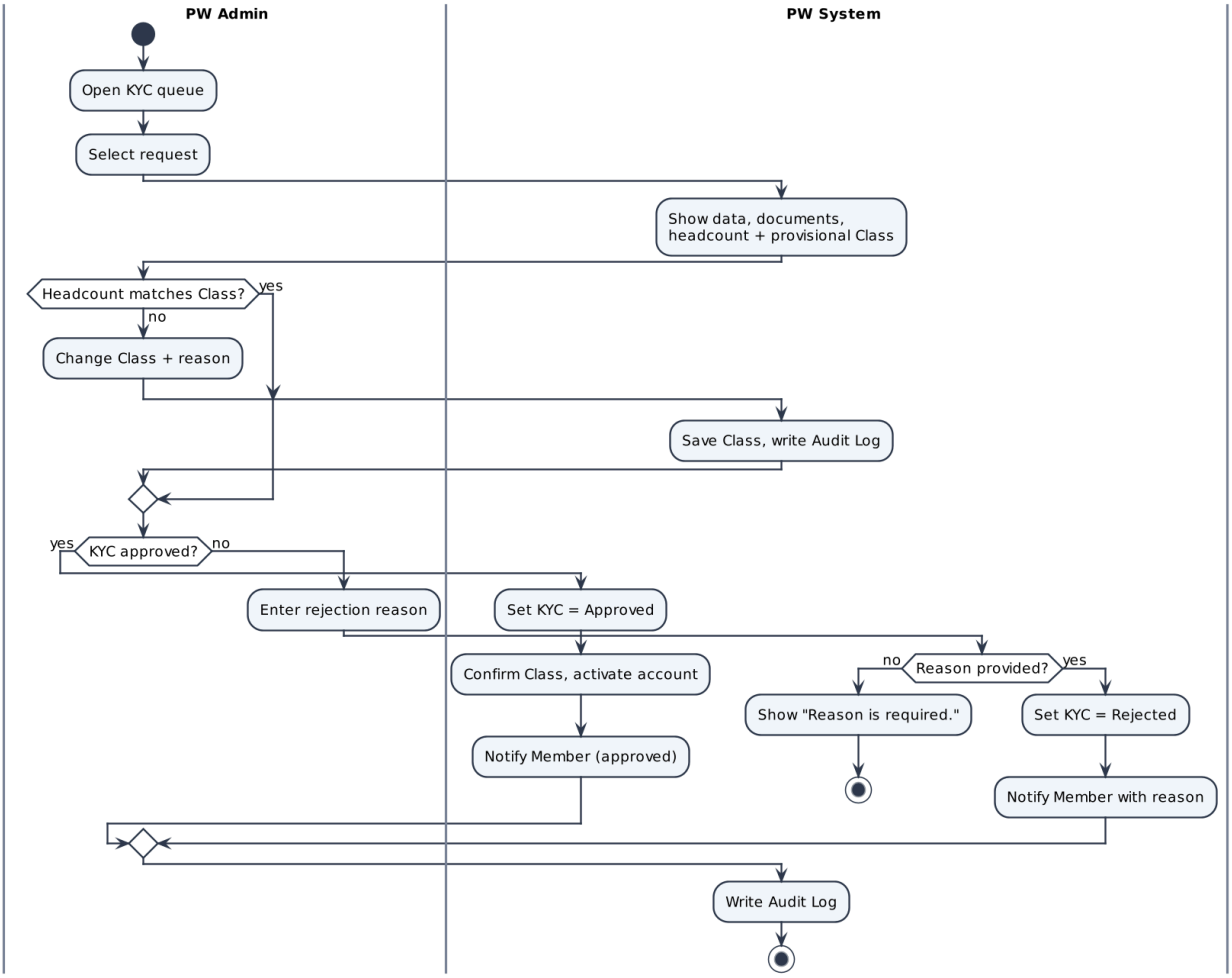
UC05: Review KYC (Approve / Reject)

Use Case Description

UC ID	UC05
UC Name	Review KYC (Approve / Reject)
Objective	Allow a PW Admin to review pending KYC requests and approve or reject them so that only verified entities can own and receive NFTs. On approval, the Company Class is confirmed and the account is activated.
Actor	PW Admin (R04); PW System (R05); Member (notified)
Trigger	The Admin opens a request from the KYC queue.
Pre-condition	- The Admin is logged in with the Admin role. - The request has KYC status = "Pending review".
Post-condition	**Approved:** • KYC status = "Approved"; Class confirmed (Company) – UC06; account activated. • Member notified (in-app + email) and able to purchase Titles. • Audit Log entry written. **Rejected:** • KYC status = "Rejected"; reason stored; Member notified with the reason and able to re-submit. • Audit Log entry written.
Basic flow	1. Admin opens "KYC Review" in the Admin Portal. 2. System lists requests with status "Pending review" (oldest first) showing reference number, name, account type, submission date. 3. Admin selects a request. 4. System displays the submitted information, documents, declared headcount band and provisional Class. 5. Admin reviews the documents and clicks "Approve". 6. System sets KYC status = "Approved". 7. System confirms the Class (Company) and activates the account (UC06). 8. System notifies the Member ("Your identity has been verified") and writes an Audit Log entry (admin, action, target, time).
Alternative flow	**A1 – Reject** 1. At step 5 Admin clicks "Reject". 2. System displays the Reason field (required). 3. Admin selects/enters the reason and confirms. 4. System sets KYC status = "Rejected", notifies the Member with the reason and writes an Audit Log entry.

	A2 – Change Class during review 1. At step 4 the documents show a different headcount. 2. Admin selects the correct Class and enters a mandatory reason. 3. System saves the new Class and records old Class, new Class, admin and time in the Audit Log; continue at step 5.
Exception flow	**E1 – Reject without reason** 1. In A1 no reason is provided. 2. System blocks the action with "Reason is required." **E2 – Request already processed** 1. Another Admin has already decided the request. 2. System displays "This request has already been processed." and refreshes the queue.
Business rule	• BR01 – Only the Admin role can approve or reject KYC. • BR02 – A rejection reason and a Class-change reason are mandatory. • BR03 – Every KYC decision and Class change is written to the Audit Log (NFR-002). • BR04 – The Class becomes final only on approval (BR-1).
Non-functional requirement	• NFR01 – Documents open in an in-browser viewer without downloading (proposed). • NFR02 – Review target: 1–2 business days per request (communicated to Members).
Acceptance criteria	**AC01 – Approve** Given a request is Pending review, when the Admin clicks "Approve", then KYC = Approved, Class confirmed, Member notified, Audit Log written. **AC02 – Reject** Given a request is Pending review, when the Admin rejects with a reason, then KYC = Rejected, Member notified with reason, Audit Log written. **AC03 – Reject without reason** Given the Admin clicks "Reject", when no reason is given, then "Reason is required." is displayed. **AC04 – Change Class** Given the documents show another headcount, when the Admin changes the Class with a reason, then the new Class is saved and the change is logged.
Traceability	BRD: FR-004, FR-005 · BPMN: D1 – Review KYC / KYC approved?

Activity Diagram



Screen Description

Screen 1: KYC Review – queue

[Screenshot: Figma frame "Admin Portal (KYC Review)"]

#	Name	Control type	Data type	Required	Description
1	Pending requests table	Table / Grid	Mixed		Columns: KYC reference, Name, Account type, Declared headcount, Provisional Class, Submitted on. Sorted oldest first.
2	Search	Textbox	String		Search by name or reference.
3	Open	Row action	N/A		Opens the request detail.

Screen 2: KYC Review – request detail

[Screenshot: Figma frame "Admin Portal (KYC Review – detail)"]

#	Name	Control type	Data type	Required	Description
1	Applicant information	Label group	String		Name, account type, email, country, headcount band, provisional Class, KYC reference.
2	Documents	Viewer / List	File		Opens each uploaded document.
3	Company Class	Dropdown	Enum 0–7		Pre-set to the provisional Class; editable by Admin (A2).
4	Class change reason	Textarea	String	Cond.	Required when the Class is changed.
5	Approve	Button	N/A		Approves the KYC (steps 5–8).
6	Reject	Button	N/A		Opens the rejection reason (A1).
7	Rejection reason	Dropdown + Textarea	String	Cond.	Required on reject; shown to the Member.

UC06: Classify Business into Class 0–7

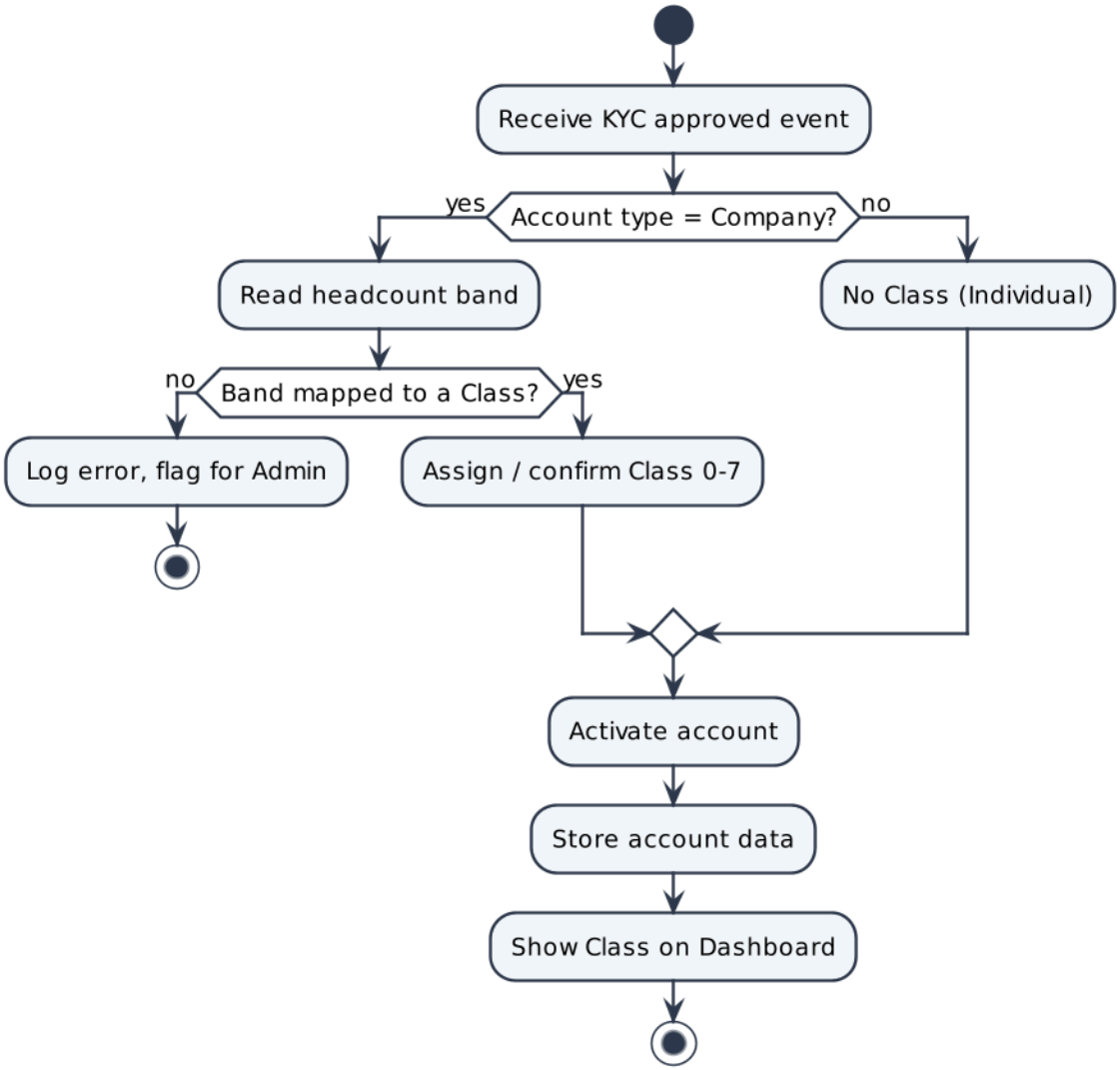
Use Case Description

UC ID	UC06
UC Name	Classify Business into Class 0–7
Objective	Automatically assign each Company to one of eight Classes (0–7) from its declared headcount band, confirm the Class when KYC is approved and activate the account. The Class determines the Title price (UC16).
Actor	PW System (R05); Company (R02) declares headcount; PW Admin (R04) may correct the Class (UC05-A2)
Trigger	(a) A headcount band is selected at registration (preview); (b) KYC is approved (confirmation).
Pre-condition	- Account type = Company.

Post-condition	**Success:** • The account holds exactly one current Class (0–7) and is Active. • The Dashboard shows "Class X · Auto-assigned" (or the Admin-corrected Class).
Basic flow	1. System receives the KYC approval event for a Company account (UC05). 2. System reads the stored headcount band. 3. System maps the band to exactly one Class using the Class table. 4. System stores the Class as confirmed, together with the date and source (Auto-assigned / Admin). 5. System activates the account (status = Active) and stores the account data. 6. System displays "Class X · Auto-assigned" on the Member Dashboard.
Alternative flow	**A1 – Live preview at registration** 1. During UC02, when a band is selected, System maps it to a Class and shows "Your class: Class X (based on headcount)" without saving it as final. **A2 – Admin correction** 1. During UC05 the Admin changes the Class with a reason. 2. System stores the corrected Class (source = Admin) and logs old/new Class, admin and time. **A3 – Individual account** 1. Account type = Individual → no Class is assigned; the account is activated at step 5 without a Class.
Exception flow	**E1 – Band not mapped** 1. The band has no Class in the Class table (configuration error). 2. System does not activate the account, logs the error and shows the request as "Needs attention" in the Admin queue.
Business rule	• BR01 – 8 Classes (0–7) by headcount → pricing tier (BR-1). • BR02 – One band ↔ exactly one Class; one account ↔ one current Class. • BR03 – The Class is provisional until KYC approval; only the Admin can change it, with a reason. • BR04 – Price calculation always uses the current Class (UC16). • BR05 – **TBD: headcount range of each Class; pricing basis for Individuals (Class 0 or a separate table).**
Non-functional requirement	• NFR01 – The Class table is configuration data (changeable without code release).
Acceptance criteria	**AC01 – Preview** Given a Company user selects a band, when the band changes, then "Your class: Class X (based on headcount)" is shown. **AC02 – Confirm on approval**

	Given a Company has a provisional Class, when KYC is approved, then the Class is confirmed, the account is Active and the Dashboard shows "Class X · Auto-assigned". **AC03 – Admin correction** Given the Admin changes the Class with a reason, when the change is saved, then the new Class applies and the change is logged. **AC04 – Individual** Given account type = Individual, when KYC is approved, then the account is activated with no Class.
Traceability	BRD: FR-005 · BPMN: D1 – Classify Business / Active Account

Activity Diagram



Screen Description

Screen 1: Member Dashboard – Class card

[Screenshot: Figma frame "PW — KYC Verification (Approved) / Member Dashboard"]

#	Name	Control type	Data type	Required	Description
1	Company Class	Label / Badge	String		"Class X · Auto-assigned" (or "Class X · Set by PW" after Admin correction). Hidden for Individuals.
2	KYC status	Badge	Enum		"Verified" after approval.

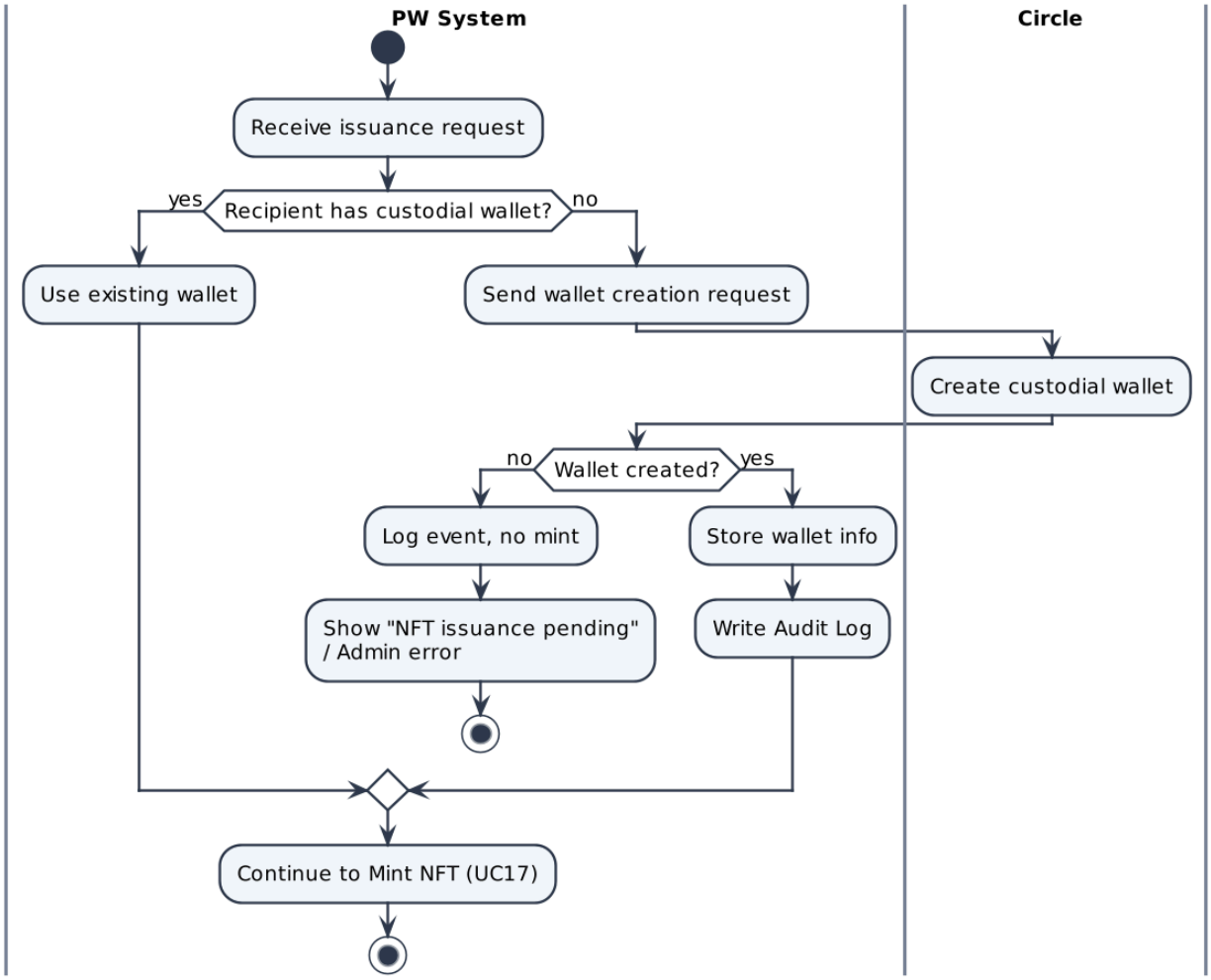
UC07: Create Custodial Wallet on Demand

Use Case Description

UC ID	UC07
UC Name	Create Custodial Wallet on Demand
Objective	Automatically create one custodial wallet per account through Circle at the moment it is first needed – when an NFT is about to be issued to a recipient who has no wallet (paid purchase or gift in D2, Award grant in D3).
Actor	PW System (R05); Circle (external); Member (wallet owner)
Trigger	Payment is confirmed (UC14 / UC15) or an Award grant is confirmed (UC29), and the recipient has no custodial wallet.
Pre-condition	- The recipient's KYC status = "Approved". - The recipient has no custodial wallet.
Post-condition	**Success:** • A custodial wallet (address, type = "Custodial", provider = Circle) is stored on the recipient's account. • Minting continues to that wallet (UC17). • Audit Log entry written (WALLET_CREATE). **Failure:** • No wallet record is stored; the NFT is not minted; the event is logged for follow-up.
Basic flow	1. System receives the issuance request with the recipient account. 2. System checks whether the recipient already has a custodial wallet.

	3. System sends a wallet-creation request to Circle. 4. Circle creates the custodial wallet and returns the wallet information (address). 5. System stores the wallet (address, type = "Custodial") on the recipient's account. 6. System writes an Audit Log entry and continues with minting (UC17).
Alternative flow	**A1 – Wallet already exists** 1. At step 2 the recipient has a custodial wallet. 2. System skips wallet creation and continues with minting to the existing wallet.
Exception flow	**E1 – Circle error or timeout** 1. At step 3/4 Circle returns an error or does not respond within the timeout. 2. System stores no wallet record and does not mint. 3. Purchase / gift: the order shows "NFT issuance pending"; the buyer is informed. Grant: the Admin sees "Award could not be issued because the recipient's wallet could not be created. Please try again." 4. **Retry / refund rule when payment succeeded is TBD.**
Business rule	• BR01 – No manual "Create wallet" action exists for users; no wallet is created at registration. • BR02 – One custodial wallet per account. • BR03 – A wallet is created only when an NFT is about to be issued to a KYC-approved recipient without a wallet (self-purchase, gift recipient per BPMN D2, Award recipient per D3). • BR04 – PW never stores or exposes private keys; keys are managed by Circle.
Non-functional requirement	• NFR01 – Circle calls use a timeout and retry policy; failures leave no partial records (NFR-004). • NFR02 – Wallet creation completes within 10 seconds under normal conditions (proposed).
Acceptance criteria	**AC01 – First issuance, no wallet** Given a KYC-approved recipient has no wallet, when payment or grant is confirmed, then Circle creates a wallet, it is stored as Custodial and the NFT is minted to it. **AC02 – Existing wallet** Given the recipient already has a wallet, when a new NFT is issued, then no new wallet is created. **AC03 – Circle failure** Given Circle errors or times out, when wallet creation is requested, then no wallet is stored, no NFT is minted and the event is logged.
Traceability	BRD: FR-007 · BPMN: D2, D3 – Does recipient already have a wallet? / Circle

Activity Diagram



Screen Description

N/A – background system process with no dedicated screen. Results are shown in the screens of the calling use cases.

UC08: View Wallet

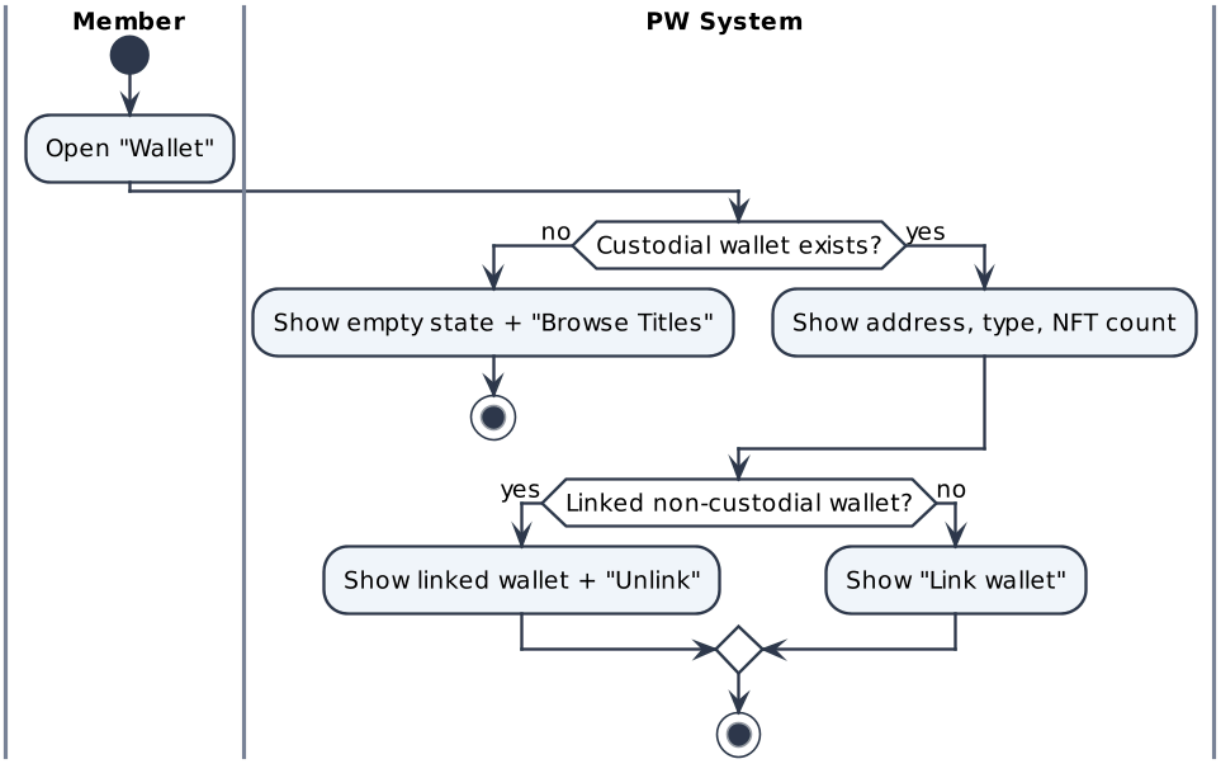
Use Case Description

UC ID	UC08
UC Name	View Wallet

Objective	Show the Member's wallet details: custodial wallet address, custody type, number of NFTs held and the linked non-custodial wallet (if any).
Actor	Company (R02), Individual (R03)
Trigger	The Member opens "Wallet".
Pre-condition	- The Member is logged in.
Post-condition	**Success:** • Wallet information or the empty state is displayed. No data is changed.
Basic flow	1. Member opens "Wallet". 2. System retrieves the custodial wallet and the linked non-custodial wallet of the account. 3. System displays the custodial wallet address (with copy action), type "Custodial" and the number of NFTs held. 4. System displays the linked non-custodial wallet with status "Linked – Verified" and its number of NFTs, with an "Unlink" action (UC11). 5. If no non-custodial wallet is linked, System shows the "Link wallet" action (UC10).
Alternative flow	**A1 – No wallet yet** 1. At step 2 the account has no custodial wallet. 2. System shows "Your wallet will be created automatically when you buy your first Title or receive an Award." and a "Browse Titles" action. **A2 – Copy address** 1. Member clicks the copy icon. 2. System copies the address and shows "Copied".
Exception flow	**E1 – Data cannot be loaded** 1. System shows "Wallet information is temporarily unavailable. Please try again."
Business rule	• BR01 – The same linked-wallet list is shown on Dashboard, Wallet and Transfer screens. • BR02 – "Link wallet" is hidden while a wallet is already linked (1 linked wallet per account).
Non-functional requirement	• NFR01 – Screen loads within 2 seconds (proposed).
Acceptance criteria	**AC01 – Has wallet** Given the Member has a custodial wallet, when "Wallet" is opened, then address (copyable), type and number of NFTs are shown, plus the linked wallet if any. **AC02 – No wallet**

	Given the Member has no wallet, when "Wallet" is opened, then the empty-state message and "Browse Titles" are shown.
Traceability	BRD: FR-008 · BPMN: —

Activity Diagram



Screen Description

Screen 1: Wallet

[Screenshot: Figma frame "PW — Wallet / Dashboard (wallet card)"]

#	Name	Control type	Data type	Required	Description
I. Custodial wallet					
1	Wallet address	Label + Copy icon	String		Shortened address (0x1234…abcd); full address copied on click.
2	Custody type	Badge	Enum		"Custodial (managed by PW via Circle)".

#	Name	Control type	Data type	Required	Description
3	NFTs held	Label	Integer		Number of NFTs in this wallet.
II. Linked non-custodial wallet					
1	Linked wallet address	Label	String		Shown when linked; status "Linked – Verified".
2	Unlink	Button	N/A		Starts UC11.
3	Link wallet	Button	N/A		Starts UC10; shown only when no wallet is linked.
III. Empty state					
1	Empty message	Label	String		"Your wallet will be created automatically when you buy your first Title or receive an Award."
2	Browse Titles	Button	N/A		Navigates to UC13.

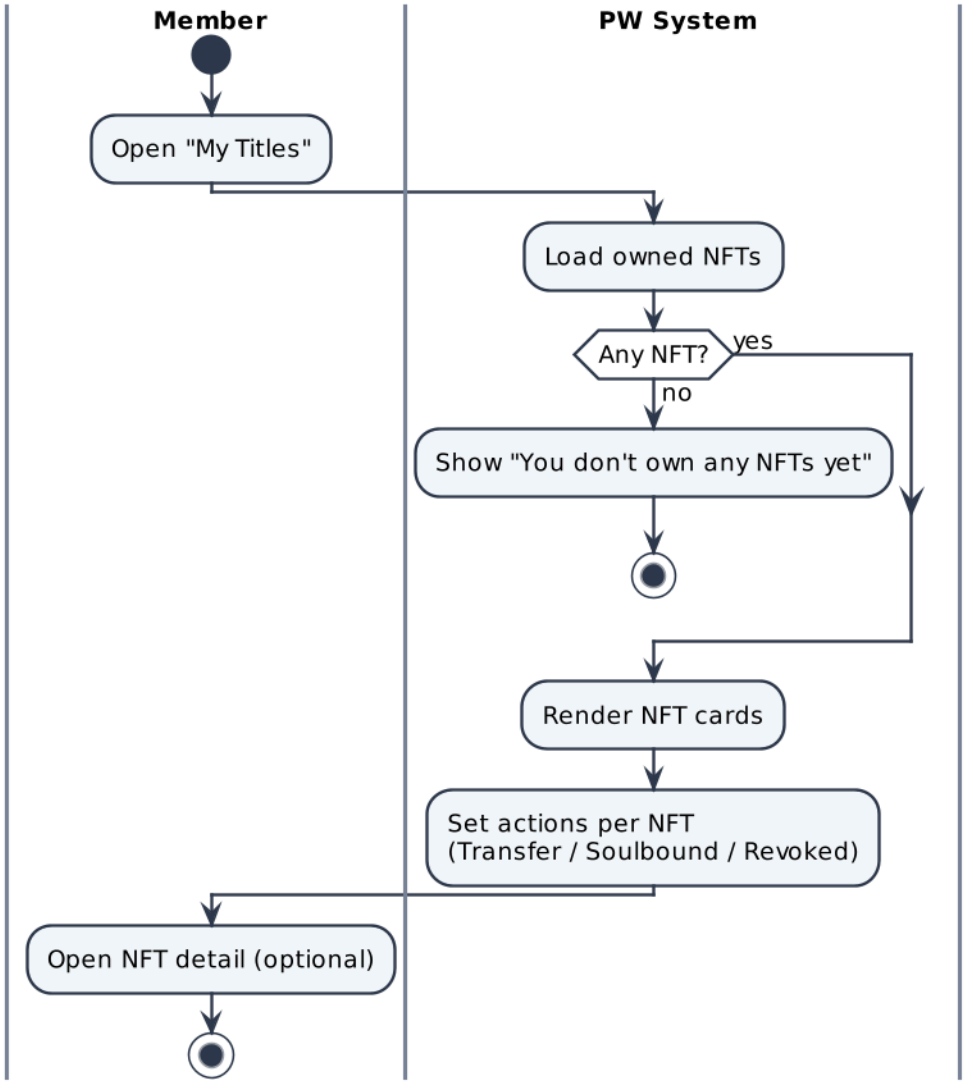
UC09: View My NFTs

Use Case Description

UC ID	UC09
UC Name	View My NFTs
Objective	List all NFTs (Distinctions and Awards) owned by the Member with their status, custody and remaining transfer rights.
Actor	Company (R02), Individual (R03)
Trigger	The Member opens "My Titles" (My NFTs).
Pre-condition	- The Member is logged in.
Post-condition	**Success:** • The NFT list or the empty state is displayed. No data is changed.
Basic flow	1. Member opens "My Titles". 2. System retrieves the NFTs owned by the account (custodial and linked wallets). 3. System displays one card per NFT: name, category (Philanthropist /

	Investor / Bridge), type (Distinction / Award), status (Active / Revoked), custody (Custodial / Transfer pending / Self-custody (verified)), token ID and issue date. 4. For each Distinction, System shows "One-time transfer available" with a "Transfer" action, or "Transfer used". 5. For each Award, System shows "Soulbound – non-transferable" and no transfer action. 6. Member selects an NFT to open its detail (token ID, transaction link, verification link – UC27).
Alternative flow	**A1 – No NFTs** 1. At step 2 no NFT is found. 2. System shows "You don't own any NFTs yet" with a "Browse Titles" action. **A2 – Revoked NFT** 1. An NFT has status Revoked. 2. System shows the badge "Revoked" and no transfer action.
Exception flow	**E1 – Data cannot be loaded** 1. System shows a retry message.
Business rule	• BR01 – Award = soulbound; Distinction = transferable once (BR-6). • BR02 – NFTs transferred to another party are no longer listed for the former owner.
Non-functional requirement	• NFR01 – List loads within 2 seconds for up to 100 NFTs (proposed).
Acceptance criteria	**AC01 – Owns NFTs** Given the Member owns NFTs, when "My Titles" is opened, then each NFT shows name, category, type, status, custody, token ID, issue date and its transfer availability. **AC02 – No NFTs** Given the Member owns none, when "My Titles" is opened, then "You don't own any NFTs yet" with "Browse Titles" is shown. **AC03 – Revoked** Given an NFT is revoked, when the list is shown, then it shows "Revoked" and no transfer action.
Traceability	BRD: FR-009 · BPMN: —

Activity Diagram



Screen Description

Screen 1: My Titles (My NFTs)

[Screenshot: Figma frame "PW — My Titles / Title detail"]

#	Name	Control type	Data type	Required	Description
1	NFT card – Name & category	Card	String		e.g. "Investor" + type label "Distinction" or "Award".
2	Status	Badge	Enum		Active | Revoked.
3	Custody	Badge	Enum		Custodial | Transfer pending |

#	Name	Control type	Data type	Required	Description
					Self-custody (verified).
4	Token ID	Label	String		On-chain token ID.
5	Issue date	Label	Date		DD/MM/YYYY.
6	Transfer availability	Label	String		"One-time transfer available" / "Transfer used" / "Soulbound – non-transferable".
7	Transfer	Button	N/A		Shown only for Active Distinctions with the transfer unused (UC19/UC20).
8	View on-chain / Verify	Link	N/A		Opens the verification result (UC27).

UC10: Link Non-custodial Wallet

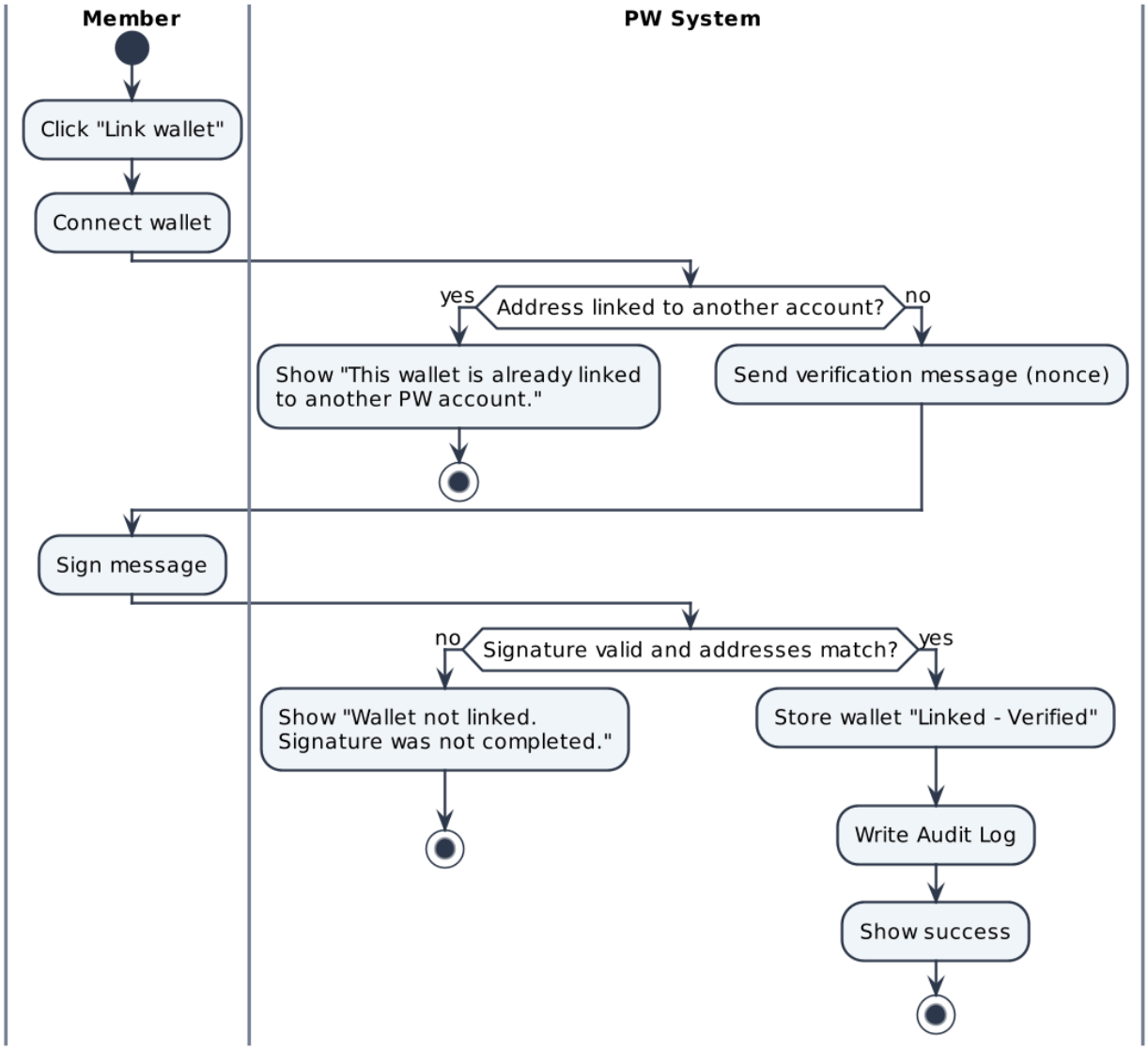
Use Case Description

UC ID	UC10
UC Name	Link Non-custodial Wallet
Objective	Allow a Member to prove ownership of a non-custodial wallet by connecting it and signing a verification message, so that NFTs moved there keep counting toward Ranking.
Actor	Company (R02), Individual (R03); user's non-custodial wallet
Trigger	The Member clicks "Link wallet" (or ticks "This is my wallet" during a transfer – UC19-A1).
Pre-condition	- The Member is logged in and KYC-approved. - The account has no linked non-custodial wallet.
Post-condition	**Success:** • The wallet is stored as "Linked – Verified" and appears in the shared linked-wallet list. • Audit Log entry written (WALLET_LINK). **Failure:** • The wallet is not stored.

Basic flow	1. Member clicks "Link wallet". 2. System opens Step 1 "Connect wallet". 3. Member connects the wallet; the wallet returns its address. 4. System checks that the address is not linked to another PW account. 5. System opens Step 2 "Sign message" and sends a one-time verification message (includes account ID and a nonce). 6. Member signs the message in the wallet. 7. System verifies the signature and that the signed address equals the connected address. 8. System stores the wallet as "Linked – Verified" and writes an Audit Log entry. 9. System displays "Wallet linked successfully."
Alternative flow	**A1 – Cancel** 1. Member closes the modal before signing → nothing is stored.
Exception flow	**E1 – Signature rejected or invalid** 1. At step 6/7 the Member rejects the request or the signature is invalid. 2. System displays "Wallet not linked. Signature was not completed." and stores nothing. **E2 – Wallet linked to another account** 1. At step 4 the address is already linked to another account. 2. System displays "This wallet is already linked to another PW account." and refuses the link. **E3 – Account already has a linked wallet** 1. At step 1 a wallet is already linked. 2. System asks the Member to unlink the current wallet first (UC11).
Business rule	• BR01 – 1 wallet ↔ 1 account; max one linked non-custodial wallet per account. • BR02 – Ownership is proven only by a valid message signature; the nonce is single-use. • BR03 – Only NFTs in the custodial wallet or in a linked, signature-verified wallet count toward Ranking.
Non-functional requirement	• NFR01 – The signing message is human-readable and states that it does not trigger any transaction or fee.
Acceptance criteria	**AC01 – Link successfully** Given no wallet is linked, when the Member connects and signs, and the addresses match, then the wallet is saved as "Linked – Verified" and appears in the linked-wallet list. **AC02 – Signature rejected** Given a wallet is connected, when the signature is rejected or invalid, then "Wallet not linked. Signature was not completed." is displayed. **AC03 – Linked elsewhere**

	Given the address is linked to another account, when the Member tries to link it, then "This wallet is already linked to another PW account." is displayed. **AC04 – Already linked** Given the account has a linked wallet, when the Member tries to link another, then the Member is asked to unlink first.
Traceability	BRD: FR-031 · BPMN: D4 (supporting)

Activity Diagram



Screen Description

Screen 1: Link wallet – Step 1 Connect / Step 2 Sign

[Screenshot: Figma frame "PW — Link Wallet (modal)"]

#	Name	Control type	Data type	Required	Description
1	Stepper	Stepper	N/A		Step 1 Connect wallet → Step 2 Sign message.
2	Connect wallet	Button	N/A		Opens the wallet connection.
3	Connected address	Label	String		Shown after connection.
4	Message preview	Label	String		Text to be signed incl. account ID and nonce; states that no transaction/fee is involved.
5	Sign message	Button	N/A		Requests the signature in the wallet.
6	Error message	Alert	String		E1/E2 messages.
7	Cancel	Button	N/A		Closes without linking.

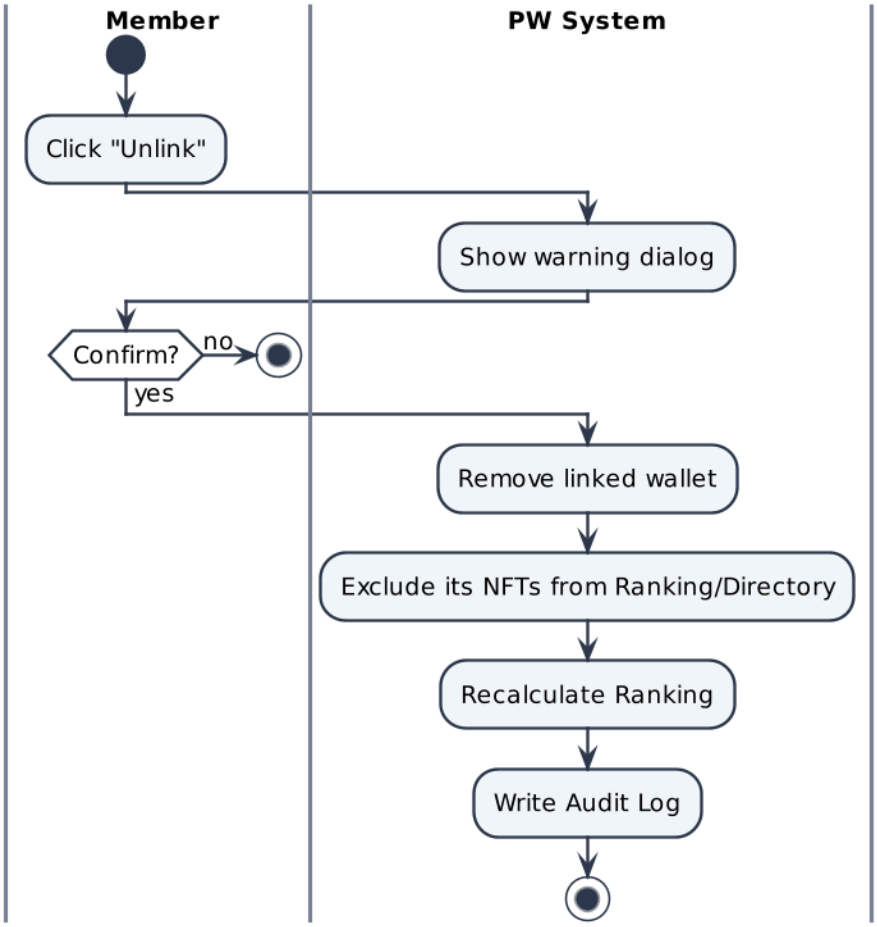
UC11: Unlink Wallet

Use Case Description

UC ID	UC11
UC Name	Unlink Wallet
Objective	Allow a Member to unlink the non-custodial wallet after a warning; NFTs in that wallet stop counting toward Ranking and the Directory profile until the wallet is linked again.
Actor	Company (R02), Individual (R03)
Trigger	The Member clicks "Unlink" on the linked wallet.
Pre-condition	- The Member has a linked non-custodial wallet.
Post-condition	**Confirmed:** • The wallet is removed from the linked list; its NFTs are excluded from Ranking and the Directory profile; Ranking is recalculated

	(UC28); Audit Log written (WALLET_UNLINK). **Cancelled:** • Nothing changes.
Basic flow	1. Member clicks "Unlink". 2. System displays the confirmation dialog: "NFTs in this wallet will no longer count toward your ranking." 3. Member clicks "Unlink wallet". 4. System removes the wallet from the linked list. 5. System excludes the NFTs held in that wallet from the Member's Ranking and Directory profile and triggers a Ranking recalculation. 6. System writes an Audit Log entry and displays "Wallet unlinked."
Alternative flow	**A1 – Cancel** 1. At step 3 Member clicks "Cancel" → nothing changes.
Exception flow	**E1 – System error** 1. The unlink cannot be saved → the wallet stays linked and an error is shown.
Business rule	• BR01 – NFTs in an unlinked wallet are not counted until the wallet is linked again (re-sign required). • BR02 – Unlinking does not move any NFT on-chain.
Non-functional requirement	• NFR01 – Ranking reflects the unlink within 1 minute (proposed).
Acceptance criteria	**AC01 – Confirm unlink** Given a wallet is linked, when the Member confirms the warning, then the wallet is removed and its NFTs are removed from Ranking and the Directory profile. **AC02 – Cancel** Given the warning is shown, when the Member clicks "Cancel", then nothing changes.
Traceability	BRD: FR-032 · BPMN: —

Activity Diagram



Screen Description

Screen 1: Unlink wallet – confirmation

[Screenshot: Figma frame "PW — Unlink Wallet (dialog)"]

#	Name	Control type	Data type	Required	Description
1	Warning text	Label	String		"NFTs in this wallet will no longer count toward your ranking."
2	Wallet address	Label	String		Address to be unlinked.
3	Unlink wallet	Button (destructive)	N/A		Confirms the unlink.
4	Cancel	Button	N/A		Closes the dialog.

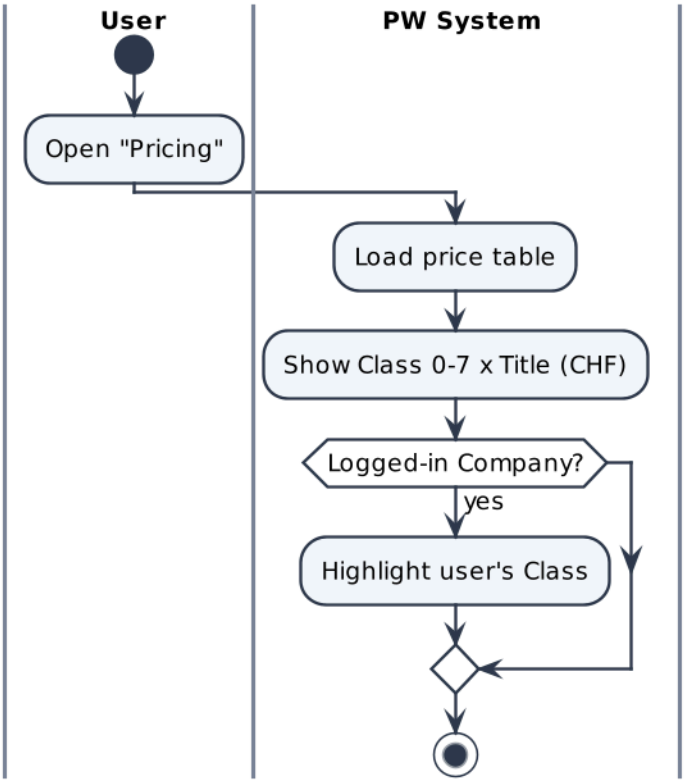
UC12: View Pricing by Class

Use Case Description

UC ID	UC12
UC Name	View Pricing by Class
Objective	Display the Title price table for Classes 0–7 in CHF so that users understand the price before purchasing.
Actor	Visitor (R01), Company (R02), Individual (R03)
Trigger	The user opens "Pricing".
Pre-condition	- None (public page).
Post-condition	**Success:** • The price table is displayed. No data is changed.
Basic flow	1. User opens "Pricing". 2. System loads the active price table. 3. System displays one row per Class (0–7) and one column per Title (Philanthropist, Investor, Bridge), amounts in CHF. 4. If the user is a logged-in Company, System highlights the user's Class and shows "Your class".
Alternative flow	**A1 – Visitor or Individual** 1. The table is shown without a highlighted Class; Individuals see the note for their pricing basis (**TBD**).
Exception flow	**E1 – Price table not configured** 1. System shows "Pricing is being updated. Please check back soon."
Business rule	• BR01 – Price depends on Title × Class (BR-2). • BR02 – Prices are shown in CHF; 1 CHF = 1 Point for rewards (BR-4). • BR03 – **TBD: price values per Class × Title; pricing basis for Individuals.**
Non-functional requirement	• NFR01 – Prices are configuration data; a change is effective without code release.
Acceptance criteria	**AC01 – Logged-in Company** Given a logged-in Company of Class X, when "Pricing" is opened, then all Classes 0–7 × 3 Titles are shown in CHF and Class X is

	highlighted. **AC02 – Visitor** Given not logged in, when "Pricing" is opened, then the full table is shown without highlight.
Traceability	BRD: FR-015 · BPMN: —

Activity Diagram



Screen Description

Screen 1: Pricing

[Screenshot: Figma frame "PW — Pricing (to be designed)"]

#	Name	Control type	Data type	Required	Description
1	Price table	Table	Decimal (CHF)		Rows Class 0–7; columns Philanthropist, Investor, Bridge.
2	Your class	Highlight / Badge	String		Only for logged-in Companies.

#	Name	Control type	Data type	Required	Description
3	Buy a Title	Button	N/A		Navigates to UC13 (login required).

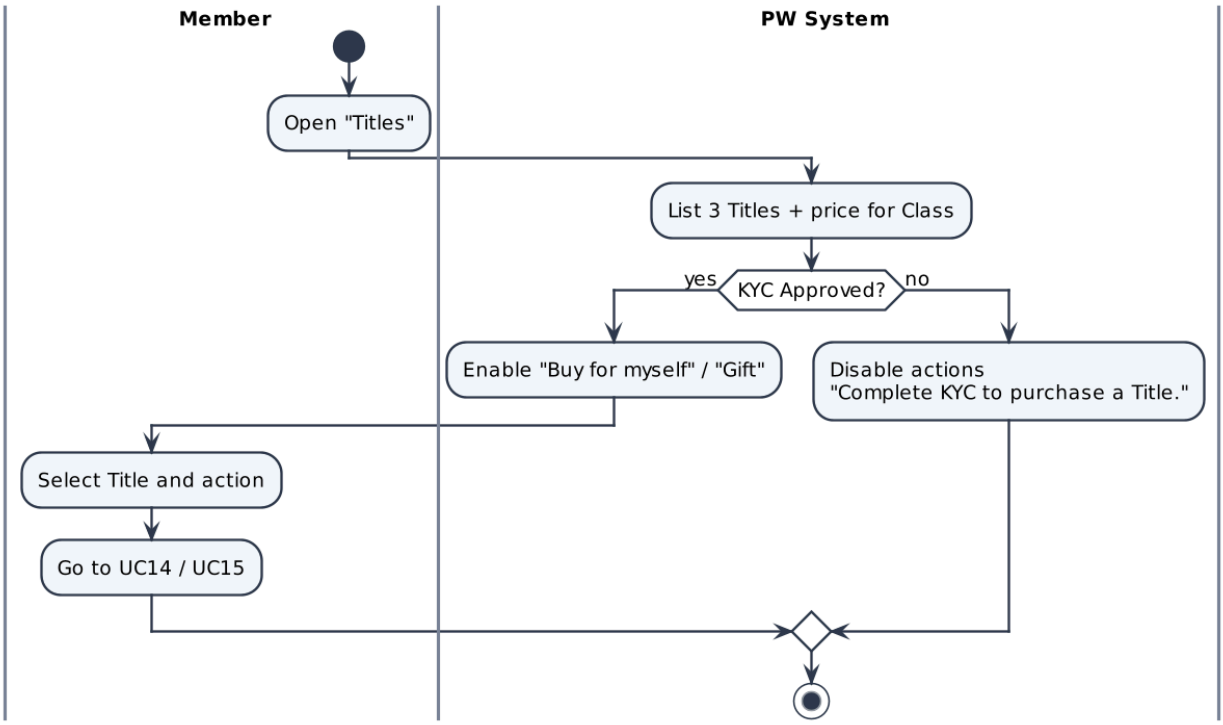
UC13: Browse Titles

Use Case Description

UC ID	UC13
UC Name	Browse Titles
Objective	Display the three purchasable Titles with description and the price for the Member's Class, and let the Member start a purchase for themselves or a gift.
Actor	Company (R02), Individual (R03)
Trigger	The Member opens "Titles".
Pre-condition	- The Member is logged in.
Post-condition	**Success:** • The Title catalogue is displayed; purchase actions are enabled only if KYC = Approved.
Basic flow	1. Member opens "Titles". 2. System lists the 3 Titles: Philanthropist, Investor, Bridge – each with description and the price for the Member's Class (UC16). 3. System shows "Buy for myself" and "Gift" on each Title. 4. Member selects a Title and an action → UC14 (Buy for myself) or UC15 (Gift).
Alternative flow	**A1 – KYC not approved** 1. At step 3 KYC ≠ Approved. 2. Titles and prices are visible; purchase actions are disabled with "Complete KYC to purchase a Title." and a link to UC04.
Exception flow	**E1 – Price not available** 1. The Class/price cannot be resolved → price shows "—" and purchase actions are disabled.
Business rule	• BR01 – Exactly three Titles: Philanthropist, Investor, Bridge. • BR02 – Purchasing a Title issues a Distinction NFT.

	• BR03 – **TBD: price shown to Individuals (no Class).**
Non-functional requirement	• NFR01 – Catalogue loads within 2 seconds (proposed).
Acceptance criteria	**AC01 – KYC approved** Given the Member is KYC-approved, when "Titles" is opened, then 3 Titles with description, price for the Member's Class and "Buy for myself" / "Gift" are shown. **AC02 – KYC not approved** Given KYC ≠ Approved, when "Titles" is opened, then actions are disabled with "Complete KYC to purchase a Title.".
Traceability	BRD: FR-011 · BPMN: D2 – Browse & Select Title

Activity Diagram



Screen Description

Screen 1: Select Title (Step 1)

[Screenshot: Figma frame "PW — Acquisition & Gifting (Step 1: Select Title)"]

#	Name	Control type	Data type	Required	Description
1	Title card	Card	N/A		One per Title: Philanthropist / Investor / Bridge (no Class label on the card).
2	Description	Label	String		Short meaning of the Title.
3	Price	Label	Decimal (CHF)		Price for the Member's Class, e.g. "3,500 CHF".
4	Buy for myself	Button	N/A		Starts UC14.
5	Gift	Button	N/A		Starts UC15.
6	KYC notice	Alert	String		"Complete KYC to purchase a Title." when KYC ≠ Approved.

UC14: Purchase Title for Myself

Use Case Description

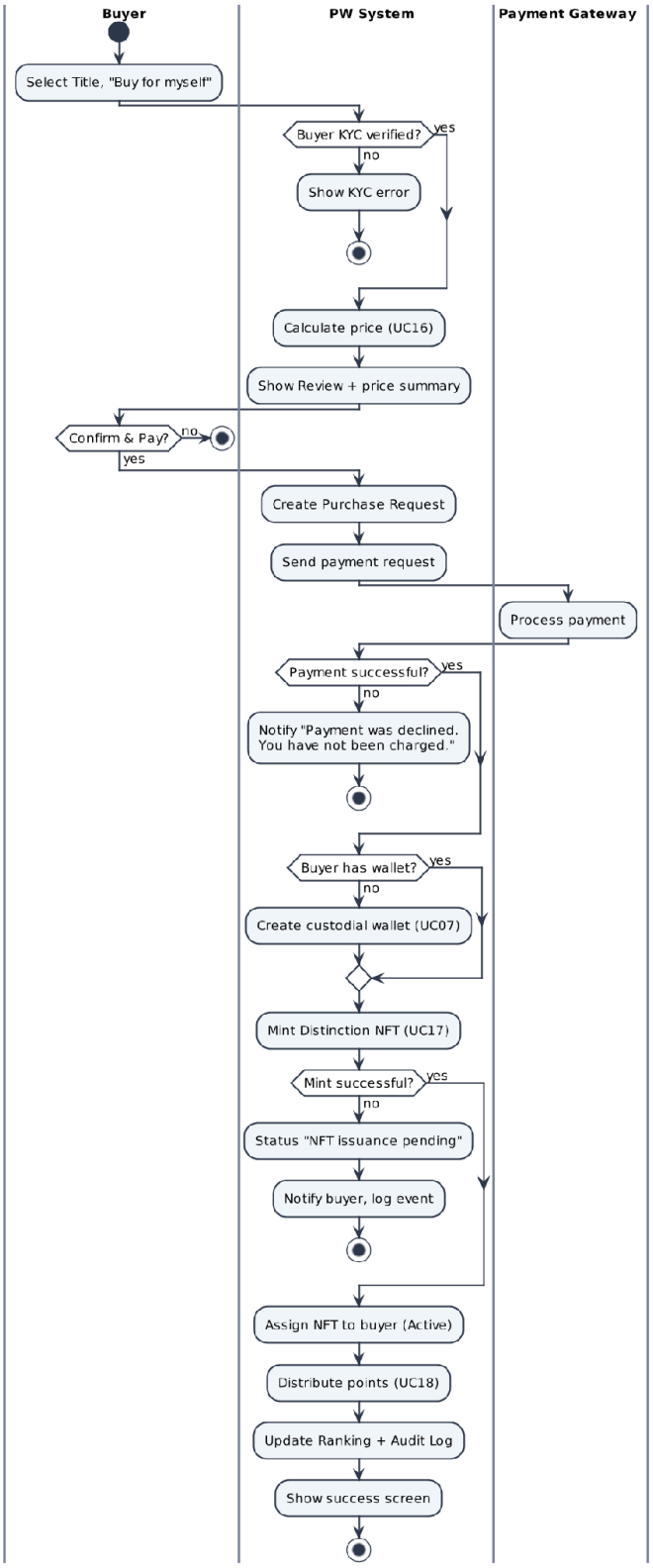
UC ID	UC14
UC Name	Purchase Title for Myself
Objective	Allow a KYC-approved Member to buy a Title for themselves. After the Payment Gateway confirms payment, PW creates a custodial wallet if needed, mints a Distinction NFT directly to the buyer (Active immediately), distributes points, updates Ranking and records the audit trail.
Actor	Buyer – Company (R02) / Individual (R03); PW System (R05); Payment Gateway, Circle, Blockchain (external)
Trigger	The Member clicks "Buy for myself" on a Title.
Pre-condition	- The buyer is logged in and KYC = "Approved". - The Title price for the buyer's Class is configured.
Post-condition	**Success:**

	• Payment captured; Purchase Request status = Completed. • Distinction NFT minted to the buyer's custodial wallet, status Active, one-time transfer available. • Points credited (UC18); Ranking updated (UC28); Audit Log written; success notification sent. **Failure:** • Payment declined/cancelled: no NFT, no points, status = Failed/Cancelled. • Payment OK but issuance failed: status = "NFT issuance pending" (see E3).
Basic flow	1. Buyer selects a Title and clicks "Buy for myself" (recipient = self). 2. System confirms that the recipient (buyer) is KYC-verified. 3. System calculates the price summary (UC16): base price for the buyer's Class, Year-2 discount if eligible, points used if any (UC24), final price in CHF. 4. System displays Step 2 "Review": Title, recipient = "Myself", price summary, estimated points to be earned. 5. Buyer clicks "Confirm & Pay". 6. System creates the Purchase Request (status = Pending payment) and sends the payment request to the Payment Gateway. 7. Payment Gateway processes the payment and returns the result. 8. System receives "payment successful" and sets the Purchase Request to Paid. 9. System checks whether the buyer has a custodial wallet; if not, creates it via Circle (UC07). 10. System prepares the NFT issuance and mints the Distinction NFT to the buyer's wallet (UC17). 11. System receives the mint result = success, assigns wallet & NFT to the buyer and sets the NFT status = Active. 12. System distributes points 10/10/10 (UC18). 13. System updates Ranking (UC28) and records the Audit Log (PURCHASE, MINT). 14. System notifies the buyer and displays the success screen "Your Title has been issued" with the NFT details (Title, token ID, issue date, links to My Titles and Verify).
Alternative flow	**A1 – Buyer cancels** 1. At step 5 Buyer clicks "Cancel" → no Purchase Request is created; return to UC13. **A2 – Buyer already has a wallet** 1. At step 9 the wallet exists → no wallet is created; continue at step 10.
Exception flow	**E1 – Buyer not KYC-verified** 1. At step 2 KYC ≠ Approved → System shows the KYC error "Complete KYC to purchase a Title."; the flow ends. **E2 – Payment declined**

	1. At step 7 the Payment Gateway declines. 2. System sets the Purchase Request = Failed and displays "Payment was declined. You have not been charged." 3. No NFT is minted; no points are credited; used points (if any) are returned. **E3 – Wallet creation or minting fails after payment** 1. At step 9 or 10–11 Circle or the blockchain fails. 2. System sets status = "NFT issuance pending", notifies the buyer "Payment received – your Title is being issued" and logs the event for follow-up. 3. **Retry / refund rule is TBD.** **E4 – Payment Gateway timeout** 1. No result within the timeout → status = Pending payment; System re-checks the payment status before any retry to avoid double charge.
Business rule	• BR01 – Only KYC-approved Members can purchase (BR-8). • BR02 – The recipient type (self / other) is fixed before payment (BR-7). • BR03 – Mint only after payment confirmation; the NFT is minted directly to the recipient and is Active immediately – no accept step. • BR04 – One Purchase Request can be paid only once (idempotent payment reference). • BR05 – Points: 30% of the transaction value, 10% buyer / 10% recipient / 10% system (BR-3); **TBD: buyer = recipient → 10% or 20%?** • BR06 – A purchased Title is a Distinction NFT: transferable once (BR-6).
Non-functional requirement	• NFR01 – Price summary displayed within 1 second after the Title is selected (proposed). • NFR02 – Payment data is handled by the Payment Gateway; PW stores no card data. • NFR03 – Minting runs asynchronously; the buyer sees a progress state and is never charged twice (NFR-004, NFR-010).
Acceptance criteria	**AC01 – Purchase successfully** Given a KYC-approved buyer selects "Buy for myself", when the buyer confirms and the payment succeeds, then a wallet is created if needed, a Distinction NFT is minted Active to the buyer, points are credited, Ranking and Audit Log are updated and the success screen is shown. **AC02 – Payment declined** Given the buyer is on payment, when the gateway declines, then "Payment was declined. You have not been charged." is shown and no NFT/points are issued. **AC03 – Cancel** Given the buyer is on Review, when "Cancel" is clicked, then no

	Purchase Request is created. **AC04 – Issuance fails** Given payment succeeded, when minting fails, then status = "NFT issuance pending", the buyer is informed and the event is logged.
Traceability	BRD: FR-012, FR-013, FR-016, FR-017, FR-018 · BPMN: D2 – Purchase (self)

Activity Diagram



Screen Description

Screen 1: Review & pay (Step 2 – For myself)

[Screenshot: Figma frame "PW — Acquisition & Gifting (Step 2: For Myself)"]

#	Name	Control type	Data type	Required	Description
I. Order					
1	Title	Label	String		Selected Title (e.g. Investor) with type "Distinction NFT".
2	Recipient	Label	String		"Myself" + account name.
II. Price summary					
1	Base price	Label	Decimal (CHF)		Price of the Title for the buyer's Class.
2	Year-2 loyalty discount	Label	Decimal (CHF)		"–20%" line shown only when eligible (UC16).
3	Points used	Numeric input / Toggle	Integer		Shown only if points redemption is enabled (**TBD – UC24**).
4	Final price	Label	Decimal (CHF)		Amount charged.
5	Points you will earn	Label	Integer		10% of the final price, rounded up.
III. Actions					
1	Confirm & Pay	Button	N/A		Creates the Purchase Request and redirects to the Payment Gateway.
2	Cancel	Button	N/A		Returns to Step 1 without creating a request.

Screen 2: Purchase result

[Screenshot: Figma frame "PW — Title issued (success) / Payment declined / Issuance pending"]

#	Name	Control type	Data type	Required	Description
1	Result title	Label	String		"Your Title has been issued" | "Payment was declined. You

#	Name	Control type	Data type	Required	Description
					have not been charged." | "Payment received – your Title is being issued".
2	NFT details	Card	Mixed		Title, type Distinction, token ID, issue date, status Active.
3	Points earned	Label	Integer		Credited points.
4	View My Titles	Button	N/A		Opens UC09.
5	Verify on-chain	Link	N/A		Opens UC27.

UC15: Gift Title to Another Member

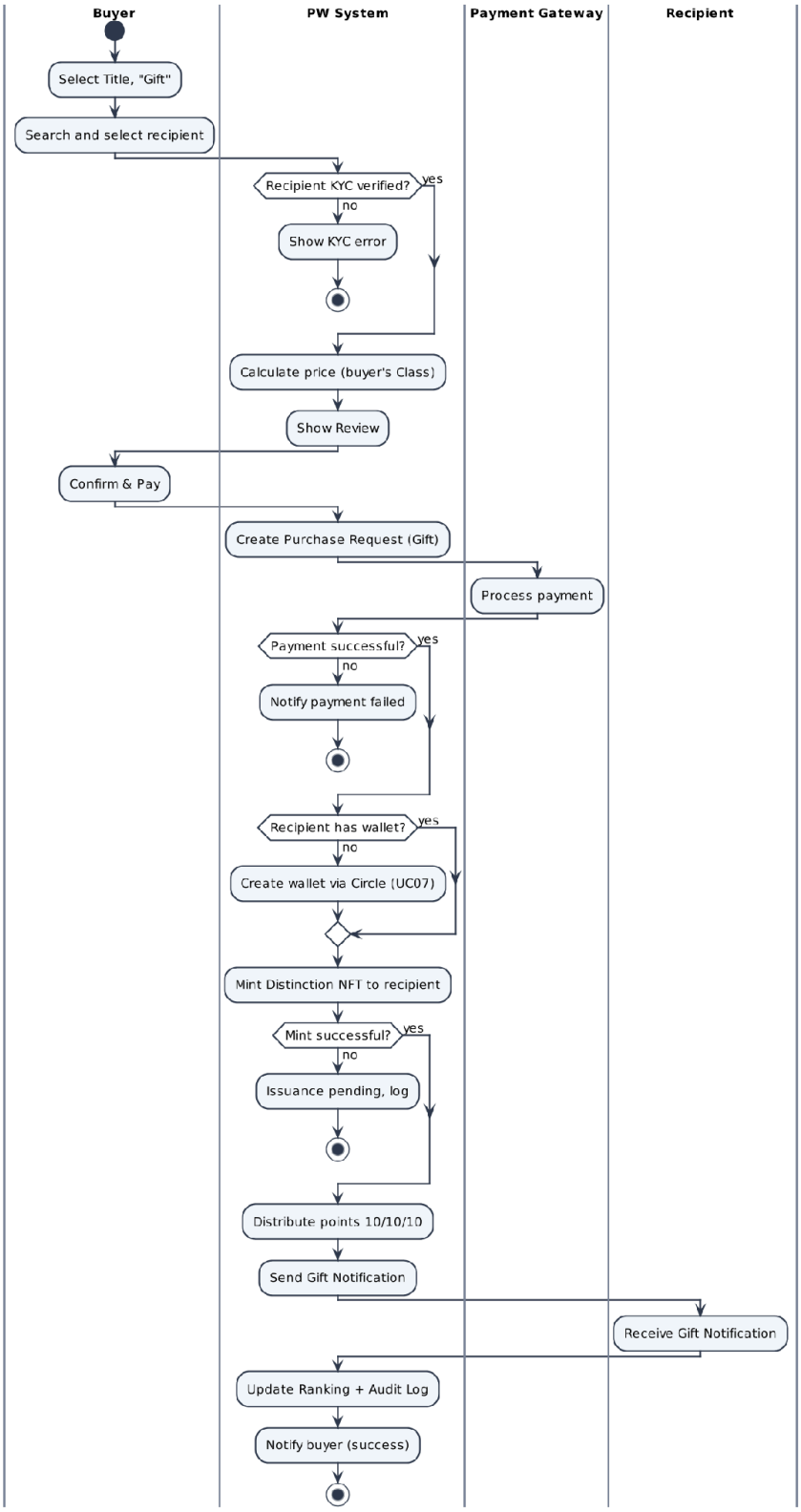
Use Case Description

UC ID	UC15
UC Name	Gift Title to Another Member
Objective	Allow a KYC-approved buyer to buy a Title for another KYC-verified PW Member. The Distinction NFT is minted directly to the recipient (wallet created via Circle if missing), is Active immediately and the recipient receives a one-way Gift Notification.
Actor	Buyer – Company (R02) / Individual (R03); Recipient (Member, notified); PW System (R05); Payment Gateway, Circle, Blockchain
Trigger	The Member clicks "Gift" on a Title.
Pre-condition	- The buyer is logged in and KYC = "Approved".
Post-condition	**Success:** • Distinction NFT minted Active to the recipient's custodial wallet; the recipient's one-time transfer is NOT consumed. • Points: 10% buyer, 10% recipient, 10% system; Gift Notification sent; Ranking and Audit Log updated. **Failure:** • As in UC14 (declined / cancelled / issuance pending).
Basic flow	1. Buyer selects a Title and clicks "Gift". 2. System displays Step 2 "Choose recipient".

	3. Buyer searches for the recipient by registered name, email or PW member ID (search keys – assumption) and selects one. 4. System checks that the recipient is KYC-verified. 5. System calculates the price summary on the BUYER's Class (assumption) (UC16) and displays Review: Title, recipient name, price summary. 6. Buyer clicks "Confirm & Pay". 7. System creates the Purchase Request (type = Gift) and sends the payment request to the Payment Gateway. 8. Payment Gateway returns "payment successful". 9. System checks whether the recipient has a custodial wallet; if not, creates it via Circle (UC07). 10. System mints the Distinction NFT directly to the recipient's wallet (UC17) and sets it Active. 11. System distributes points: 10% buyer, 10% recipient, 10% system (UC18). 12. System sends the Gift Notification to the recipient (in-app + email): Title, buyer's name, link to the NFT – no action required. 13. System updates Ranking for the recipient, records the Audit Log (GIFT, MINT) and shows the buyer the success screen "Your gift has been issued to [Recipient]".
Alternative flow	**A1 – Recipient = buyer** 1. At step 3 the buyer selects their own account → the order is treated as "Buy for myself" (UC14). **A2 – Change recipient** 1. Before step 6 the buyer clicks "Change" → back to step 3. **A3 – Cancel** 1. At step 6 the buyer cancels → no Purchase Request is created.
Exception flow	**E1 – Recipient not KYC-verified** 1. At step 4 → System shows "This member must complete identity verification before receiving a Title." and does not allow Confirm & Pay. **E2 – Recipient not found** 1. At step 3 no Member matches → "No verified member found." **E3 – Payment declined / timeout** 1. Same as UC14-E2 / E4. **E4 – Wallet creation or minting fails** 1. Same as UC14-E3; the recipient is not notified until the NFT is Active.
Business rule	• BR01 – The recipient must be a KYC-verified PW Member (locked rule; BPMN D2 "Is Recipient KYC Verified?"). • BR02 – A gift is a variation of Purchase, not a Transfer: it does NOT consume the recipient's one-time transfer. • BR03 – No accept/decline step: the NFT is Active on issue; the notification is one-way. • BR04 – If the recipient has no wallet, PW creates one via Circle

	(BPMN D2). • BR05 – Price is based on the BUYER's Class (assumption). • BR06 – **Alignment: US-TN03 (enter a wallet address) and the US-AW01 rule "no wallet for gift recipient" differ from BRD/BPMN D2 used here – US to be updated.**
Non-functional requirement	• NFR01 – Recipient search returns results within 2 seconds (proposed). • NFR02 – Gift Notification delivered within 1 minute after mint confirmation (proposed).
Acceptance criteria	**AC01 – Gift successfully** Given the recipient is a KYC-verified Member, when the buyer pays successfully, then the NFT is minted Active to the recipient (wallet created if needed), 10%/10%/10% points are credited, the recipient gets the Gift Notification and the recipient's transfer stays available. **AC02 – Recipient not verified** Given the selected member is not KYC-verified, when the buyer tries to continue, then the KYC error is shown and Confirm & Pay is unavailable. **AC03 – Self as recipient** Given the buyer selects their own account, when the recipient is confirmed, then the order is processed as "Buy for myself".
Traceability	BRD: FR-012, FR-013, FR-018 · BPMN: D2 – Purchase (gift) / Notify Recipient

Activity Diagram



Screen Description

Screen 1: Choose recipient & review (Step 2 – Gift)

[Screenshot: Figma frame "PW — Acquisition & Gifting (Step 2: Gift)"]

#	Name	Control type	Data type	Required	Description
I. Recipient					
1	Search recipient	Search box (autocomplete)	String	Yes	Search KYC-verified Members by name, email or member ID (assumption); min. 3 characters.
2	Selected recipient	Card	Mixed		Name, type (Company/Individual), "Verified" badge.
3	Change	Link	N/A		Clears the selection (A2).
4	KYC error	Alert	String		"This member must complete identity verification before receiving a Title."
II. Price summary					
1	Base price / discount / final price	Label	Decimal (CHF)		Same rules as UC14, calculated on the buyer's Class.
2	Points	Label	Integer		"You earn X points · Recipient earns X points".
III. Actions					
1	Confirm & Pay	Button	N/A		Enabled only when a verified recipient is selected.
2	Cancel	Button	N/A		Leaves without creating a request.

Screen 2: Gift Notification (recipient)

[Screenshot: Figma frame "PW — Notification (Gift received)"]

#	Name	Control type	Data type	Required	Description
1	Title	Label	String		"You have received the [Title] Title from [Buyer name]".

#	Name	Control type	Data type	Required	Description
2	NFT link	Link	N/A		Opens the NFT in My Titles (UC09).
3	Channel	In-app + Email	N/A		One-way; no action required.

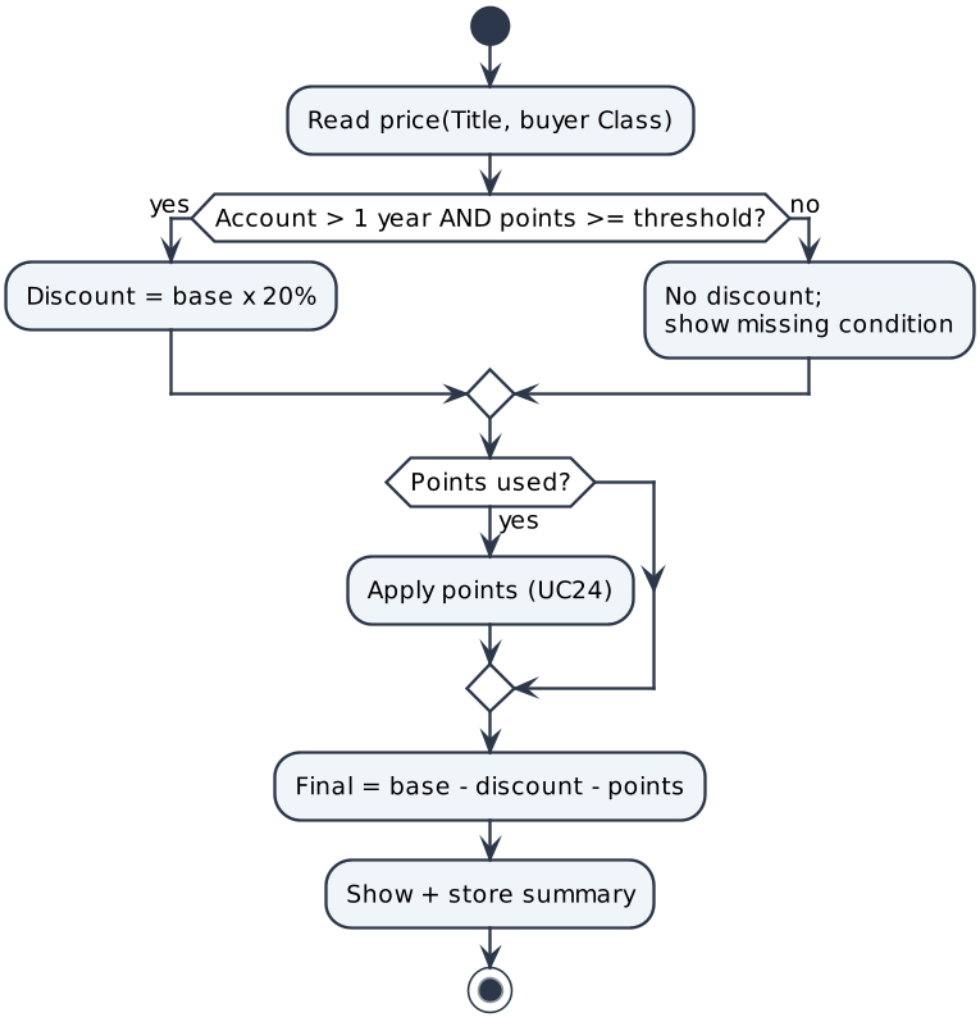
UC16: Calculate Price and Apply Discount

Use Case Description

UC ID	UC16
UC Name	Calculate Price and Apply Discount
Objective	Calculate the amount payable for a Title from the buyer's Class, apply the 20% Year-2 loyalty discount when both conditions are met and (if enabled) points used, and present a transparent price summary.
Actor	PW System (R05)
Trigger	The buyer reaches the Review step of UC14/UC15, or the Purchase Request is received.
Pre-condition	- The buyer's Class is known. - The price table is configured.
Post-condition	**Success:** • Base price, discount, points used and final price (CHF) are calculated, displayed and stored on the Purchase Request.
Basic flow	1. System reads the price of the selected Title for the buyer's current Class (base price). 2. System checks Year-2 eligibility: account active for more than one year AND point balance ≥ threshold. 3. If eligible, System applies discount = base price × 20%. 4. If points redemption is enabled and the buyer chooses to use points, System applies them (UC24). 5. System calculates final price = base price − discount − points value (never below 0). 6. System displays the summary and stores all amounts on the Purchase Request.
Alternative flow	**A1 – Not eligible for discount**

	1. At step 2 a condition is not met → no discount; System shows what is still required (time remaining or points missing). **A2 – Gift** 1. The Class used is the BUYER's Class (assumption).
Exception flow	**E1 – Price missing for the Class** 1. System blocks the purchase and logs a configuration error.
Business rule	• BR01 – Base price = price(Title, buyer's Class) (BR-2). • BR02 – Year-2 discount = 20% when account age > 1 year AND point balance ≥ threshold (BR-5); **TBD: threshold and how "Year 2" is counted.** • BR03 – Amounts in CHF with 2 decimals; points conversion 1 CHF = 1 Point, rounded up (BR-4). • BR04 – The price is re-calculated when the Purchase Request is created; if it differs from the displayed price, the buyer must confirm again. • BR05 – **TBD: pricing basis for Individuals; points redemption mechanism.**
Non-functional requirement	• NFR01 – Calculations are exact and auditable (NFR-010).
Acceptance criteria	**AC01 – No discount** Given the buyer is not eligible, when the summary is calculated, then final price = base price for the buyer's Class. **AC02 – With discount** Given the buyer is eligible and base price = 1,000 CHF, when the summary is calculated, then discount = 200 CHF and final price = 800 CHF, labelled "Year-2 loyalty discount –20%". **AC03 – Not eligible – hint** Given the account is younger than 1 year, when the summary is shown, then the missing condition (time remaining) is displayed.
Traceability	BRD: FR-016, FR-017 · BPMN: D2 – Calculate Price + Discount/Points

Activity Diagram



Screen Description

N/A – background system process with no dedicated screen. Results are shown in the screens of the calling use cases.

UC17: Mint NFT

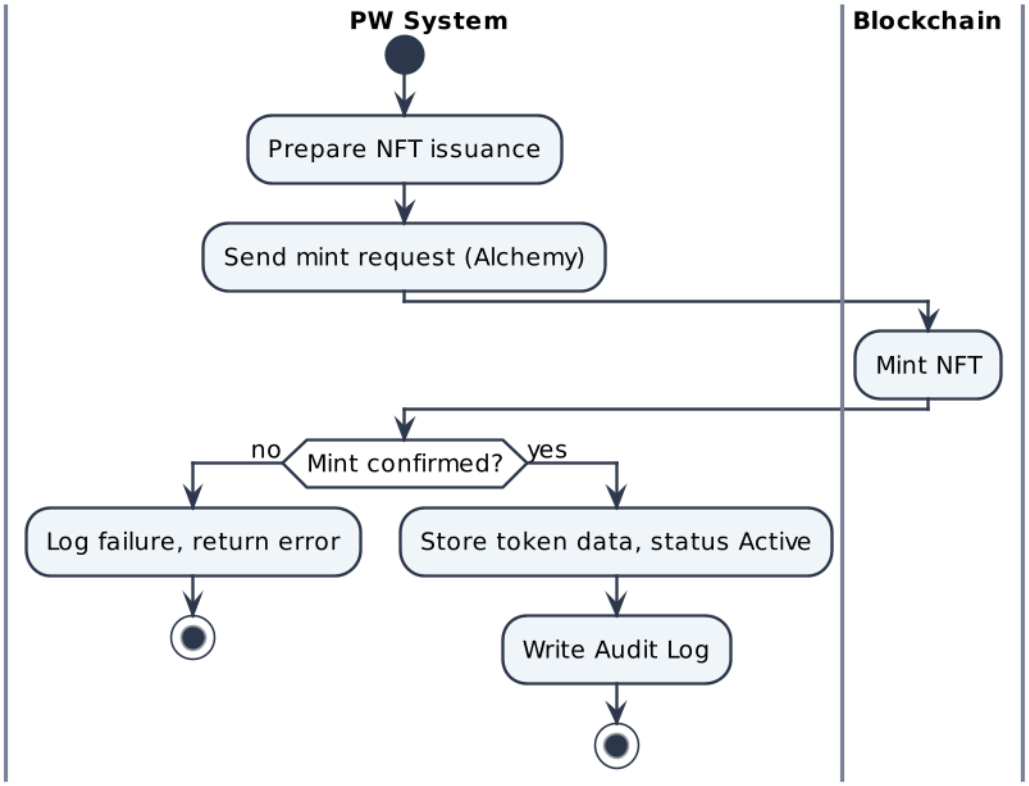
Use Case Description

UC ID	UC17
UC Name	Mint NFT
Objective	Mint an NFT on-chain through the PW smart contract (via Alchemy)

	whenever a purchase is paid (Distinction NFT) or an Award is granted (soulbound Award NFT). There is no manual mint screen.
Actor	PW System (R05); Blockchain / Smart Contract (external)
Trigger	Payment confirmed (UC14/UC15) or Award grant confirmed (UC29), after the recipient wallet is available.
Pre-condition	- The recipient wallet address is known. - For purchases: payment is confirmed.
Post-condition	**Success:** • NFT on-chain; token ID, transaction hash, owner, type, category, issue date stored; status Active; Audit Log (MINT / AWARD_MINT). **Failure:** • No Active NFT; event logged; caller flow handles the error.
Basic flow	1. System prepares the issuance: type (Distinction for purchase, Award for grant), category, recipient wallet, metadata. 2. System sends the mint request to the smart contract via Alchemy. 3. Blockchain executes the mint and returns the transaction hash. 4. System waits for on-chain confirmation. 5. System stores token ID, transaction hash, owner, type, category, issue date and sets status = Active. 6. System writes the Audit Log entry and returns success to the calling use case.
Alternative flow	**A1 – Award** 1. For a grant, the token is minted as soulbound (non-transferable).
Exception flow	**E1 – Mint failed / timeout** 1. At step 3/4 the transaction fails or is not confirmed within the timeout. 2. System does not set the NFT Active, logs the event and returns failure (purchase → "NFT issuance pending"; grant → Admin error).
Business rule	• BR01 – Minting is triggered only by a confirmed payment or an Admin grant; no manual mint function (BR-9). • BR02 – PW is the sole issuing authority; the NFT is minted directly to the recipient and is Active immediately. • BR03 – Every mint is recorded in the Audit Log.
Non-functional requirement	• NFR01 – NFTs are immutable and publicly verifiable on-chain (NFR-005). • NFR02 – Mint confirmation is typically obtained within 2 minutes (proposed; depends on chain).
Acceptance criteria	**AC01 – Mint on purchase** Given payment is confirmed, when the mint succeeds, then a

	Distinction NFT is Active for the recipient with token ID, tx hash and issue date stored. **AC02 – Mint on grant** Given a grant is confirmed, when the mint succeeds, then a soulbound Award NFT is Active for the recipient. **AC03 – Mint fails** Given a mint is requested, when the blockchain call fails, then no Active NFT exists and the event is logged.
Traceability	BRD: FR-013 · BPMN: D2, D3 – Prepare NFT Issuance / Mint

Activity Diagram



Screen Description

N/A – background system process with no dedicated screen. Results are shown in the screens of the calling use cases.

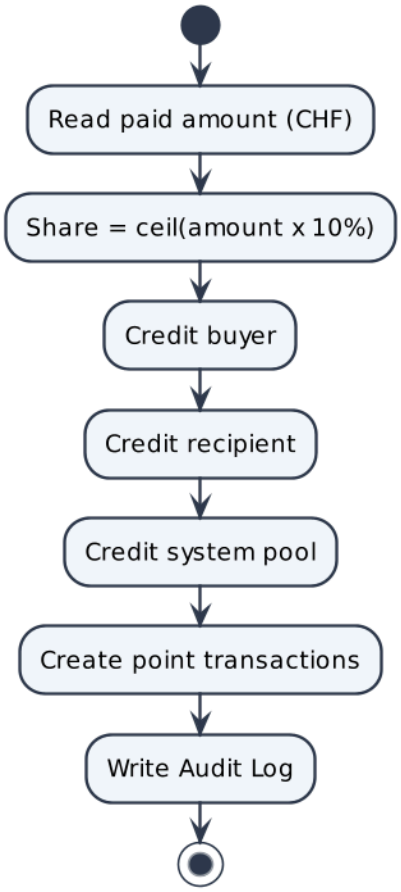
UC18: Distribute Reward Points

Use Case Description

UC ID	UC18
UC Name	Distribute Reward Points
Objective	After a paid transaction, award points equal to 30% of the transaction value: 10% to the buyer, 10% to the recipient and 10% to the system pool, using 1 CHF = 1 Point rounded up.
Actor	PW System (R05)
Trigger	The NFT of a paid Purchase Request is Active (UC14/UC15).
Pre-condition	- The transaction is paid (not an Admin grant).
Post-condition	**Success:** • Point transactions created (account, amount, type Earned, reason, reference, time); balances updated.
Basic flow	1. System reads the final paid amount (CHF) of the Purchase Request. 2. System calculates each share = amount × 10%, rounded UP to a whole point. 3. System credits the buyer share. 4. System credits the recipient share. 5. System credits the system-pool share. 6. System creates one point transaction per credit, linked to the Purchase Request, and writes an Audit Log entry (POINTS_EARN).
Alternative flow	**A1 – Admin grant** 1. For an Award grant no points are credited (0 points). **A2 – Buyer = recipient** 1. **TBD: buyer receives 10% or 20% (buyer + recipient share).**
Exception flow	**E1 – Credit fails** 1. The whole credit is rolled back and retried; no partial credit is kept.
Business rule	• BR01 – 30% of the transaction value: 10/10/10 (BR-3); paid transactions only. • BR02 – 1 CHF = 1 Point, rounded up (BR-4). • BR03 – Points are credited only once per Purchase Request.
Non-functional requirement	• NFR01 – Exact, auditable calculation (NFR-010).
Acceptance criteria	**AC01 – Gift 1,255 CHF** Given a gift is paid 1,255 CHF, when points are distributed, then

	buyer 126, recipient 126, system 126 points (125.5 → 126) and 3 point transactions exist. **AC02 – Admin grant** Given an Award is granted, when the NFT is issued, then no points are credited.
Traceability	BRD: FR-018 · BPMN: D2 – Distribute Points 10/10/10

Activity Diagram



Screen Description

N/A – background system process with no dedicated screen. Results are shown in the screens of the calling use cases.

UC19: Transfer NFT to Own Non-custodial Wallet (Self-custody)

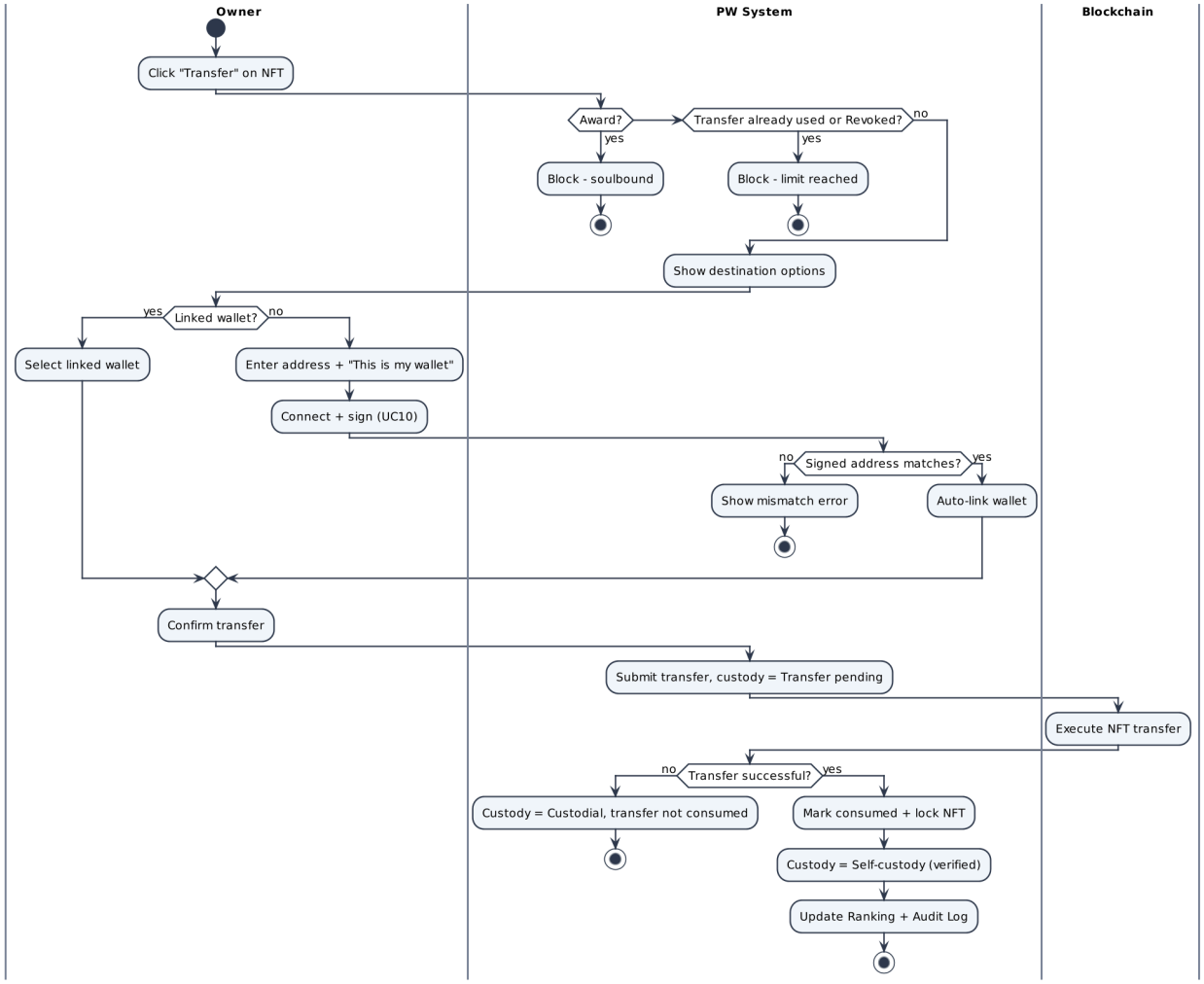
Use Case Description

UC ID	UC19
UC Name	Transfer NFT to Own Non-custodial Wallet (Self-custody)
Objective	Allow the owner to move an Active Distinction NFT from the PW custodial wallet to their own verified non-custodial wallet. This uses the one-time transfer; the NFT is then locked in that wallet and keeps counting toward Ranking.
Actor	NFT owner – Company (R02) / Individual (R03); PW System (R05); Blockchain
Trigger	The owner clicks "Transfer" on a Distinction NFT and chooses "To my own wallet".
Pre-condition	- The NFT is an Active Distinction in the custodial wallet. - Its one-time transfer is unused.
Post-condition	**Success:** • NFT transferred on-chain; one-time transfer consumed; custody = "Self-custody (verified)"; NFT locked; still counted in Ranking; Audit Log (TRANSFER_SELF). **Failure:** • No transfer; custody stays Custodial; one-time transfer NOT consumed.
Basic flow	1. Owner opens the NFT and clicks "Transfer". 2. System validates the NFT: type = Distinction, status = Active, transfer not yet used. 3. System displays "Choose destination" with the shared linked-wallet list and the option "Another party" (UC20). 4. Owner selects the linked, verified wallet. 5. System displays the Review: NFT, destination, notice "This uses your one-time transfer. The NFT will be locked in the destination wallet." 6. Owner clicks "Confirm transfer". 7. System submits the on-chain transfer (no new signature needed) and sets custody = "Transfer pending" (UC21). 8. Blockchain executes the transfer. 9. System receives the confirmation, marks the one-time transfer as consumed, locks the NFT and sets custody = "Self-custody (verified)". 10. System keeps the NFT counted in Ranking (UC28), writes the Audit Log and notifies "Transfer completed."

Alternative flow	**A1 – New wallet "This is my wallet"** 1. At step 4 the destination is not linked. 2. Owner enters the address and ticks "This is my wallet". 3. System runs Connect + Sign (UC10); if the signed address equals the destination, the wallet is linked automatically. 4. Continue at step 5. **A2 – Cancel** 1. At step 6 Owner cancels → no transfer.
Exception flow	**E1 – Award NFT** 1. At step 2 type = Award → "Award NFTs are soulbound and cannot be transferred." (Transfer action is normally hidden). **E2 – Transfer already used** 1. At step 2 → "This NFT has already used its one-time transfer." **E3 – Revoked NFT** 1. At step 2 status = Revoked → transfer blocked. **E4 – Signature mismatch** 1. In A1 the signed address differs → "The signed wallet doesn't match the destination address."; no transfer. **E5 – On-chain failure** 1. At step 8/9 the transaction fails → custody back to Custodial; transfer NOT consumed; "Transfer failed. Your NFT is still in your PW wallet."
Business rule	• BR01 – Award = soulbound; Distinction = one transfer only; Revoked = no transfer (BR-6). • BR02 – Self-custody OR transfer to another party – the same single transfer; not both. • BR03 – After the transfer the NFT is locked in the destination wallet. • BR04 – Only a linked, signature-verified wallet keeps the NFT in Ranking; 1 wallet ↔ 1 account. • BR05 – The transfer is consumed only after on-chain success.
Non-functional requirement	• NFR01 – Status shows "Transfer pending" until on-chain confirmation; no duplicate submission possible.
Acceptance criteria	**AC01 – To linked wallet** Given an Active Distinction with transfer unused and a linked verified wallet, when the owner confirms, then the NFT moves on-chain without a new signature, transfer is consumed, custody = Self-custody (verified) and it still counts in Ranking. **AC02 – New wallet signed** Given destination not linked, "This is my wallet" ticked, when the owner signs and addresses match, then the wallet is auto-linked and the transfer proceeds. **AC03 – Signature mismatch** Given "This is my wallet" ticked, when the signed address differs, then "The signed wallet doesn't match the destination address." and

	no transfer. **AC04 – Award** Given the NFT is an Award, when a transfer is requested, then "Award NFTs are soulbound and cannot be transferred.".
Traceability	BRD: FR-010, FR-014, FR-031, FR-034 · BPMN: D4 – Transfer (to non-custodial)

Activity Diagram



Screen Description

Screen 1: Transfer – choose destination & review

[Screenshot: Figma frame "PW — Transfer NFT (Choose destination / Review)"]

#	Name	Control type	Data type	Required	Description
I. Destination					
1	Destination type	Radio	Enum	Yes	"My linked wallet" | "Another party" (UC20).
2	Linked wallet list	Radio list	String	Cond.	Shared linked-wallet list; shows address and "Verified".
3	Wallet address	Textbox	String	Cond.	For a new own wallet; format/network validated.
4	This is my wallet	Checkbox	Boolean		Triggers Connect + Sign (A1).
II. Review					
1	One-time transfer notice	Alert	String		"This uses your one-time transfer. The NFT will be locked in the destination wallet."
2	Confirm transfer	Button	N/A		Submits the transfer.
3	Cancel	Button	N/A		Leaves without transfer.
4	Status	Badge	Enum		Transfer pending → Self-custody (verified) / Failed.

UC20: Transfer Distinction NFT to Another Party

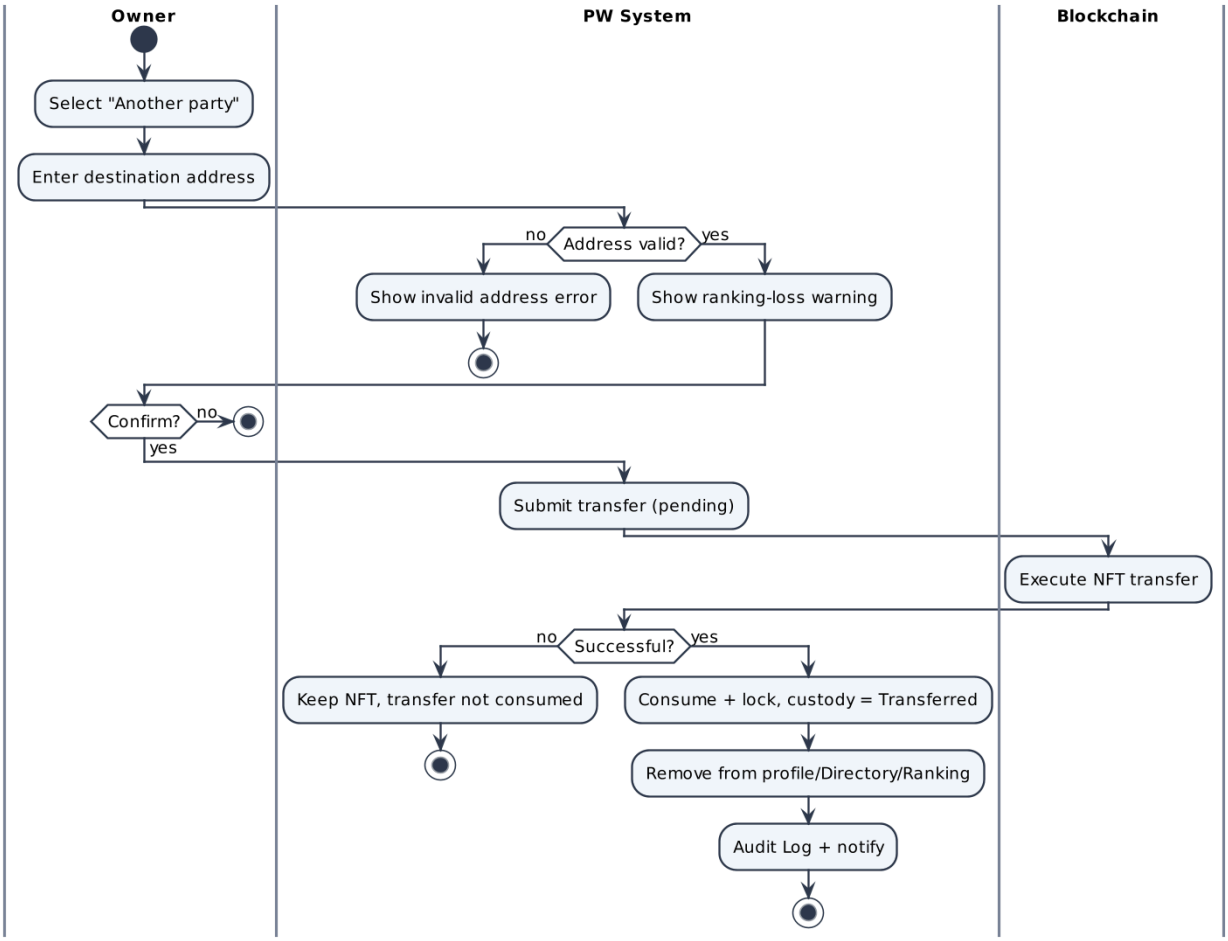
Use Case Description

UC ID	UC20
UC Name	Transfer Distinction NFT to Another Party
Objective	Allow the owner to transfer an Active Distinction NFT to another party's wallet after an explicit warning. This uses the one-time transfer; the NFT is removed from the owner's profile, Directory and Ranking.
Actor	NFT owner – Company (R02) / Individual (R03); PW System (R05);

	Blockchain
Trigger	The owner chooses "Another party" as destination.
Pre-condition	- The NFT is an Active Distinction whose one-time transfer is unused.
Post-condition	**Success:** • NFT transferred on-chain; transfer consumed; NFT locked in destination; custody = "Transferred"; removed from the owner's profile, Directory and Ranking; Audit Log (TRANSFER_OUT). **Failure / cancel:** • No transfer; transfer not consumed.
Basic flow	1. Owner clicks "Transfer" and selects "Another party". 2. System validates the NFT (as UC19 step 2). 3. Owner enters the destination wallet address ("This is my wallet" not ticked). 4. System validates the address format and network. 5. System displays the warning "After this transfer the NFT will be removed from your profile, the Directory and your Ranking. This cannot be undone." 6. Owner ticks "I understand" and clicks "Confirm transfer". 7. System submits the on-chain transfer; custody = "Transfer pending". 8. Blockchain executes the transfer. 9. System marks the transfer consumed, locks the NFT, sets custody = "Transferred" and removes the NFT from the owner's profile, Directory and Ranking (UC28). 10. System writes the Audit Log and notifies the owner "Transfer completed."
Alternative flow	**A1 – Cancel at warning** 1. At step 6 Owner clicks "Cancel" → no transfer.
Exception flow	**E1 – Invalid address** 1. At step 4 → "Invalid wallet address. Please check and re-enter."; transfer blocked. **E2 – Award / used / revoked** 1. As UC19-E1/E2/E3. **E3 – On-chain failure** 1. As UC19-E5; NFT stays with the owner.
Business rule	• BR01 – One-time transfer; then locked (BR-6). • BR02 – Transfers to another party are not counted in the original owner's Ranking or profile. • BR03 – The warning must be acknowledged before confirmation.
Non-functional requirement	• NFR01 – Address validation runs on input with feedback within 1 second (proposed).

Acceptance criteria	**AC01 – Transfer with warning** Given an Active Distinction with transfer unused, when the owner enters a valid address, acknowledges the warning and confirms, then the NFT is transferred, transfer consumed, and removed from profile, Directory and Ranking. **AC02 – Invalid address** Given a wrong-format address, when it is entered, then an error is shown and transfer is blocked. **AC03 – Cancel** Given the warning is shown, when "Cancel" is clicked, then no transfer happens.
Traceability	BRD: FR-033, FR-014, FR-034 · BPMN: D4 – Transfer (to another owner)

Activity Diagram



Screen Description

Screen 1: Transfer to another party – warning

[Screenshot: Figma frame "PW — Transfer NFT (Another party – warning)"]

#	Name	Control type	Data type	Required	Description
1	Destination wallet address	Textbox	String	Yes	Validated for format and network.
2	Warning	Alert (danger)	String		"After this transfer the NFT will be removed from your profile, the Directory and your Ranking. This cannot be undone."
3	I understand	Checkbox	Boolean	Yes	Enables "Confirm transfer".
4	Confirm transfer	Button	N/A		Submits the transfer.
5	Cancel	Button	N/A		No transfer.

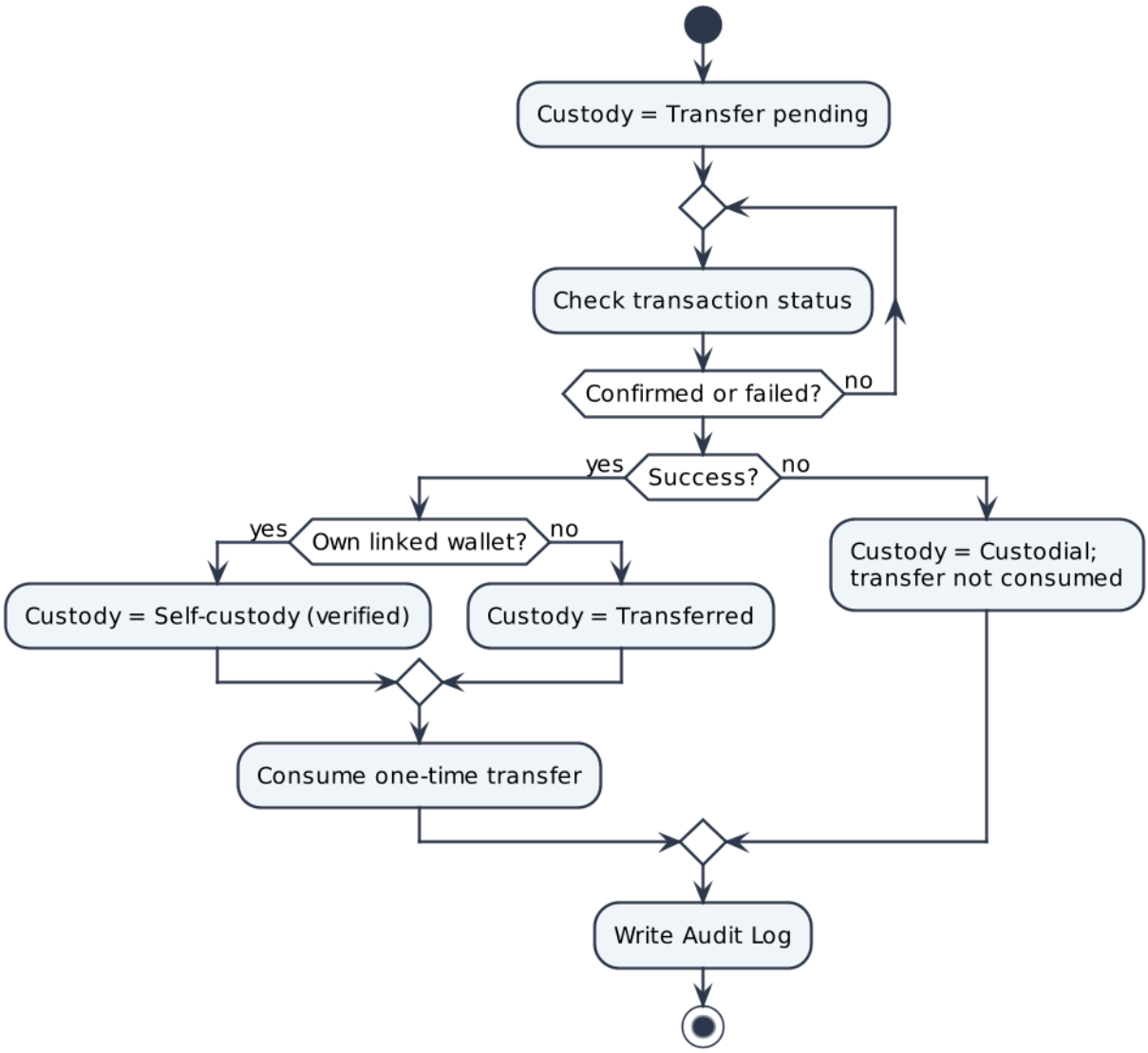
UC21: Track NFT Custody Status

Use Case Description

UC ID	UC21
UC Name	Track NFT Custody Status
Objective	Keep each NFT's custody status in sync with the blockchain: Custodial → Transfer pending → Self-custody (verified) or Transferred; on failure the NFT stays Custodial and the one-time transfer is not used.
Actor	PW System (R05); Blockchain
Trigger	A transfer transaction is submitted (UC19/UC20).
Pre-condition	- A transfer transaction has been submitted.
Post-condition	**Success:** • Custody status reflects the on-chain result; Audit Log entry per change.
Basic flow	1. System sets custody = "Transfer pending" when the transaction is submitted. 2. System monitors the transaction status on-chain.

	3. On confirmation, System sets custody = "Self-custody (verified)" (own linked wallet) or "Transferred" (another party). 4. System marks the one-time transfer as consumed and writes an Audit Log entry.
Alternative flow	N/A
Exception flow	**E1 – Transaction failed or dropped** 1. System sets custody back to "Custodial", does NOT consume the transfer, logs the event and notifies the owner.
Business rule	• BR01 – Custody values: Custodial, Transfer pending, Self-custody (verified), Transferred. • BR02 – The transfer counter changes only on on-chain success.
Non-functional requirement	• NFR01 – Status polling interval ≤ 30 seconds while pending (proposed).
Acceptance criteria	**AC01 – Confirmed** Given a transfer to a linked wallet is submitted, when it is confirmed on-chain, then custody = Self-custody (verified) and an Audit Log entry exists. **AC02 – Pending** Given the transaction is unconfirmed, when the owner views the NFT, then "Transfer pending" is shown. **AC03 – Failed** Given the transaction fails, when the failure is received, then custody = Custodial and the transfer is not consumed.
Traceability	BRD: FR-034 · BPMN: D4 – Receive Transfer Result / Update Owner

Activity Diagram



Screen Description

N/A – background system process with no dedicated screen. Results are shown in the screens of the calling use cases.

UC22: Reconcile Linked Wallets On-chain

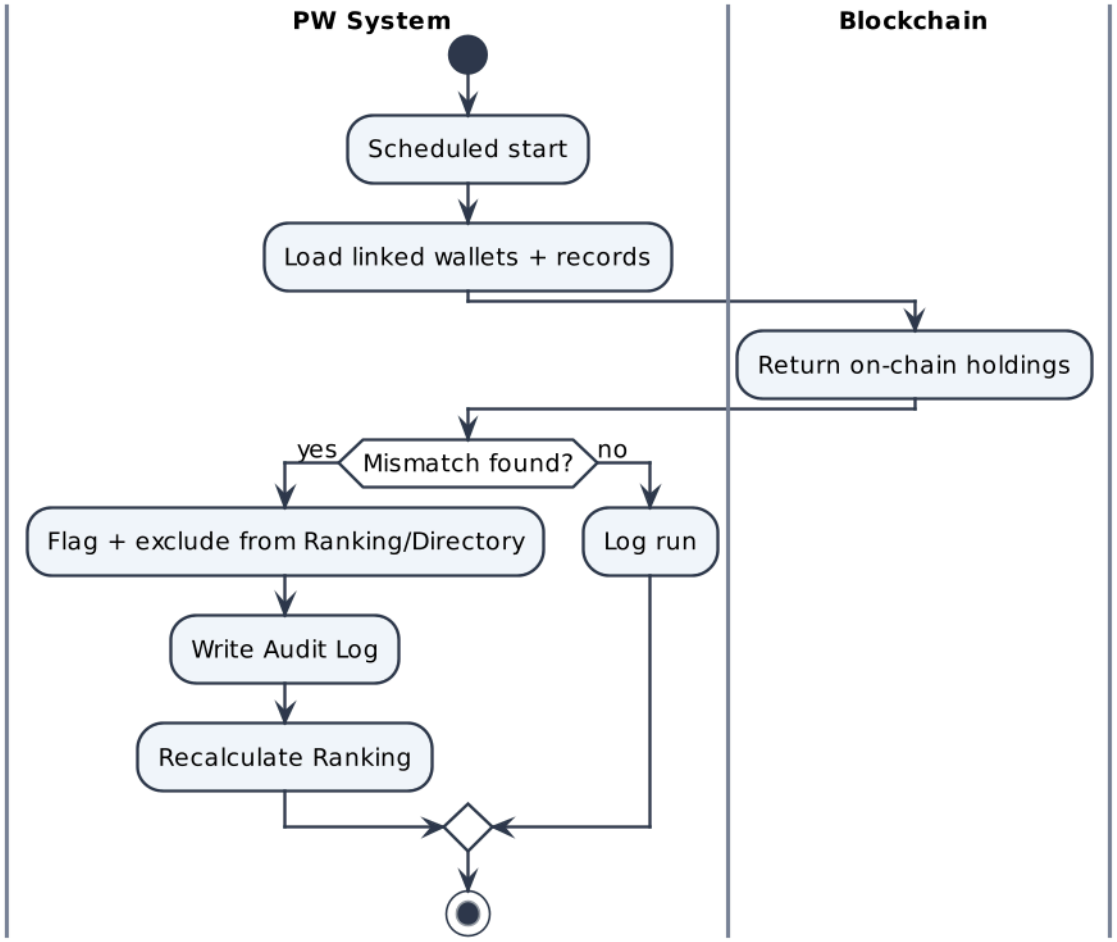
Use Case Description

UC ID	UC22

UC Name	Reconcile Linked Wallets On-chain
Objective	Periodically compare the on-chain holdings of every linked non-custodial wallet with platform records; NFTs no longer held there are removed from the owner's Ranking and Directory profile and each finding is logged.
Actor	PW System (R05, scheduler); Blockchain
Trigger	Scheduled job.
Pre-condition	- At least one linked non-custodial wallet exists.
Post-condition	**Success:** • Mismatches flagged, excluded from Ranking/Directory, written to the Audit Log (RECONCILE).
Basic flow	1. Scheduler starts the reconciliation job. 2. System reads the list of linked wallets and the NFTs recorded in each. 3. System queries the on-chain holdings of each wallet. 4. System compares on-chain holdings with the records. 5. For each NFT no longer in the linked wallet, System flags the mismatch, removes it from the owner's Ranking and Directory profile and writes an Audit Log entry. 6. System triggers a Ranking recalculation if any change was found (UC28).
Alternative flow	**A1 – No mismatch** 1. No change is made; the job run is logged.
Exception flow	**E1 – Blockchain query fails** 1. The wallet is skipped and retried in the next run; the error is logged.
Business rule	• BR01 – **TBD: check frequency.** • BR02 – The job never moves NFTs; it only updates records and Ranking.
Non-functional requirement	• NFR01 – The job must not degrade user-facing performance (runs in background).
Acceptance criteria	**AC01 – Mismatch** Given an NFT recorded in a linked wallet is no longer there on-chain, when the job runs, then it is removed from Ranking and Directory profile and logged. **AC02 – No mismatch** Given all records match, when the job runs, then no ranking change is made.

Traceability	BRD: FR-035 · BPMN: —

Activity Diagram



Screen Description

N/A – background system process with no dedicated screen. Results are shown in the screens of the calling use cases.

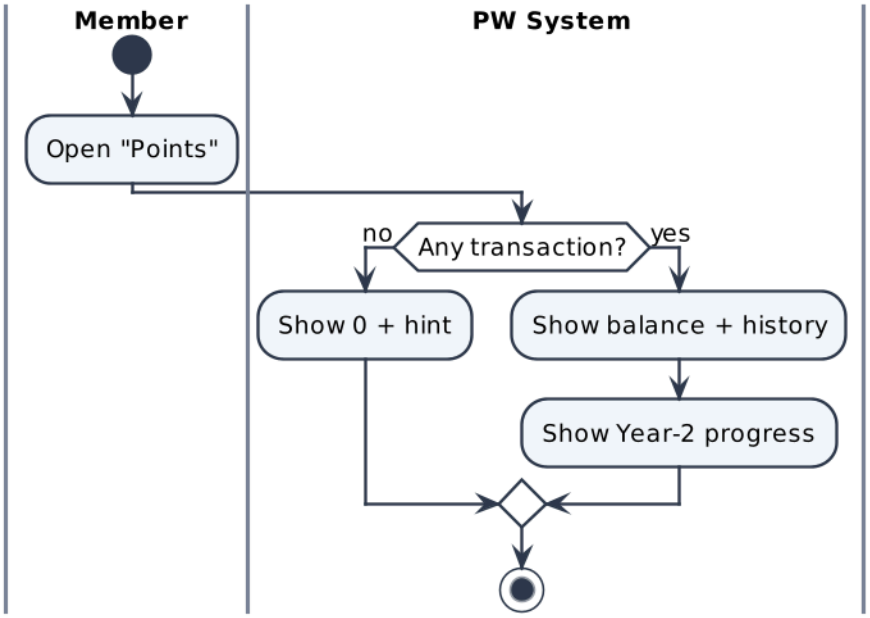
UC23: View Point Balance and History

Use Case Description

UC ID	UC23

UC Name	View Point Balance and History
Objective	Show the Member's current point balance and the history of point transactions.
Actor	Company (R02), Individual (R03)
Trigger	The Member opens "Points".
Pre-condition	- The Member is logged in.
Post-condition	**Success:** • Balance and history displayed. No data is changed.
Basic flow	1. Member opens "Points". 2. System displays the current balance. 3. System displays the history (date, type Earned / Used / Adjusted, amount ±, reference, balance after), newest first, 20 rows per page (assumption). 4. System shows the Year-2 discount progress (account age and points vs threshold).
Alternative flow	**A1 – No points** 1. Balance 0 and "Earn points when you buy or receive a Title."
Exception flow	**E1 – Load error** 1. Retry message.
Business rule	• BR01 – Balance = sum of all point transactions; never below 0.
Non-functional requirement	• NFR01 – History loads within 2 seconds (proposed).
Acceptance criteria	**AC01 – Has points** Given the Member has transactions, when "Points" is opened, then balance and history (newest first) are shown. **AC02 – No points** Given no transactions, when "Points" is opened, then balance 0 and the hint message are shown.
Traceability	BRD: FR-019 · BPMN: —

Activity Diagram



Screen Description

Screen 1: Points

[Screenshot: Figma frame "PW — Points (to be designed)"]

#	Name	Control type	Data type	Required	Description
1	Current balance	Label / KPI card	Integer		Total points.
2	Year-2 discount progress	Progress bar	Mixed		Account age and points vs threshold (**threshold TBD**).
3	History table	Table	Mixed		Date, Type (Earned/Used/Adjusted), Amount (±), Reference (order/NFT/admin), Balance after.
4	Pagination	Pager	N/A		20 rows per page (assumption).

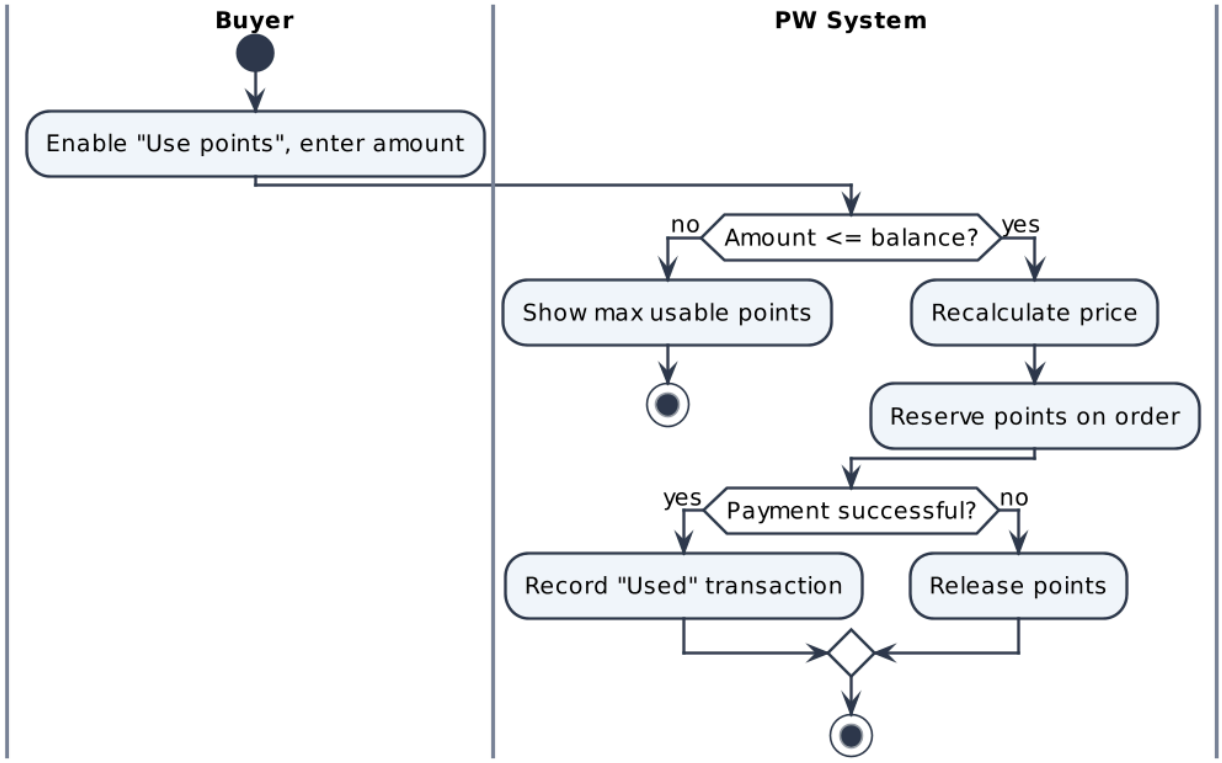
UC24: Use Points

Use Case Description

UC ID	UC24
UC Name	Use Points
Objective	Allow the Member to use earned points during a purchase within the available balance; used points are linked to the order and returned if the order is cancelled or payment fails.
Actor	Company (R02), Individual (R03); PW System (R05)
Trigger	The buyer chooses to use points on the Review step of UC14/UC15.
Pre-condition	- The buyer has a positive point balance. - **Points redemption mechanism is confirmed (TBD).**
Post-condition	**Success:** • A "Used" point transaction linked to the order; balance reduced. **Order cancelled / payment failed:** • Used points returned ("Returned" transaction).
Basic flow	1. Buyer enables "Use points" and enters the number of points. 2. System checks that the amount ≤ available balance. 3. System recalculates the price summary (UC16). 4. On "Confirm & Pay", System reserves the points on the order. 5. On payment success, System records a "Used" transaction linked to the order.
Alternative flow	**A1 – Order cancelled / payment failed** 1. System releases the reserved points back to the balance.
Exception flow	**E1 – Amount exceeds balance** 1. "You can use up to [balance] points."; input rejected.
Business rule	• BR01 – Balance never below 0. • BR02 – **TBD: are points only a discount-eligibility gate, or deductible from the price, and at what rate?**
Non-functional requirement	• NFR01 – Reservation and release are atomic with the order status (NFR-010).
Acceptance criteria	**AC01 – Within balance** Given balance 500, when the buyer uses 200 and pays, then a Used transaction of 200 is linked to the order and balance = 300. **AC02 – Exceeds balance** Given balance 100, when the buyer enters 150, then the input is

	rejected. **AC03 – Payment fails** Given points reserved, when payment fails, then points are returned.
Traceability	BRD: FR-020 · BPMN: D2 – Discount/Points

Activity Diagram



Screen Description

N/A – background system process with no dedicated screen. Results are shown in the screens of the calling use cases.

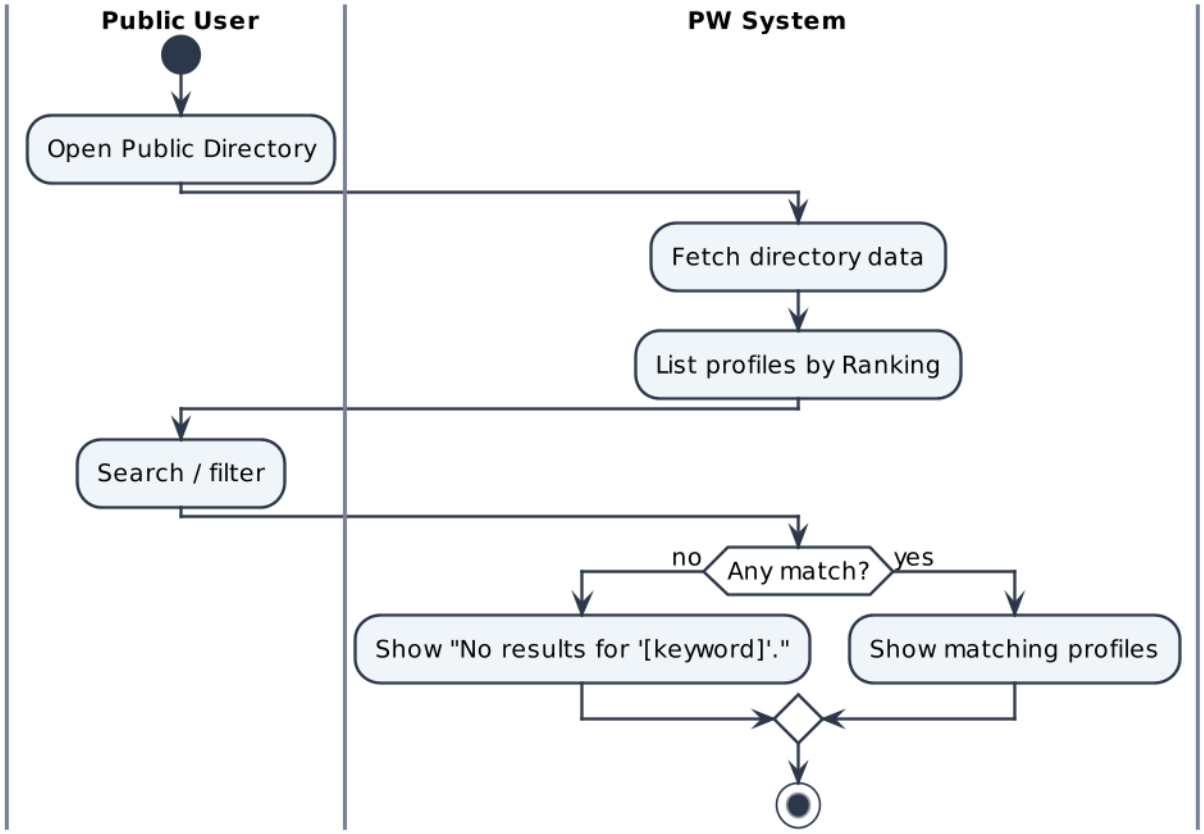
UC25: Browse Public Directory

Use Case Description

UC ID	UC25
UC Name	Browse Public Directory

Objective	Provide a public, login-free directory of KYC-approved Companies and Individuals recognized by PW, ordered by Ranking, with search and filters.
Actor	Public Verifying User / Visitor (R01) (also Members and Admin)
Trigger	The user opens the Public Directory.
Pre-condition	- None.
Post-condition	**Success:** • Matching profiles displayed. No data is changed.
Basic flow	1. User opens the Public Directory. 2. System fetches directory data and lists KYC-approved public profiles: rank, name, type (Company / Individual), Class, number of counted NFTs, ordered by Ranking. 3. User searches by name or filters by type, Class or Title category. 4. System displays only matching profiles (paginated). 5. User selects a profile → UC26.
Alternative flow	**A1 – No result** 1. "No results for '[keyword]'." with a "Clear filters" action.
Exception flow	**E1 – Load error** 1. Retry message.
Business rule	• BR01 – Only KYC-approved entities are listed. • BR02 – Order = Ranking (UC28).
Non-functional requirement	• NFR01 – Directory first page loads within 2 seconds (proposed). • NFR02 – Public data only; no personal contact data.
Acceptance criteria	**AC01 – Browse** Given the Directory is opened without login, when the page loads, then approved profiles are listed by Ranking. **AC02 – Filter** Given the user filters by Title category, when the filter is applied, then only matching profiles are shown. **AC03 – No result** Given nothing matches, when the search runs, then "No results for '[keyword]'." is shown.
Traceability	BRD: FR-021 · BPMN: D6 – Browse Public Directory

Activity Diagram



Screen Description

Screen 1: Public Directory

[Screenshot: Figma frame "PW — Public Directory"]

#	Name	Control type	Data type	Required	Description
1	Search	Textbox	String		Search by name.
2	Type filter	Dropdown	Enum		All / Company / Individual.
3	Class filter	Multi-select	Enum		Class 0–7.
4	Title category filter	Multi-select	Enum		Philanthropist / Investor / Bridge.
5	Directory list	Table / Cards	Mixed		Rank, name, type, Class, number of NFTs; row click → UC26.
6	Empty state	Label	String		"No results for '[keyword]'."

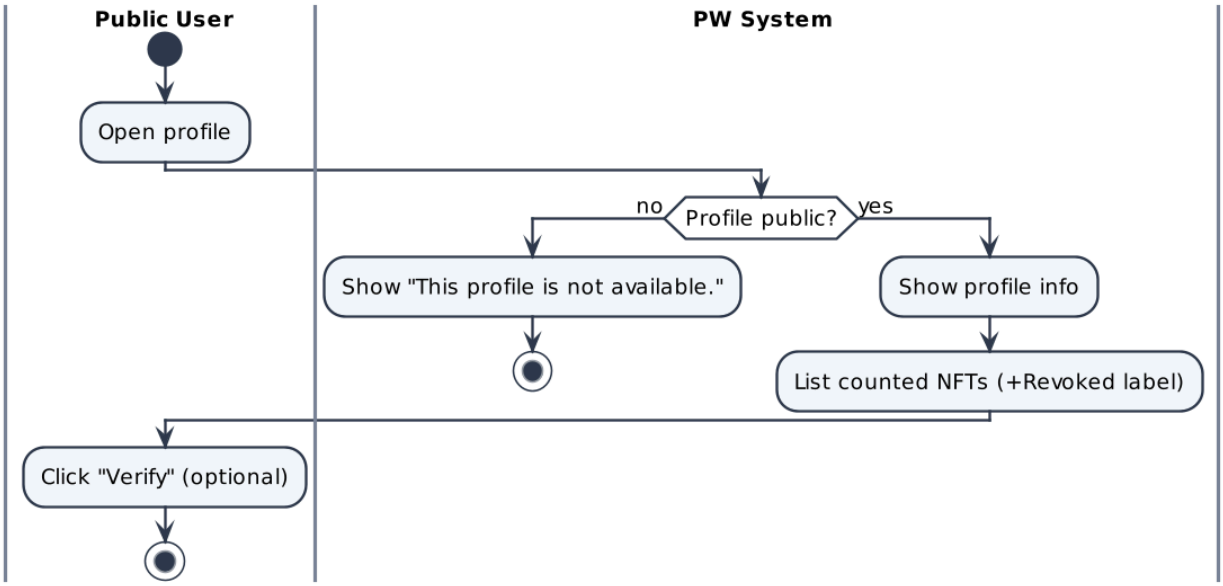
UC26: View Profile and Recognitions

Use Case Description

UC ID	UC26
UC Name	View Profile and Recognitions
Objective	Show the public profile of a Company or Individual and the NFTs counted for it, so that third parties can see which recognitions it holds.
Actor	Public Verifying User / Visitor (R01)
Trigger	The user selects a profile in the Directory or opens a profile link.
Pre-condition	- The profile belongs to a KYC-approved account.
Post-condition	**Success:** • Profile and NFT list displayed. No data is changed.
Basic flow	1. User opens a profile. 2. System displays public information: name, type, Class, rank. 3. System lists the counted NFTs: name, category, type (Distinction / Award), "Issued by PW", issue date, status. 4. User clicks "Verify" on an NFT → UC27.
Alternative flow	**A1 – Revoked NFT** 1. The NFT is shown with the label "Revoked". **A2 – No NFTs** 1. "No recognitions yet."
Exception flow	**E1 – Profile not found / not public** 1. "This profile is not available."
Business rule	• BR01 – Only public data is shown (no email, KYC data or point balance). • BR02 – NFTs transferred to another party or in an unlinked wallet are not shown.
Non-functional requirement	• NFR01 – Profile loads within 2 seconds (proposed).
Acceptance criteria	**AC01 – View** Given a profile with NFTs, when it is opened, then name, type, Class, rank and each NFT's name, category, type, issuer, date and status are shown. **AC02 – Revoked** Given one NFT is revoked, when the profile is shown, then that NFT is labelled "Revoked".

	AC03 – No NFTs Given no NFT, when the profile is opened, then "No recognitions yet." is shown.
Traceability	BRD: FR-022, FR-023 · BPMN: D6 – View Profile / View NFT

Activity Diagram



Screen Description

Screen 1: Public profile

[Screenshot: Figma frame "PW — Public Profile"]

#	Name	Control type	Data type	Required	Description
1	Profile header	Label group	String		Name, type, Company Class (Companies only), rank.
2	Recognitions list	Cards	Mixed		Name, category, type, "Issued by PW", issue date, status.
3	Verify	Button	N/A		Opens UC27 for the NFT.
4	Empty state	Label	String		"No recognitions yet."

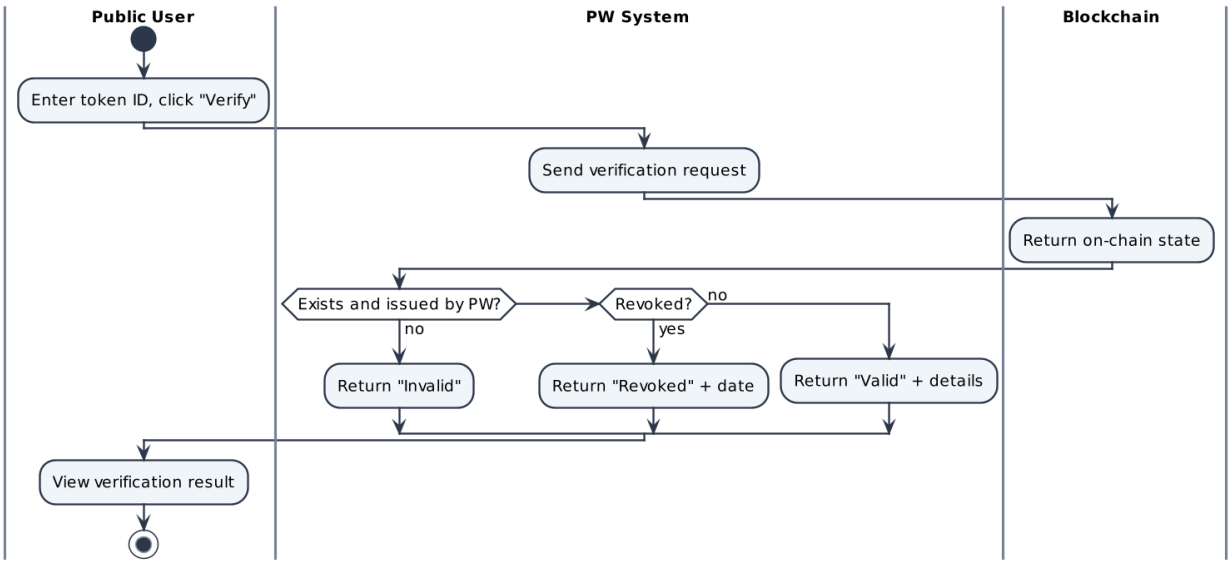
UC27: Verify NFT On-chain

Use Case Description

UC ID	UC27
UC Name	Verify NFT On-chain
Objective	Allow anyone to verify an NFT against the blockchain and PW's contract and receive one of three results: Valid, Revoked or Invalid.
Actor	Public Verifying User / Visitor (R01); PW System (R05); Blockchain
Trigger	The user enters a token ID on "Verify" or clicks "Verify" on a profile/NFT.
Pre-condition	- None.
Post-condition	**Success:** • Exactly one result is displayed. No data is changed.
Basic flow	1. User opens "Verify" and enters a token ID (or it is pre-filled from a profile). 2. User clicks "Verify". 3. System sends the verification request and queries the on-chain state (existence, issuing contract, revocation state). 4. System determines the status. 5. Valid: System shows "Valid" with owner, type, category, issue date and transaction link.
Alternative flow	**A1 – Revoked** 1. Token issued by PW but revoked → "Revoked" with the revocation date. **A2 – Invalid** 1. Token does not exist or was not issued by PW's contract → "Invalid – this NFT was not issued by PW."
Exception flow	**E1 – Empty token ID** 1. "Please enter a token ID." **E2 – Blockchain unavailable** 1. "Verification is temporarily unavailable. Please try again."
Business rule	• BR01 – Exactly one of Valid / Revoked / Invalid. • BR02 – The on-chain state is the source of truth (NFR-005).
Non-functional requirement	• NFR01 – Result within 3 seconds under normal conditions (proposed).
Acceptance criteria	**AC01 – Valid** Given an existing PW token not revoked, when it is verified, then

	"Valid" with owner, type, category, issue date and tx link. **AC02 – Revoked** Given a revoked PW token, when it is verified, then "Revoked" with revocation date. **AC03 – Invalid** Given a non-PW or non-existent token, when it is verified, then "Invalid – this NFT was not issued by PW.".
Traceability	BRD: FR-024 · BPMN: D6 – Request NFT Verification / NFT status?

Activity Diagram



Screen Description

Screen 1: Verify NFT

[Screenshot: Figma frame "PW — Verify / Verification Result"]

#	Name	Control type	Data type	Required	Description
1	Token ID	Textbox	String	Yes	Token ID to verify.
2	Verify	Button	N/A		Runs the verification.
3	Result badge	Badge	Enum		Valid (green) / Revoked (red) / Invalid (grey).
4	Details	Label group	Mixed		Owner, type, category, issue date, transaction link (Valid);

#	Name	Control type	Data type	Required	Description
					revocation date (Revoked).

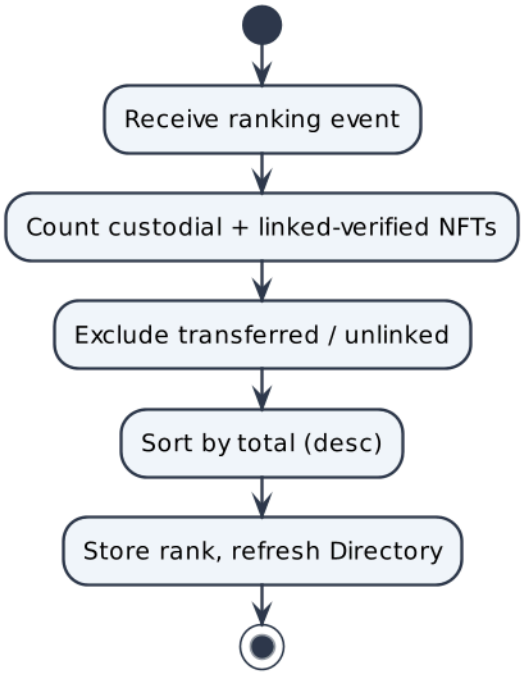
UC28: Calculate Ranking

Use Case Description

UC ID	UC28
UC Name	Calculate Ranking
Objective	Automatically rank accounts by the total number of counted NFTs and recalculate after every event that changes holdings.
Actor	PW System (R05)
Trigger	Mint, transfer, revoke, link/unlink or reconciliation event.
Pre-condition	- At least one KYC-approved account exists.
Post-condition	**Success:** • Every account has an up-to-date rank used by the Directory.
Basic flow	1. System receives a ranking event. 2. System counts, per account, NFTs in the PW custodial wallet + NFTs in the linked signature-verified wallet. 3. System excludes NFTs transferred to another party and NFTs in unlinked wallets. 4. System orders accounts by the counted total, highest first, and stores the rank. 5. System refreshes the Directory order.
Alternative flow	N/A
Exception flow	**E1 – Calculation error** 1. The previous ranking is kept and the error is logged; recalculation is retried.
Business rule	• BR01 – Ranking by total counted NFTs (BR-11). • BR02 – **TBD: Award vs Distinction weighting; tie-break; whether revoked NFTs are excluded.**
Non-functional requirement	• NFR01 – Ranking reflects an event within 1 minute (proposed).

Acceptance criteria	**AC01 – Order** Given A has 3 counted NFTs, B has 5, when ranking runs, then B ranks above A. **AC02 – Transfer out** Given A transfers one NFT to another party, when ranking runs, then A's count decreases by 1. **AC03 – Self-custody** Given A moves an NFT to its linked wallet, when ranking runs, then A's count is unchanged.
Traceability	BRD: FR-025 · BPMN: D2, D3, D4 – Update Ranking

Activity Diagram



Screen Description

N/A – background system process with no dedicated screen. Results are shown in the screens of the calling use cases.

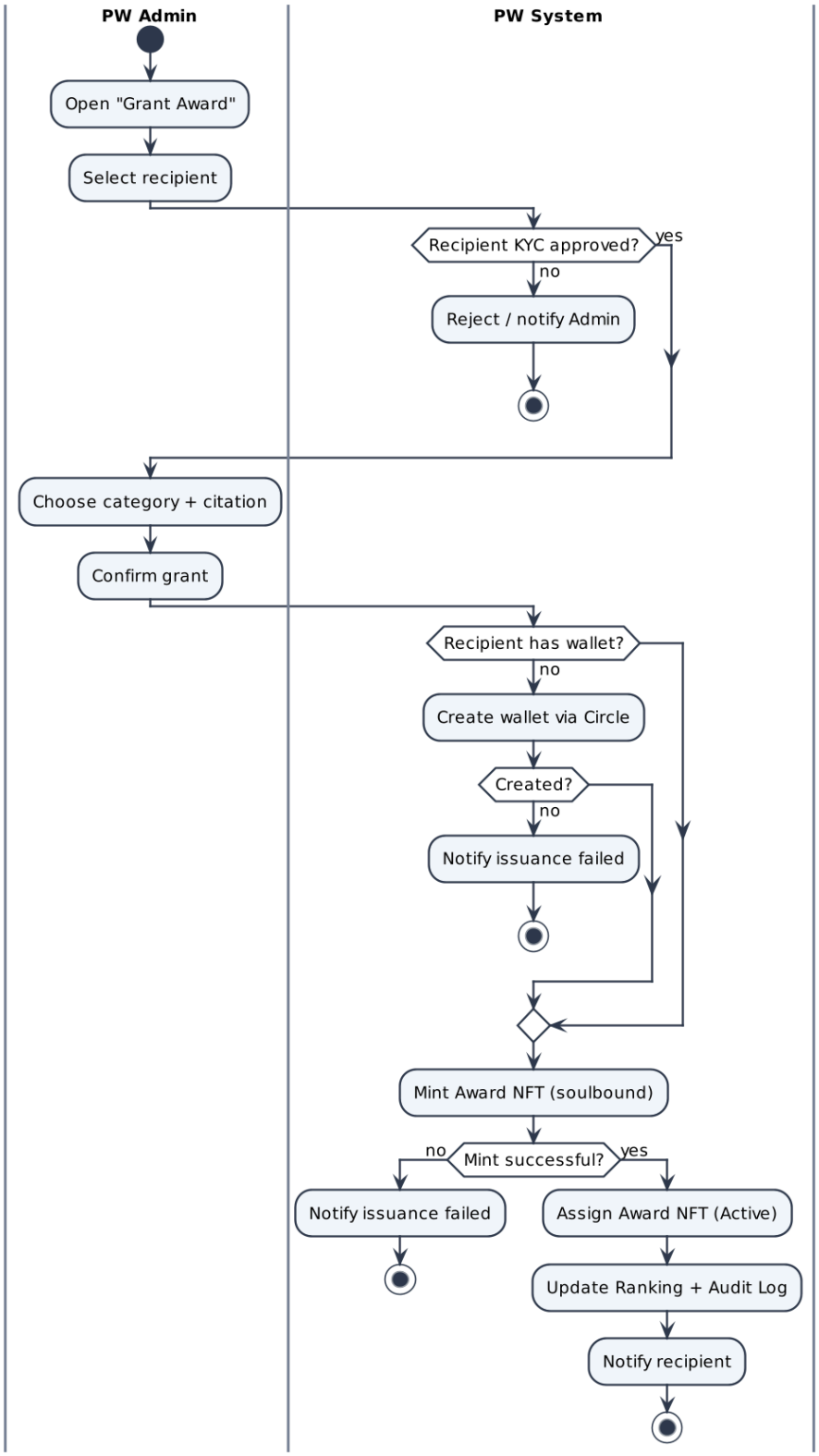
UC29: Grant Award

Use Case Description

UC ID	UC29
UC Name	Grant Award
Objective	Allow a PW Admin to grant a free Award (Investor / Philanthropist / Bridge) to a KYC-verified Member. A soulbound Award NFT is minted directly to the recipient; a custodial wallet is created first if needed.
Actor	PW Admin (R04); PW System (R05); Circle; Blockchain; Recipient (notified)
Trigger	The Admin clicks "Grant Award".
Pre-condition	- The Admin is logged in with the Admin role.
Post-condition	**Success:** • Soulbound Award NFT Active for the recipient; 0 points; Ranking updated; recipient notified; Audit Log (AWARD_MINT). **Failure:** • No Active Award; Admin sees the error and can retry.
Basic flow	1. Admin opens "Grant Award". 2. Admin searches and selects the recipient. 3. System validates the recipient: KYC = Approved. 4. Admin selects the Award category (Investor / Philanthropist / Bridge), enters the citation and clicks "Grant Award". 5. System displays the confirmation "Grant a soulbound [category] Award to [recipient]? 0 points will be credited."; Admin confirms. 6. System checks whether the recipient has a custodial wallet (creates it via Circle if not – UC07). 7. System prepares the issuance and mints the soulbound Award NFT (UC17). 8. System assigns the Award NFT (Soulbound, Active) to the recipient. 9. System updates Ranking, records the Audit Log and notifies the recipient ("PW has granted you the [category] Award"). 10. System shows "Award granted successfully."
Alternative flow	**A1 – Recipient has no wallet** 1. At step 6 System creates the wallet via Circle, stores it and continues at step 7. **A2 – Cancel** 1. At step 5 Admin cancels → nothing is issued.
Exception flow	**E1 – Recipient not KYC-verified**

	1. At step 3 → "Recipient must be KYC-verified before receiving an Award."; grant blocked. **E2 – Wallet creation fails** 1. "Award could not be issued because the recipient's wallet could not be created. Please try again." **E3 – Mint fails** 1. "Award could not be issued. Please try again."; no Active Award. **E4 – Missing category or citation** 1. Inline error; "Grant Award" disabled.
Business rule	• BR01 – Only the Admin role can grant. • BR02 – Award uses the same 3 categories as Titles but is free and soulbound (e.g. "Investor Award"). • BR03 – Recipient must be KYC-verified. • BR04 – Grants credit 0 points. • BR05 – No accept step; Active immediately.
Non-functional requirement	• NFR01 – Issuance progress shown until mint confirmation.
Acceptance criteria	**AC01 – Grant with wallet** Given a KYC-approved recipient with a wallet, when the Admin confirms, then a soulbound Award NFT is Active, 0 points, Ranking updated, recipient notified, Audit Log written. **AC02 – Grant without wallet** Given a KYC-approved recipient without a wallet, when the Admin confirms, then a custodial wallet is created and the Award is minted to it. **AC03 – Not KYC-verified** Given the recipient is not approved, when the Admin tries to grant, then the grant is blocked with the KYC message. **AC04 – Mint fails** Given the grant is confirmed, when the mint fails, then "Award could not be issued. Please try again.".
Traceability	BRD: FR-026 · BPMN: D3 – Grant Award

Activity Diagram



Screen Description

Screen 1: Grant Award

[Screenshot: Figma frame "Admin Portal (Grant Award)"]

#	Name	Control type	Data type	Required	Description
1	Recipient	Search / Select	String	Yes	KYC-approved Members only; shows name, type, KYC badge, wallet status.
2	Award category	Radio / Dropdown	Enum	Yes	Investor / Philanthropist / Bridge.
3	Citation	Textarea	String	Yes	Reason for the Award shown on the NFT (max 500 characters – assumption).
4	What happens	Info panel	String		Soulbound Award NFT · 0 points · Ranking updated · recipient notified.
5	Preview	Card	N/A		Award badge preview.
6	Grant Award	Button	N/A		Opens the confirmation dialog.
7	Cancel	Button	N/A		Leaves without granting.
8	Result message	Toast / Alert	String		Success or E1–E3 messages.

UC30: Revoke NFT

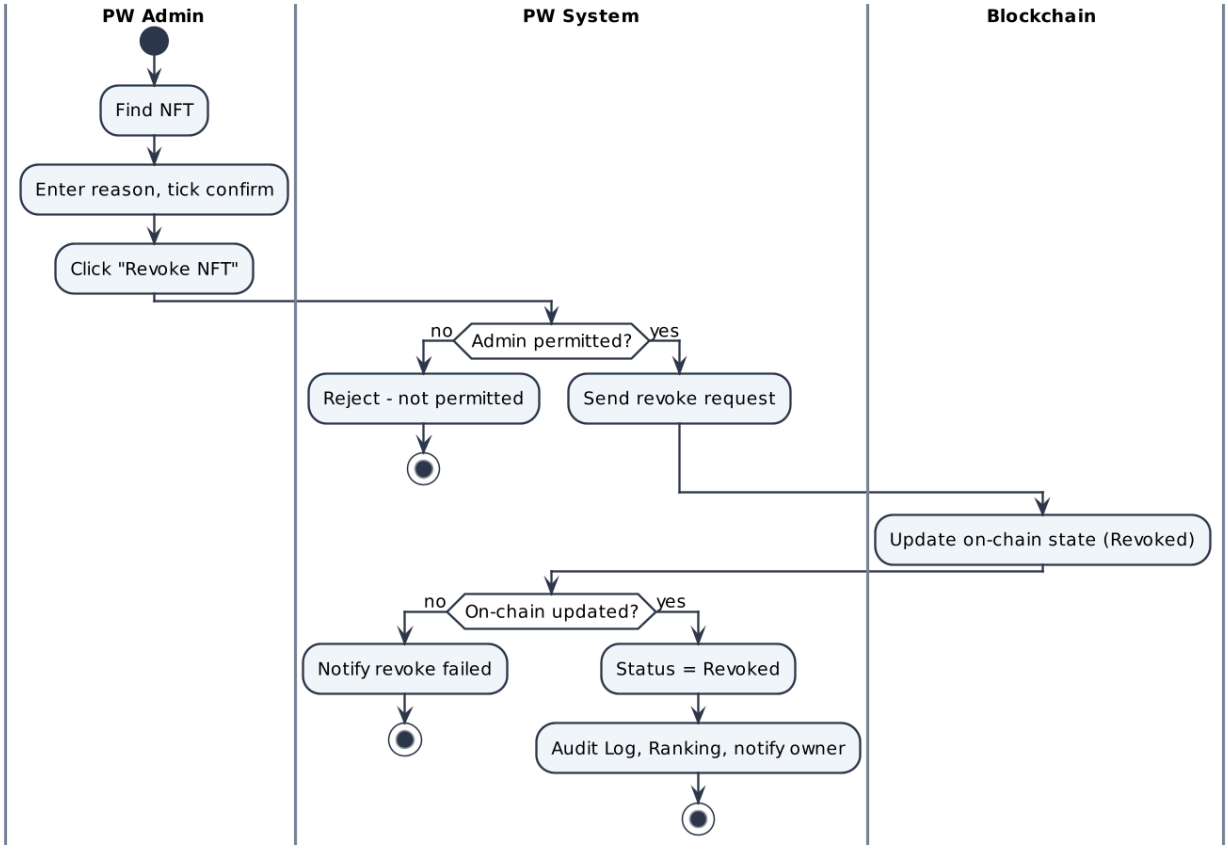
Use Case Description

UC ID	UC30
UC Name	Revoke NFT
Objective	Allow a PW Admin to irreversibly revoke a Distinction or Award NFT so that it is no longer presented as valid; the revocation is written on-chain.
Actor	PW Admin (R04); PW System (R05); Blockchain; NFT owner (notified)
Trigger	The Admin clicks "Revoke NFT".

Pre-condition	- The Admin is logged in. - The NFT status = Active.
Post-condition	**Success:** • NFT status = Revoked on-chain and in PW; Verify returns Revoked; owner notified; Ranking recalculated; Audit Log (NFT_REVOKE). **Failure:** • Status unchanged; Admin notified "Revoke failed".
Basic flow	1. Admin opens "Revoke NFT" and finds the NFT by token ID or owner. 2. System displays the NFT record: name, type (Distinction / Award), owner, current wallet, token ID, issue date. 3. Admin selects a reason (required), optionally adds a note (max 500 characters) and ticks "I understand this action is irreversible". 4. Admin clicks "Revoke NFT". 5. System validates the Admin permission. 6. System sends the revoke request to the smart contract. 7. Blockchain updates the on-chain state to Revoked and returns the result. 8. System sets NFT status = Revoked, records the Audit Log, recalculates Ranking and notifies the owner. 9. System shows "NFT revoked."
Alternative flow	**A1 – Cancel** 1. Before step 4 → nothing changes.
Exception flow	**E1 – Not permitted** 1. At step 5 the user lacks the Admin role → "Access denied." **E2 – Missing reason or confirmation** 1. "Revoke NFT" stays disabled. **E3 – On-chain update fails** 1. At step 7 → status unchanged; "Revoke failed. Please try again."; event logged. **E4 – NFT not found** 1. "No NFT found for this token ID or owner."
Business rule	• BR01 – Only Admin can revoke (BR-10); applies to Distinction and Award. • BR02 – Revocation is irreversible. • BR03 – A reason is mandatory.
Non-functional requirement	• NFR01 – Revocation is visible in Verify immediately after on-chain confirmation.
Acceptance criteria	**AC01 – Revoke** Given an Active NFT, when the Admin gives a reason, ticks the confirmation and revokes, then status = Revoked, Verify returns Revoked, owner notified, Audit Log written. **AC02 – Missing reason**

	Given no reason or checkbox, when the Admin tries to revoke, then the button stays disabled. **AC03 – On-chain fails** Given the revoke is sent, when the on-chain update fails, then status unchanged and "Revoke failed" shown.
Traceability	BRD: FR-027 · BPMN: D5 – Revoke

Activity Diagram



Screen Description

Screen 1: Revoke NFT

[Screenshot: Figma frame "Admin Portal (Revoke NFT)"]

#	Name	Control type	Data type	Required	Description
1	Find NFT	Search	String	Yes	By token ID or owner name.
2	NFT record	Label	Mixed		Name, type (Distinction NFT /

#	Name	Control type	Data type	Required	Description
		group			Award NFT · Soulbound), owner, current wallet, token ID, issue date.
3	Reason	Dropdown	Enum	Yes	e.g. Fraud / misuse, Issued in error, Legal request, Other.
4	Note	Textarea	String		Max 500 characters with counter.
5	Irreversible confirmation	Checkbox	Boolean	Yes	Enables "Revoke NFT".
6	Revoke NFT	Button (destructive)	N/A		Submits the revocation.
7	Recent revocations	Table	Mixed		Last revocations (token, reason, admin, date).

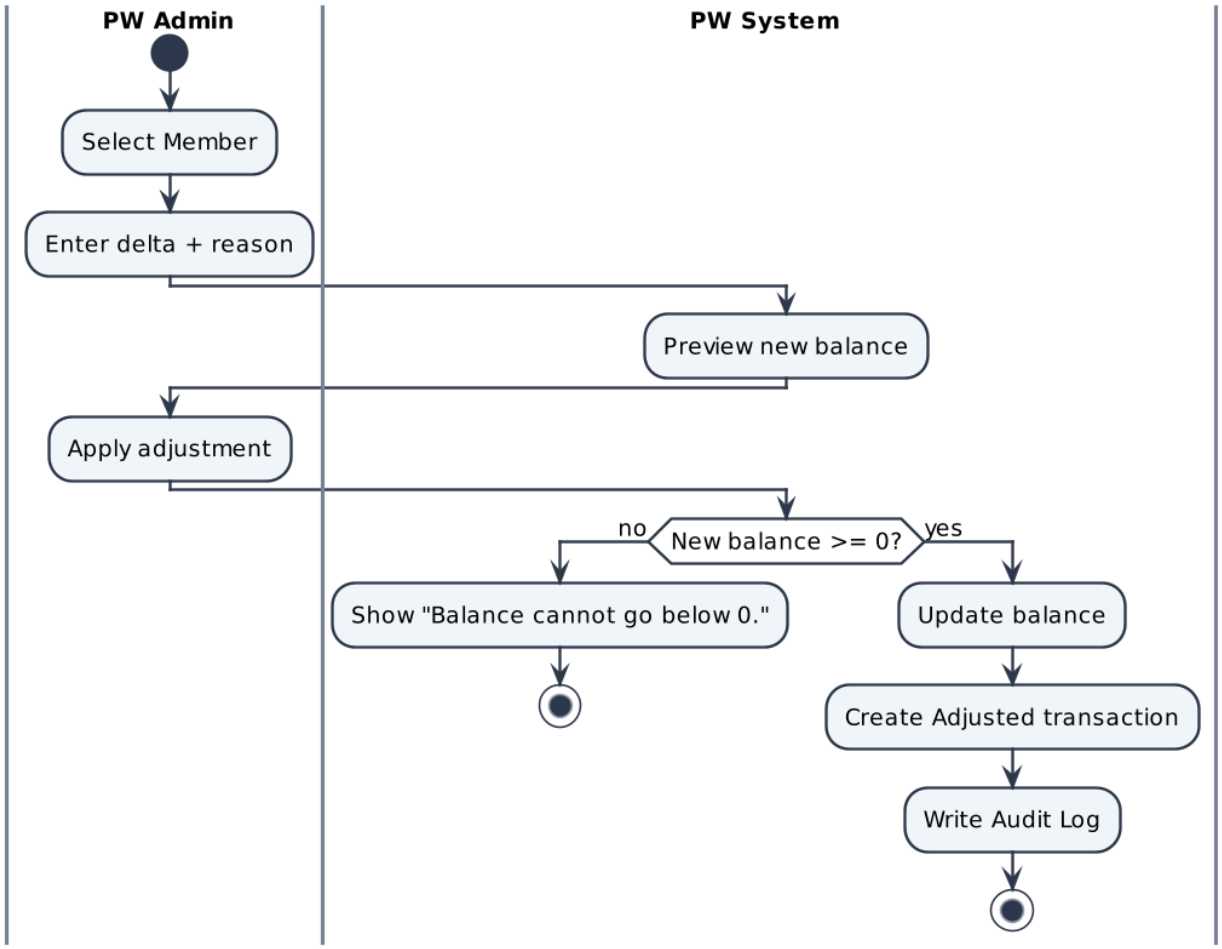
UC31: Manually Adjust Points

Use Case Description

UC ID	UC31
UC Name	Manually Adjust Points
Objective	Allow a PW Admin to add or deduct points on a Member account with a mandatory reason, for corrections and exceptions.
Actor	PW Admin (R04); PW System (R05)
Trigger	The Admin confirms an adjustment.
Pre-condition	- The Admin is logged in.
Post-condition	**Success:** • Balance updated; "Adjusted" point transaction; Audit Log (POINTS_ADJUST) with old/new balance and reason.
Basic flow	1. Admin opens "Adjust Points" and selects the Member. 2. System shows the current balance and recent point history. 3. Admin enters a delta (+ credit / − debit) and a reason.

	4. System previews the new balance. 5. Admin clicks "Apply adjustment". 6. System validates the new balance ≥ 0, updates the balance, creates an "Adjusted" transaction and writes the Audit Log.
Alternative flow	**A1 – Cancel** 1. No change.
Exception flow	**E1 – Negative balance** 1. "Balance cannot go below 0." **E2 – Missing reason or zero delta** 1. Inline error; button disabled.
Business rule	• BR01 – Only Admin. • BR02 – Reason mandatory. • BR03 – Balance never below 0.
Non-functional requirement	• NFR01 – Every adjustment traceable (NFR-002, NFR-010).
Acceptance criteria	**AC01 – Adjust** Given a Member with balance 300, when the Admin applies −100 with a reason, then balance = 200, an Adjusted transaction exists and the Audit Log shows old 300 / new 200. **AC02 – Negative** Given balance 50, when the Admin applies −100, then "Balance cannot go below 0.".
Traceability	BRD: FR-028 · BPMN: —

Activity Diagram



Screen Description

Screen 1: Adjust Points

[Screenshot: Figma frame "Admin Portal (Adjust Points)"]

#	Name	Control type	Data type	Required	Description
1	Target member	Search / Select	String	Yes	Member account.
2	Current points	Label	Integer		Current balance.
3	Adjustment type	Toggle	Enum	Yes	+ Credit / − Debit.
4	Amount	Numeric input	Integer	Yes	> 0.

#	Name	Control type	Data type	Required	Description
5	Reason	Textarea	String	Yes	Shown in the Audit Log.
6	New balance preview	Label	Integer		Calculated live.
7	Apply adjustment	Button	N/A		Applies the change.

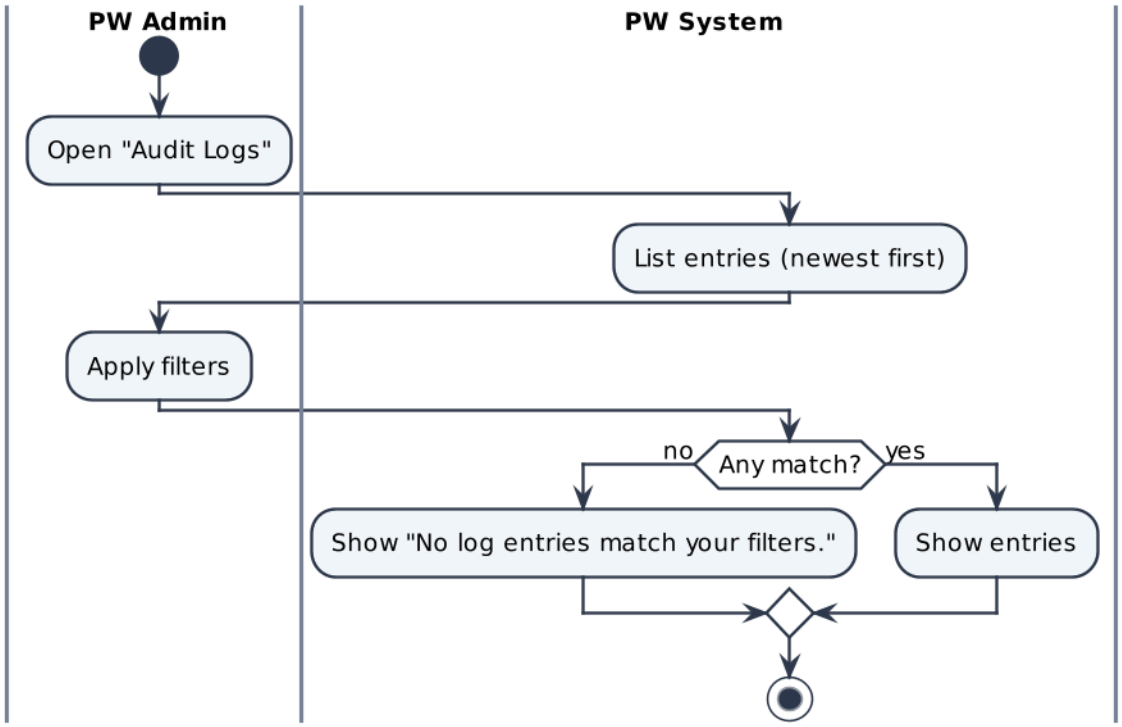
UC32: View Audit Logs

Use Case Description

UC ID	UC32
UC Name	View Audit Logs
Objective	Give the Admin a read-only, filterable trail of all important actions for compliance and traceability.
Actor	PW Admin (R04)
Trigger	The Admin opens "Audit Logs".
Pre-condition	- The Admin is logged in.
Post-condition	**Success:** • Entries displayed; nothing can be changed.
Basic flow	1. Admin opens "Audit Logs". 2. System lists entries newest first: time, actor, action, target, reference, details. 3. Admin filters by date range, actor, action type (KYC, Class, Mint, Transfer, Revoke, Points, Wallet, Reconcile) or target. 4. System shows matching entries (paginated).
Alternative flow	**A1 – No result** 1. "No log entries match your filters."
Exception flow	**E1 – Load error** 1. Retry message.
Business rule	• BR01 – Entries are immutable: no edit or delete for any role (NFR-002). • BR02 – Logged actions: account created, KYC decisions, Class changes, wallet create/link/unlink, purchase/gift, mint, transfer,

	custody change, revoke, grant, points earn/use/adjust, reconciliation findings.
Non-functional requirement	• NFR01 – Filter results within 3 seconds for 1 year of data (proposed).
Acceptance criteria	**AC01 – View** Given the Admin opens Audit Logs, when the page loads, then entries are listed newest first and cannot be edited or deleted. **AC02 – Filter** Given filters applied, when results load, then only matching entries are shown. **AC03 – No result** Given nothing matches, when filter applied, then "No log entries match your filters.".
Traceability	BRD: FR-029 · BPMN: D2–D5 – Record Audit Log

Activity Diagram



Screen Description

Screen 1: Audit Logs

[Screenshot: Figma frame "Admin Portal (Audit Logs)"]

#	Name	Control type	Data type	Required	Description
1	Date range	Date range picker	Date		From / to.
2	Actor	Search	String		Admin, Member or System.
3	Action type	Multi-select	Enum		KYC_APPROVE, KYC_REJECT, CLASS_CHANGE, WALLET_*, PURCHASE, MINT, AWARD_MINT, TRANSFER_*, NFT_REVOKE, POINTS_*, RECONCILE.
4	Target / reference	Textbox	String		Token ID, account, order.
5	Log table	Table / Grid	Mixed		Time, actor, action, target, reference, details; read-only.

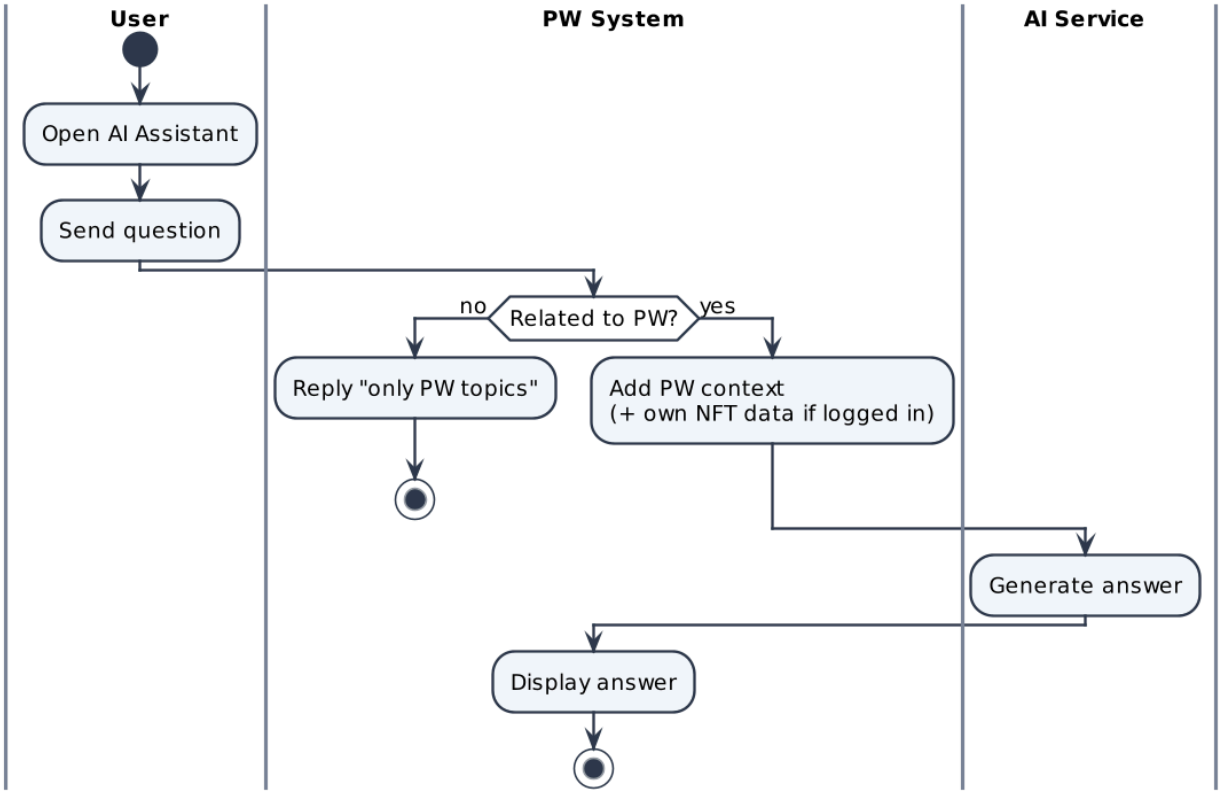
UC33: Ask AI Assistant

Use Case Description

UC ID	UC33
UC Name	Ask AI Assistant
Objective	Provide an AI Assistant that answers questions about PW, looks up the logged-in Member's own NFTs, gives platform guidance and recommends a Title category.
Actor	Visitor (R01), Company (R02), Individual (R03); AI Service (external)
Trigger	The user opens the AI Assistant and sends a message.
Pre-condition	- NFT lookup requires login.
Post-condition	**Success:** • An answer is displayed. No business data is changed.
Basic flow	1. User opens the AI Assistant.

	2. User types a question (e.g. "How do points work?") and sends it. 3. System sends the question with PW context (and, if logged in, the user's own NFT data) to the AI Service. 4. System displays the answer within 2–5 seconds.
Alternative flow	**A1 – NFT lookup** 1. A logged-in user asks about their NFT → type, status, custody and transfer availability of that NFT. **A2 – Title recommendation** 1. User describes their business → the Assistant suggests Philanthropist, Investor or Bridge with a short reason. **A3 – Out of scope** 1. "I can only help with questions about PW."
Exception flow	**E1 – AI Service unavailable** 1. "The assistant is unavailable right now. Please try again later."
Business rule	• BR01 – Never discloses another user's private data. • BR02 – The Assistant is read-only; it cannot buy, transfer or change data. • BR03 – **TBD: official scope of the AI Assistant.**
Non-functional requirement	• NFR01 – Response within 2–5 seconds under normal conditions (NFR-001).
Acceptance criteria	**AC01 – Question** Given the Assistant is open, when a PW question is sent, then an answer appears within 2–5 seconds. **AC02 – NFT lookup** Given logged in, when the user asks about their NFT, then that NFT's type, status, custody and transfer availability are returned. **AC03 – Out of scope** Given an unrelated question, when it is sent, then the Assistant replies it can only help with PW topics.
Traceability	BRD: FR-030 · BPMN: —

Activity Diagram



Screen Description

Screen 1: AI Assistant

[Screenshot: Figma frame "PW — AI Assistant (to be designed)"]

#	Name	Control type	Data type	Required	Description
1	Chat panel	Panel	N/A		Floating panel available on all pages.
2	Suggested questions	Chips	String		e.g. "How do points work?", "Which Title fits me?".
3	Message input	Textbox	String	Yes	Max 500 characters (assumption).
4	Send	Button	N/A		Sends the question.
5	Answer	Label	String		Assistant reply.

OPEN ISSUES

ID	Open issue (decision needed)	Related UC
OI-01	Headcount range of each Class and Title price per Class × Title.	UC06, UC12, UC16
OI-02	Pricing basis for Individuals (Class 0 or a separate table).	UC06, UC12, UC13, UC16
OI-03	Year-2 discount: point threshold and how "Year 2" is counted.	UC16, UC23
OI-04	Self-purchase points: buyer = recipient → 10% or 20%?	UC14, UC18
OI-05	Points redemption: discount gate only, or deductible from the price (rate)?	UC16, UC24
OI-06	Ranking: Award vs Distinction weighting, tie-break, revoked NFTs counted or not.	UC28
OI-07	Payment succeeded but wallet creation / minting failed: retry policy and refund rule.	UC07, UC14, UC15, UC17
OI-08	Align US-TN03 / US-AW01 with BRD + BPMN D2 (gift recipient = KYC-verified Member; wallet created if missing).	UC07, UC15
OI-09	Frequency of the linked-wallet reconciliation job.	UC22
OI-10	Official scope of the AI Assistant.	UC33
OI-11	Final KYC field and document list (Compliance).	UC04, UC05

Assumptions shown in gold (e.g. gift priced on the buyer's Class, recipient search keys, proposed NFR timings) must be confirmed together with these issues.